



A COMPLETE STUDY GUIDE | NO CODE, ALL CONCEPTS

Applied Machine Learning for Earth Science

Twelve NASA Earthdata projects explained in depth: the Earth science, the free datasets, the methods, the evaluation, and the Google Colab strategy.

Floods | Wildfire | Urban heat | Crop stress | Forest carbon | Landslides
Coastal erosion | Desertification | Air pollution | Glaciers | Algal blooms | Urban growth

Contents

Foundations	3
1. How to Use This Book	4
2. Earth Observation Fundamentals	7
3. NASA Earthdata and the Open-Data Ecosystem	12
4. Machine Learning for Geospatial Data	15
5. Working in Google Colab	19
 The Twelve Projects	 21
Project 1: AI Flood Extent & Damage Detector	22
Project 2: Wildfire Spread Predictor	27
Project 3: Urban Heat Island Mapper	32
Project 4: Crop Water-Stress Detector	37
Project 5: Forest Carbon Estimator	42
Project 6: Landslide Susceptibility Predictor	47
Project 7: Coastal Erosion Predictor	52
Project 8: Desertification Early-Warning System	57
Project 9: Air Pollution Hotspot Predictor	62
Project 10: Glacier Retreat & Melt Predictor	67
Project 11: Harmful-Algal-Bloom Detector	72
Project 12: Urban Expansion Forecaster	77
 Appendices	 82
Appendix A: Dataset Directory	83
Appendix B: Index Formulas and Metric Reference	86
Appendix C: Skills Map and Capstone Plan	88
Appendix D: Glossary	90
Appendix E: Responsible Use and Failure Modes	92
Appendix F: Further Learning	93

PART I

Foundations

The science, data and machine-learning ideas you need before starting any project.

1. How to Use This Book

Purpose, project map and the universal workflow

What this book is

This book is a complete, code-free study guide for twelve Earth-science machine-learning projects built on NASA Earthdata and other openly available satellite data. For every project it explains the underlying Earth science, the data you can download for free, how to prepare it, which machine-learning methods suit the problem and why, how to evaluate results honestly, and how to fit the work inside a free Google Colab session.

The book deliberately contains no solution code. Instead it gives you what you need to design the solution yourself: the reasoning, the vocabulary, the pipeline, the pitfalls and worked numerical examples. If you can read this book and then sketch the full pipeline on paper, you are ready to implement it.

Why these projects

Most beginner satellite projects stop at "classify a picture of land cover". The twelve projects here go further: each answers a question that a real agency, city or farmer might ask, such as *Which neighbourhoods will suffer the strongest heat stress?* or *Where is this fire likely to spread in the next 24 hours?* They combine remote sensing, image processing, tabular machine learning, time-series analysis and geospatial reasoning, and they can all be scaled down to small regions, short time windows, patches and lightweight models that run on a free Colab account.

The twelve projects at a glance

#	Project	Main data	ML / CV approach	Difficulty	Impact
1	AI Flood Extent & Damage Detector	HLS, Sentinel-1, GPM IMERG, DEM	Pixel classifier to U-Net	3/5	5/5
2	Wildfire Spread Predictor	VIIRS/MODIS fire, NDVI, weather, terrain	Tabular ML to CNN	4/5	5/5
3	Urban Heat Island Mapper	ECOSTRESS, Landsat, land cover	Regression to CNN	3/5	5/5
4	Crop Water-Stress Detector	ECOSTRESS, Landsat/HLS, SMAP, rainfall	Regression and forecasting	3/5	5/5
5	Forest Carbon Estimator	GEDI, Landsat/Sentinel, terrain	Regression, upscaling	3/5	4/5
6	Landslide Susceptibility Predictor	DEM, rainfall, imagery, inventories	RF / XGBoost	3/5	5/5
7	Coastal Erosion Predictor	Landsat/HLS/Sentinel-2	Classical CV + regression	4/5	4/5
8	Desertification Early-Warning System	MODIS/VIIRS, HLS, climate	Time-series ML, anomaly detection	4/5	5/5
9	Air Pollution Hotspot Predictor	MAIAC/VIIRS AOD, OMI/TROPOMI, ERA5	Regression, spatial transfer	3/5	4/5
10	Glacier Retreat & Melt Predictor	Landsat, Sentinel-2, MODIS/VIIRS snow	CV + time series	4/5	4/5

#	Project	Main data	ML / CV approach	Difficulty	Impact
11	Harmful-Algal-Bloom Detector	MODIS, VIIRS, PACE, Sentinel-3	Clustering, anomaly detection	4/5	5/5
12	Urban Expansion Forecaster	Landsat/HLS, nighttime lights, GHSL	CNN + temporal ML	4/5	4/5

Difficulty and impact are relative ratings out of 5 for a learner working in free Colab. Impact reflects real-world decision value, particularly for India.

How each project chapter is organised

- **Project at a glance** – goal, data, method, difficulty and Colab fit in one box.
- **Problem and motivation** – the real-world question and why it matters.
- **Earth-science background** – the physical or ecological processes the model must capture.
- **Data** – a table of free datasets with provider, resolution, how to access them, and the role each plays.
- **Pipeline and preparation** – the end-to-end workflow, preprocessing and feature engineering.
- **Machine-learning approach** – baselines, main models, advanced options, training choices.
- **Worked example** – a concrete numerical walk-through (hypothetical numbers, clearly labelled).
- **Evaluation** – metrics, validation design, error analysis.
- **Colab strategy, extensions, pitfalls and deliverables.**

Suggested learning paths

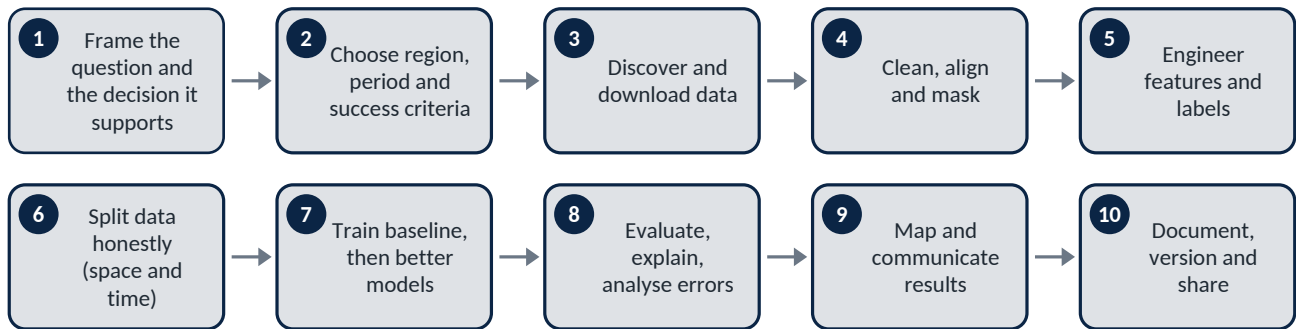
Level	Start with	Then	Skills you will build
Beginner	Project 3 (Urban Heat) or Project 6 (Landslide)	Project 1 (Flood), Project 5 (Forest Carbon)	Rasters, indices, tabular ML, validation, mapping results
Intermediate	Project 9 (Air Pollution), Project 4 (Crop Stress)	Project 7 (Coastal), Project 10 (Glacier)	Time series, classical computer vision, regression forecasting, spatial transfer
Advanced	Project 2 (Wildfire), Project 8 (Desertification)	Project 11 (Algal Bloom), Project 12 (Urban Growth)	Spatiotemporal modelling, anomaly detection, CNN/U-Net, scenario simulation

WATCH OUT A note on honesty in numbers

Whenever this book gives example performance numbers, rainfall thresholds or probabilities, they are hypothetical teaching values and are labelled as such. Real results depend on your region, your labels and your validation design. Part of being a good Earth-data scientist is reporting what the data actually shows, including when a model fails.

The universal project lifecycle

Every project in this book follows the same ten-stage lifecycle. Learn it once and you can apply it to any geospatial machine-learning problem.



2. Earth Observation Fundamentals

Sensors, resolution, rasters, radar, thermal and lidar in plain language

1. Remote sensing in one page

Remote sensing is measuring something about the Earth without touching it. A sensor on a satellite records electromagnetic energy that has been reflected, emitted or scattered by the surface and atmosphere. Different materials interact with energy in different ways, and that difference is what a machine-learning model learns to exploit.

- **Passive sensors** (Landsat, Sentinel-2, MODIS, VIIRS, ECOSTRESS, PACE) measure sunlight reflected by the surface or heat emitted by it. They usually need daylight for reflected light and are blocked by clouds.
- **Active sensors** carry their own energy source. Radar (Sentinel-1, ALOS PALSAR, NISAR) sends microwaves and measures the echo, so it works through cloud and at night. Lidar (GEDI, ICESat-2) sends laser pulses and measures return time and shape, giving heights and structure.

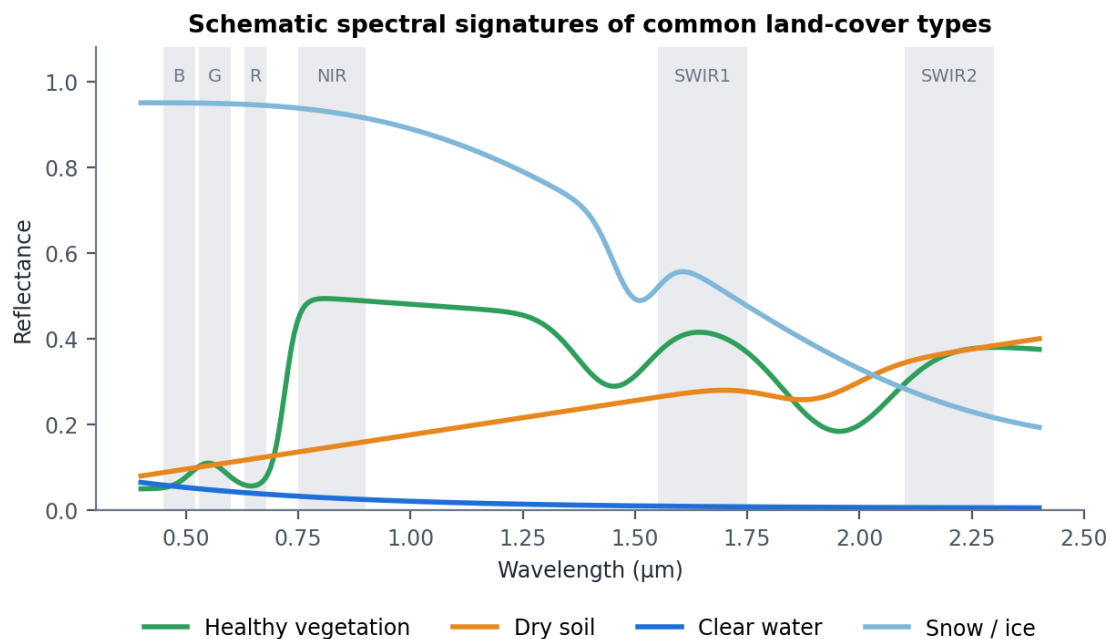


Figure 1. Schematic spectral signatures (synthetic curves for teaching). Healthy vegetation is dark in red but very bright in near-infrared (NIR); water is dark everywhere beyond the visible; snow is bright in visible light but darker in shortwave infrared (SWIR). Spectral indices are built from exactly these contrasts.

2. The four kinds of resolution

Resolution	Meaning	Typical trade-off
Spatial	Size of one pixel on the ground (e.g. 10 m, 30 m, 250 m, 1 km)	Fine pixels see detail but cover small areas and give huge files; coarse pixels cover large areas but mix several land covers in one pixel
Spectral	Number and width of wavelength bands	More bands separate materials better (hyperspectral PACE, ECOSTRESS thermal bands) but increase data volume

Resolution	Meaning	Typical trade-off
Temporal	How often the same place is observed	Daily MODIS/VIIRS catches fast change but at coarse scale; 16-day Landsat misses short events
Radiometric	How finely brightness is quantised (8, 12, 16 bit)	Higher depth detects subtle differences, e.g. dark water or thin aerosol

No sensor is best at everything. The art of Earth-data science is combining sensors so that each covers the weakness of another: for example daily coarse data to know *when* something changed and 10–30 m data to know *exactly where*.

3. Orbits and overpass time

Most land-observing satellites fly in **sun-synchronous** orbits, passing over any location at roughly the same local solar time each day (Terra about 10:30 in the morning, Aqua about 13:30 in the afternoon, Landsat about 10:00–10:30). This keeps illumination consistent. The International Space Station, which carries ECOSTRESS and GEDI, is **not** sun-synchronous: its orbit precesses, so it samples different times of day over a few weeks, which is ideal for studying the daily cycle of temperature but means the data are irregular. ISS-based instruments only cover latitudes up to roughly 52° (ECOSTRESS) or 51.6° (GEDI), which includes all of India.

WATCH OUT Local time matters

Land surface temperature at 10:30 and at 13:30 can differ by more than 10 °C. Always record the overpass time and, when comparing scenes, either compare at similar local times or include the time as a feature. Satellite files use UTC; India Standard Time is UTC+5:30.

4. Sensor cheat-sheet

Sensor / product	Measures	Pixel size	Revisit	Typical use in this book
Landsat 8 & 9 (OLI, TIRS)	Visible, NIR, SWIR, thermal	30 m (thermal 100 m, resampled to 30 m in products)	8 days combined	Heat, coast, glacier, urban, vegetation
Sentinel-2 (MSI)	Visible, red-edge, NIR, SWIR	10, 20, 60 m	about 5 days	Flood, forest, coast, glacier, urban
HLS (Harmonized Landsat Sentinel-2)	Landsat + Sentinel-2 made consistent	30 m	about 2–3 days	Flood, crops, desertification, coast
Sentinel-1 (C-band SAR)	Radar backscatter VV/VH	about 10 m	6–12 days	Flood, landslide, glacier, forest
MODIS (Terra, Aqua)	36 bands, many land/ocean/atmosphere products	250 m – 1 km	daily	Fire, NDVI, LST, AOD, ocean colour (archive since 2000)
VIIRS (Suomi-NPP, NOAA-20/21)	Successor to MODIS; active fire at 375 m	375 m – 750 m	daily	Fire, NDVI, snow, nighttime lights, AOD
ECOSTRESS	Thermal infrared, plant water use	about 70 m	irregular (ISS)	Heat, crop stress
GEDI	Waveform lidar canopy structure	25 m footprints along tracks	sparse sampling	Forest height and biomass

Sensor / product	Measures	Pixel size	Revisit	Typical use in this book
GPM IMERG	Merged satellite rainfall	0.1° (about 10 km)	30 min / daily	Flood, landslide, crop, desert
SMAP	Soil moisture (L-band)	9–36 km	2–3 days	Crop, desertification
PACE (OCI)	Hyperspectral ocean colour, UV to NIR	about 1 km	about 1–2 days	Algal blooms
OMI / TROPOMI	NO ₂ , SO ₂ , aerosols	13×24 km / about 5×3.5 km	daily	Air pollution

Specifications are summarised for orientation; check each product's user guide for exact values and current status before designing a study.

5. Processing levels and quality

Level	Description	Example
L0 / L1	Raw counts and radiances (calibrated sensor measurements)	MODIS L1B radiance, Landsat L1 top-of-atmosphere
L2	Geophysical variable at original swath/pixel grid after corrections	Surface reflectance, land surface temperature, chlorophyll-a, AOD swaths
L3	Variable mapped to a regular grid and often averaged over a day, 8 days or month	MOD13 vegetation indices, GPM daily rainfall, gridded chlorophyll
L4	Model-assisted or gap-filled products	Evapotranspiration, GEDI gridded biomass, soil-moisture analyses

For almost all machine-learning work you want **surface reflectance** (atmospherically corrected) rather than raw top-of-atmosphere values, and you must use the **quality layers** that accompany each product: cloud, cloud-shadow, snow, water, aerosol and "retrieval failed" flags. Ignoring them is the single most common reason beginner models behave strangely.

6. Working with rasters

- **Coordinate reference system (CRS):** how pixel positions map to latitude/longitude or metres. Landsat/Sentinel/HLS tiles use UTM zones; MODIS uses a sinusoidal grid; global model data use geographic degrees. Always reproject layers to one common CRS and grid before stacking.
- **Tiles:** HLS and Sentinel-2 use the MGRS tile system (about 110 km squares); MODIS uses h/v tiles of 10° by 10°. One city can straddle tile edges.
- **Scale factors and fill values:** products store integers to save space. MODIS NDVI is stored multiplied by 10,000, and land surface temperature in units of 0.02 K. Fill values (such as –3000 or 65535) mean no data. Apply the documented scale and mask fill values first.
- **Resampling:** use nearest-neighbour for categorical layers (land cover) and bilinear or area-weighted averaging for continuous layers. Resampling changes the statistics, so do it deliberately.
- **Cloud-optimised GeoTIFF (COG) and chunking:** lets you read just the window you need from a huge file over the network, which is essential in Colab.

7. The vocabulary of spectral indices

A spectral index combines two or more bands to highlight a property while cancelling out nuisances such as illumination. The most important are summarised here (a larger list is in Appendix B).

Index	Idea	Highlights
NDVI	$(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$	Green vegetation vigour; values near 0 for bare soil, 0.6–0.9 for dense vegetation, negative for water
EVI	NDVI with blue-band correction and gain	Dense canopies where NDVI saturates
NDWI (McFeeters)	$(\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$	Open water
MNDWI	$(\text{Green} - \text{SWIR1}) / (\text{Green} + \text{SWIR1})$	Water in built-up areas (suppresses buildings better than NDWI)
NDMI / NDWI (Gao)	$(\text{NIR} - \text{SWIR1}) / (\text{NIR} + \text{SWIR1})$	Vegetation water content
NBR	$(\text{NIR} - \text{SWIR2}) / (\text{NIR} + \text{SWIR2})$	Burned areas and burn severity
NDBI	$(\text{SWIR1} - \text{NIR}) / (\text{SWIR1} + \text{NIR})$	Built-up surfaces
NDSI	$(\text{Green} - \text{SWIR1}) / (\text{Green} + \text{SWIR1})$	Snow and ice

8. Radar (SAR) in plain language

A radar satellite measures **backscatter**: how much of the transmitted microwave energy returns. Smooth surfaces such as calm water reflect the beam away like a mirror, so they appear dark. Rough surfaces (crops, rocks) scatter some energy back and look brighter. Vertical structures such as buildings or flooded trees can create a "double bounce" that appears very bright. Backscatter is usually converted to decibels (dB). Two polarisations are common: VV (vertical send, vertical receive) and VH (cross-polarised, sensitive to volume scattering from vegetation).

Radar images contain **speckle**, a grainy noise that must be filtered, and geometric effects in mountains (layover, foreshortening, shadow) that make terrain correction essential. In return, radar sees through monsoon clouds, which makes it the preferred sensor for flood mapping in India.

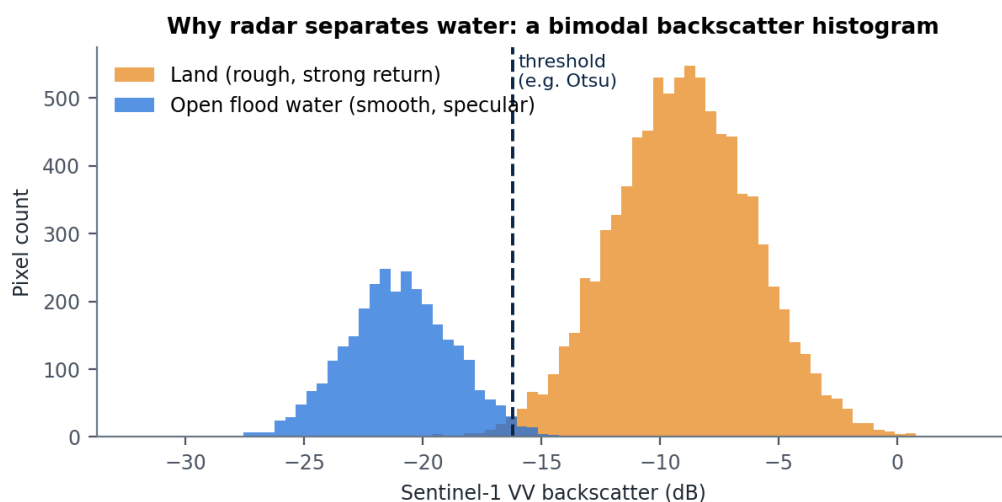


Figure 2. Open water is smooth and returns little energy to the radar, so its histogram separates from land; a threshold such as Otsu's method can split the two. (Synthetic distributions.)

9. Thermal infrared and land surface temperature

Thermal sensors measure emitted heat. The raw quantity is **brightness temperature**, which is the temperature a perfect emitter would have to produce the observed radiance. To get true **land surface temperature (LST)**, you correct for the atmosphere and for **emissivity** (how efficiently a material emits heat; vegetation is near 0.98, bare soil and concrete can be lower). Products such as ECOSTRESS LSTE, Landsat Collection 2 surface temperature and MODIS MOD11 provide LST ready to use, with different resolutions and overpass times. Note that LST is the temperature of the *skin* of the surface (roof, road, leaf), not the air temperature that a weather station reports.

10. Lidar and canopy structure

GED1 fires a laser at the forest and records the returned energy as a **waveform**: energy returned from the top of the canopy, the middle layers and the ground. From it come metrics such as **RH98** (the height above ground below which 98 percent of returned energy lies, a robust measure of canopy height), canopy cover, and plant area index. Models relate these metrics to measured biomass in field plots.

11. Common pitfalls in satellite data

- **Clouds and shadows** contaminate optical data. During monsoon months, entire scenes may be unusable.
- **Mixed pixels**: a 1 km pixel over an Indian city contains roofs, trees, roads and water together.
- **Phenology**: vegetation changes with season. Comparing January to July without accounting for the season produces false "change".
- **Sun-sensor geometry (BRDF) and topography** change apparent brightness, especially in mountains.
- **Modifiable areal unit problem**: results change when you change the grid size or region boundaries.
- **Sensor change and drift**: a time series spanning different sensors or drifting orbits may contain artificial trends.
- **Edge effects and seams** where tiles or swaths meet.

3. NASA Earthdata and the Open-Data Ecosystem

Where to find data and how to get just the part you need

1. NASA Earthdata in one picture

NASA's Earth Observing System Data and Information System (EOSDIS) distributes satellite, airborne and field data through a set of **Distributed Active Archive Centers (DAACs)**, each specialising in a science discipline. All of the data are free, but most downloads require a free **Earthdata Login** account. A single search interface (Earthdata Search) and a programmatic catalogue (Common Metadata Repository, CMR) let you discover data across all centres.

DAAC	Specialty	Datasets used in this book
LP DAAC (USGS EROS)	Land processes	HLS, MODIS/VIIRS land products (NDVI, LST, burned area, land cover, albedo), ECOSTRESS, GEDI L1-L2, NASADEM, MAIAC aerosol
GES DISC (NASA Goddard)	Atmosphere, precipitation, hydrology	GPM IMERG rainfall, OMI trace gases, MERRA-2 reanalysis, GLDAS/NLDAS land models
NSIDC DAAC	Snow, ice, cryosphere, SMAP	MODIS/VIIRS snow cover, glacier products, SMAP soil moisture, GLIMS
OB.DAAC (Ocean Biology)	Ocean colour	MODIS-Aqua, VIIRS, PACE chlorophyll and reflectance products
ORNL DAAC	Biogeochemistry, ecology	GEDI L4 biomass, global biomass and carbon maps, field campaign data
PO.DAAC	Physical oceanography	Sea surface temperature, sea level, ocean winds
ASF DAAC	Synthetic aperture radar	Sentinel-1, ALOS PALSAR, OPERA radar-derived products, NISAR
LAADS DAAC	MODIS/VIIRS atmosphere and level-1	MODIS/VIIRS L1B radiances, aerosol, cloud, and fire-related data
SEDAC (Columbia)	Human dimensions	Population grids, settlements, urban extents, PM2.5 surfaces

2. Tools for finding and downloading data

Tool	What it does	Best for
Earthdata Search	Web portal to search, preview and subset thousands of collections	Exploration, choosing products and time ranges
earthaccess (Python library)	Logs in with your Earthdata account, searches CMR, streams or downloads granules	Scripted, repeatable access in Colab
AppEEARS	Cloud service that extracts and subsets data for points or polygons (MODIS, VIIRS, ECOSTRESS, HLS and others) and returns small, analysis-ready files	Time series for your study area without downloading global tiles
Giovanni	Browser-based plotting of gridded NASA time series and maps	Quick looks at rainfall, aerosol, soil moisture
Worldview and FIRMS	Interactive global imagery browser and fire map	Finding events, fire hotspots and cloud-free dates

Tool	What it does	Best for
OPeNDAP / Harmony	Remote subsetting and format conversion services	Reading just a region and variable from big files

TIP Subset in the cloud, not on your laptop

A single global MODIS or IMERG archive is terabytes. Always ask the server for *your* region, *your* dates and *your* variables (using AppEEARS, Harmony, STAC queries or Earth Engine). This is the most important habit for working within Colab limits.

3. Other free and open data sources

Source	What you get	Access
Copernicus Data Space Ecosystem	Sentinel-1, 2, 3, 5P imagery	Free registration; browser, API, or STAC
Copernicus Climate Data Store	ERA5 and ERA5-Land reanalysis (temperature, wind, humidity, rain, boundary-layer height)	Free account and API
USGS EarthExplorer	Landsat Collection 2 archive since 1972–1984	Free account
Microsoft Planetary Computer	STAC catalogue with Landsat, Sentinel, MODIS, NASADEM, Copernicus DEM, land cover and more	Free data access via API; hub features may need an application
Google Earth Engine	Huge catalogue with server-side compute (MODIS, Landsat, Sentinel, CHIRPS, ERA5, GEDI, WorldCover, Dynamic World)	Free for non-commercial and research use after registration; Python API in Colab
AWS Open Data and Google Cloud public datasets	Mirrors of Landsat, Sentinel, HLS-related, and many gridded datasets	Public buckets, no cost for reading data in many cases
OpenStreetMap, Google Open Buildings, Microsoft Building Footprints	Roads, rivers, buildings, land use	Free downloads and APIs
India-specific: ISRO Bhuvan, NRSC, India-WRIS, IMD, CPCB, INCOIS, GSI Bhukosh, data.gov.in	Thematic atlases, gridded rainfall, river data, air-quality stations, ocean advisories, geology	Public portals; some require registration

4. File formats you will meet

Format	Typical content	Notes
GeoTIFF / COG	Single or multi-band rasters	COG allows partial reads over the network
NetCDF4 / HDF5	Multi-dimensional gridded data (time, lat, lon, variable) with metadata	Standard for climate, rainfall, aerosol; opened with xarray
HDF4 / HDF-EOS	Legacy MODIS products	Harder to read; use AppEEARS or Earth Engine instead where possible
Zarr	Chunked cloud-native arrays	Excellent for large time series
GeoJSON / GeoPackage / Shapefile / GeoParquet	Vector points, lines, polygons	Labels, boundaries, inventories
CSV	Tabular station and inventory data	Remember to document coordinate system and time zone

5. The Python geospatial toolbox (concepts, not code)

- **xarray + rioxarray + rasterio**: labelled multi-dimensional rasters and time series; the backbone of most workflows.
- **GeoPandas + Shapely + pyproj**: vector data, spatial joins, reprojection.
- **pystac-client + odc-stac or stackstac**: search STAC catalogues and build lazily loaded image stacks.
- **Dask**: process data in chunks that fit in memory.
- **earthaccess, AppEEARS API, earthengine-api**: data access.
- **scikit-learn, XGBoost/LightGBM**: tabular machine learning; **PyTorch** (with `segmentation_models_pytorch` or `TorchGeo`) or **TensorFlow/Keras**: deep learning.
- **OpenCV and scikit-image**: thresholding, edge detection, contours and morphology.
- **folium, leafmap, geemap, cartopy, matplotlib**: maps and figures.

6. Important status note: MODIS is winding down

WATCH OUT Plan for the MODIS to VIIRS transition

NASA's Terra and Aqua satellites, which carry MODIS, are far beyond their design lives, and their orbits have been drifting away from their original overpass times. NASA has advised users to move toward the equivalent VIIRS products for new work, while the 25-year MODIS archive remains a valuable historical record. Dates for the end of operations have shifted in official plans, so check the Earthdata and LP DAAC announcements before designing a study.

Practical advice: use MODIS for long historical baselines (2000–2023), use VIIRS (or Sentinel-3, HLS, Landsat) for current and future monitoring, and be careful when merging the two records, because small calibration differences can look like trends.

7. Licences, citation and ethics of data use

NASA Earth science data are open and free of charge, but you should still cite the dataset (each product page gives a DOI and recommended citation), keep track of the version you used, and read any specific licence for third-party layers such as OpenStreetMap or commercial-origin building footprints. Never publish location-specific predictions about people (for example exact flood exposure of named households) without considering privacy and the consequences of errors.

4. Machine Learning for Geospatial Data

Tasks, features, models, validation, metrics and uncertainty

1. Matching the question to a machine-learning task

Task	Question it answers	Projects
Classification	Which category is this pixel or patch?	Flood (flooded or not), landslide (susceptible or not), fire (will burn or not)
Regression	What is the numeric value?	Heat (temperature), forest carbon (biomass), air pollution (PM2.5)
Semantic segmentation	Label every pixel in an image	Flood extent, glacier outlines, shorelines
Change detection	What differs between two dates?	Flood, coast, glacier, urban growth
Anomaly detection	Is this behaviour abnormal compared with normal?	Desertification, algal blooms
Clustering and dimensionality reduction	What natural groups exist without labels?	Algal blooms, land-degradation regimes
Time-series forecasting	What will happen next?	Crop stress, erosion, glacier, urban growth, fire spread

2. Three ways to look at a pixel

- **Pixel-based:** each pixel is a row in a table, described by its band values and derived features. Simple, fast, great with Random Forest, but blind to neighbours.
- **Patch/image-based:** a small window (for example 64×64 or 256×256 pixels) is given to a CNN so it can learn texture, shape and context.
- **Object-based:** pixels are first grouped into meaningful objects (fields, buildings, lakes) and the model works on object statistics.

3. Feature engineering for Earth data

Good features usually matter more than a fancier algorithm. Typical feature families:

- **Spectral:** raw bands, indices (NDVI, NDWI, NBR, NDBI), band ratios.
- **Terrain:** elevation, slope, aspect, curvature, topographic wetness index, terrain ruggedness, height above nearest drainage, flow accumulation.
- **Texture and neighbourhood:** local mean, standard deviation, gradient, grey-level co-occurrence statistics, moving-window percentiles.
- **Temporal:** seasonal mean, minimum, maximum, trend slope, amplitude, number of days above a threshold, **lags** (value 7 or 30 days earlier) and **anomalies** (departure from the long-term mean for that time of year).
- **Meteorological:** accumulated and maximum rainfall, antecedent wetness, temperature, humidity, wind speed and direction.

- **Distance and context:** distance to river, coast, road, city centre, previous fire; land-cover fractions in a surrounding radius.
- **Human:** population density, building density, nighttime light.

4. Model families and when to use them

Model	Strengths	Limits	Use when
Linear / logistic regression	Transparent, fast, good baseline	Misses non-linear and interaction effects	First baseline; small data
Random Forest	Robust, few settings, handles mixed features, built-in importance	Cannot extrapolate beyond training range; no spatial context	Default for tabular geospatial data
Gradient boosting (XGBoost, LightGBM, HistGradientBoosting)	Often the most accurate tabular model; handles missing values	More tuning; can overfit noise	When you want the best tabular accuracy
k-means, DBSCAN, Gaussian mixtures	No labels needed	Cluster meaning must be interpreted by an expert	Exploratory classification, bloom detection
PCA	Compresses correlated bands; reveals main variability	Components are not always physically interpretable	Preprocessing, anomaly detection
Isolation Forest / one-class methods	Learns what "normal" looks like	Needs a clean notion of normal	Early-warning and anomaly problems
CNN	Learns spatial patterns from patches	Needs more data and a GPU	Texture and shape matter
U-Net (encoder-decoder)	Pixel-wise segmentation of images	Needs masks for training; memory-hungry at high resolution	Flood, glacier, shoreline, burned area
LSTM / ConvLSTM / temporal CNN	Sequences and spatiotemporal data	More complex to train; needs long records	Forecasting spread, growth, stress
Pretrained geospatial foundation models (for example IBM-NASA Prithvi-EO, trained on HLS imagery)	Strong features learned from huge unlabelled archives; small labelled sets suffice	Heavier; check licence and input requirements	Advanced extension with limited labels

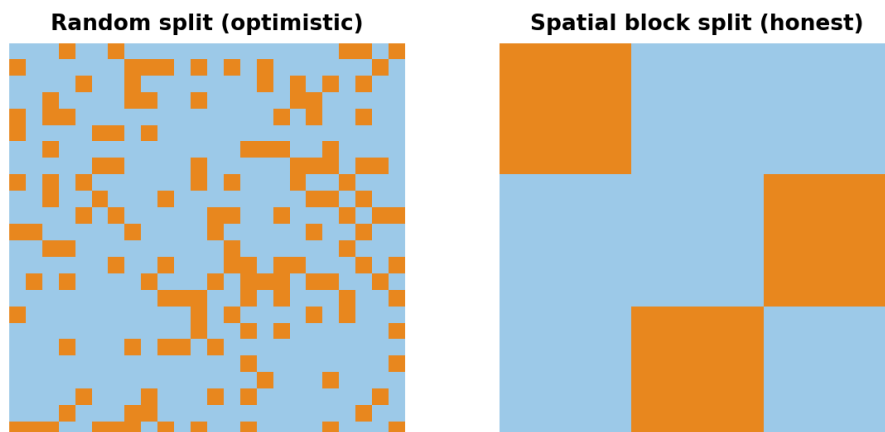
5. Dealing with imbalanced classes

Floods, fires, landslides and algal blooms are rare: often under 5 percent of pixels. A model that predicts "no event" everywhere would score 95 percent accuracy and be useless. Remedies:

- Choose metrics that focus on the rare class: precision, recall, F1, IoU, PR-AUC (not plain accuracy).
- Use class weights or a loss designed for imbalance (Dice, focal, Tversky).
- Sample the training set so rare and common classes are both well represented, but **evaluate on the natural distribution**.
- Tune the decision threshold to the cost of a miss versus a false alarm. For early-warning systems, recall is usually prioritised.

6. Validation: the most important section of this book

Earth data are **spatially and temporally autocorrelated**: nearby pixels and consecutive days are very similar (Tobler's first law of geography: near things are more related than distant things). If you split pixels randomly into training and test sets, the test pixels sit next to training pixels with almost the same values. The model appears excellent but has really only memorised the neighbourhood.



Blue = training pixels, orange = test pixels. Neighbouring pixels are similar, so random splits leak information.

Figure 3. Random splits scatter test pixels among training pixels (optimistic). Block splits keep whole regions separate and give a more honest estimate.

Validation design	How it works	Use for
Spatial block / tile cross-validation	Divide the map into blocks; train on some blocks, test on others	Mapping and susceptibility problems
Leave-one-region-out	Hold out an entire city, basin or event	Testing whether a model generalises (Delhi to Mumbai; one flood event to another)
Forward-chaining (time-series split)	Train on the past, test on the future, then roll forward	Forecasting, early warning
Leave-one-year-out	Hold out whole years	Seasonal and climate-sensitive models
Independent external validation	Compare with an independent reference (field data, official maps)	Final credibility check

WATCH OUT Data leakage checklist

Leakage means information from the test set sneaks into training. Typical examples: random pixel splits; computing normalisation statistics on all data; using a feature that is computed from the label (for example a fire-severity index derived after the fire); using future weather to "predict" past events; using overlapping image patches in both train and test.

7. Metrics cheat-sheet

Metric	Meaning	Best for
Precision	Of the pixels predicted positive, the share that really are	Costly false alarms
Recall (sensitivity)	Of the real positives, the share found	Costly misses (disasters)
F1 score	Harmonic mean of precision and recall	Balanced summary
IoU (Jaccard) and Dice	Overlap between predicted and true area	Segmentation and mapping
ROC-AUC and PR-AUC	Ranking quality across all thresholds (PR-AUC is more informative when positives are rare)	Probability models
Cohen's kappa	Agreement beyond chance	Multi-class land cover
RMSE, MAE, bias, R^2	Error size, average error, systematic over/under-prediction, explained variance	Regression
Skill versus a baseline	Improvement over persistence or climatology	Forecasting
Brier score, reliability diagram	Are predicted probabilities honest (30 percent really means 30 percent)?	Risk maps

8. Explaining and trusting a model

- **Permutation importance:** shuffle one feature and see how much performance drops. More reliable than built-in impurity importance, which favours features with many distinct values.
- **SHAP values:** attribute each individual prediction to features; excellent for explaining why a particular pixel is high risk.
- **Partial dependence and ICE curves:** show how predictions change as one variable changes, for example risk versus rainfall.
- **Error maps:** map where the model is wrong and ask why (land cover, terrain, season, sensor).

9. Uncertainty

A single number or map without uncertainty is incomplete. Practical options include ensembles of models trained on different samples, quantile regression or quantile forests for prediction intervals, conformal prediction for distribution-free intervals, Monte Carlo dropout for neural networks, and calibration of probabilities. Always propagate known measurement uncertainty, such as a ± 1 pixel (30 m) shoreline-detection error, into the final result.

10. Small-data strategies

- Start with simple models and strong features rather than deep networks.
- Use data augmentation for imagery: flips, rotations, small crops, mild noise (avoid unrealistic colour changes).
- Use transfer learning: pretrained encoders, geospatial foundation models, or models trained on similar regions.
- Use weak labels from existing products (for example an official water mask), then clean them manually on a small subset.
- Add physical knowledge as features or constraints (for example flood water cannot sit on a steep hillside).

5. Working in Google Colab

Making ambitious projects fit inside a free notebook

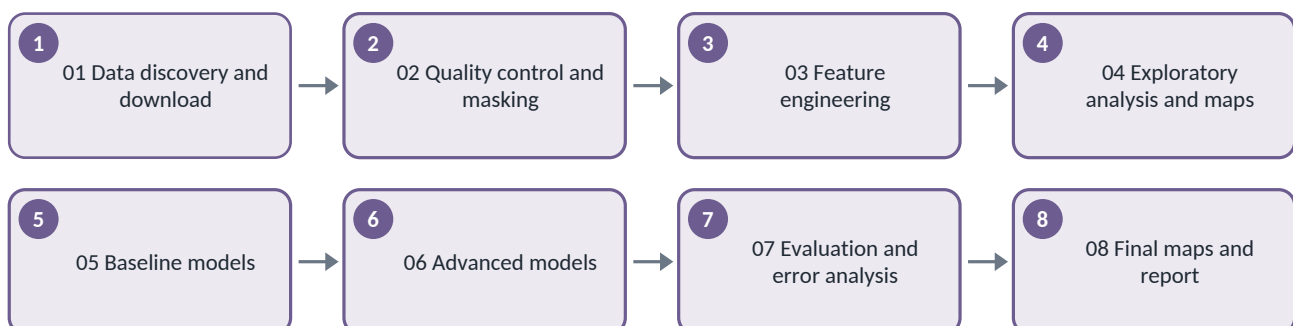
1. What the free tier gives you (and takes away)

Google Colab provides a free hosted notebook with a few gigabytes of RAM, a temporary disk, and sometimes a GPU. Exact limits change over time and are not guaranteed, but you should plan for: roughly 12 GB of RAM, a disk that is wiped when the session ends, sessions that disconnect after inactivity or after several hours, and GPU availability that varies by day. Therefore the winning strategy is to **make the problem small** and **save progress often**.

2. Design rules that make projects fit

- **Small area of interest:** a city, a district, a river basin or a glacier catchment, not a country.
- **Short, smart time windows:** one or two events, one or two seasons, or a few years of monthly composites.
- **Cloud subsetting:** use AppEEARS, STAC with windowed reads, or Earth Engine to download only your area and bands.
- **Compact data types:** store float32 or integers with a scale factor instead of float64; use compressed GeoTIFF, Parquet or NPZ.
- **Patch datasets:** cut imagery into small tiles (for example 128 or 256 pixels) and train in batches.
- **Subsample pixels** for tabular models: a few hundred thousand well-chosen pixels usually beat millions of redundant ones.
- **Checkpoint everything:** save intermediate rasters, feature tables and trained models to Google Drive.
- **Use the GPU only for deep learning;** Random Forests and gradient boosting rarely need it.
- **Protect secrets:** store Earthdata credentials in Colab's secrets manager or a credentials file, never in a shared notebook.

3. A recommended notebook structure



Splitting work into numbered notebooks means a crashed session only costs you one stage. Each notebook reads inputs from Drive and writes outputs back to Drive, so you can restart cleanly.

4. Reproducibility and documentation

- Record dataset names, versions, DOIs and download dates.
- Fix random seeds and log library versions.
- Keep a short **data card** (what each layer is, units, scale factor, resolution, known issues) and a **model card** (purpose, training data, metrics, limits, intended use).
- Use version control (GitHub) for notebooks and a README that explains the order to run them.

5. A report skeleton you can reuse

Section	What to write
Abstract	Question, data, method, headline result, limitation
Introduction	Why the problem matters; what is already known
Study area and data	Map, table of datasets, time period
Methods	Preprocessing, features, models, validation design
Results	Maps, metrics, comparison with baselines
Discussion	Where it works, where it fails, why
Limitations and ethics	Uncertainty, bias, misuse risks
Conclusion and future work	Next improvements

PART II

The Twelve Projects

Each chapter explains one complete project: problem, Earth science, data, pipeline, machine learning, evaluation and extensions.

PROJECT 01

AI Flood Extent & Damage Detector

From before-and-after satellite views and rainfall to a flood probability map

PROJECT AT A GLANCE

Goal	Estimate which areas became flooded after a heavy-rain event and quantify what was affected (cropland, buildings, people).
Core data	HLS optical imagery, Sentinel-1 radar, GPM IMERG rainfall, NASADEM or Copernicus DEM, JRC permanent-water mask
ML / CV	Rule-based thresholds, Random Forest or gradient boosting on pixels, then a small U-Net on patches
Difficulty / Colab fit	3/5 – good fit when you work on one 40–60 km window and one or two events
Output	Flood probability map, flooded area in km ² , exposure statistics, comparison with an official map

1. Problem and motivation

Floods are among the most frequent and costly natural disasters worldwide, and in India they recur every monsoon: the Brahmaputra basin in Assam, the Kosi and Gandak rivers in Bihar, Kerala in 2018, Chennai in 2015 and Mumbai in 2005 are only a few well-known examples. After a flood, responders need to know within hours which areas are underwater, where roads and fields are cut off, and how many people live there. Ground surveys are slow and often impossible; satellites can map the whole region.

The objective is not just "is there a flood?" but a **spatial answer**: *given observations before and after a rainfall event, estimate which pixels became flooded, with a probability and an uncertainty*. From that map you can compute flooded area, flooded cropland, affected roads and exposed population.

2. Earth-science background

How floods form

A **riverine flood** occurs when a river exceeds the capacity of its channel and spills onto its floodplain, usually after prolonged rain upstream. A **flash flood** follows intense rain on steep or saturated ground. An **urban flood** occurs when drainage cannot remove runoff from paved surfaces. A **coastal flood** comes from storm surge or high tides. Rainfall alone does not decide the outcome: **antecedent soil moisture** (how wet the ground already was), terrain, land cover, river level and drainage all matter. Water collects in low, flat areas close to rivers, so terrain-derived measures such as **HAND (Height Above Nearest Drainage)** are powerful predictors of where flooding is physically possible.

How water appears to satellites

- **Optical sensors**: water absorbs near-infrared and shortwave-infrared light, so it is dark in those bands. Indices such as NDWI (Green and NIR) and MNDWI (Green and SWIR) highlight it. The big problem: heavy rain means heavy cloud, and cloud blocks optical sensors exactly when you need them.
- **Radar (SAR)**: calm open water acts as a mirror and sends the radar beam away, producing a very low backscatter (dark pixel). Radar works through cloud and at night, which is why it is the workhorse of

operational flood mapping. Complications: wind-roughened water can look brighter; flooded vegetation and flooded buildings produce bright double-bounce returns; radar shadow behind steep terrain also looks dark and can be mistaken for water.

- **Permanent versus flood water:** rivers and lakes are always water. A flood map should show *new* water, so you subtract a permanent-water reference such as the JRC Global Surface Water layer or a pre-event image.

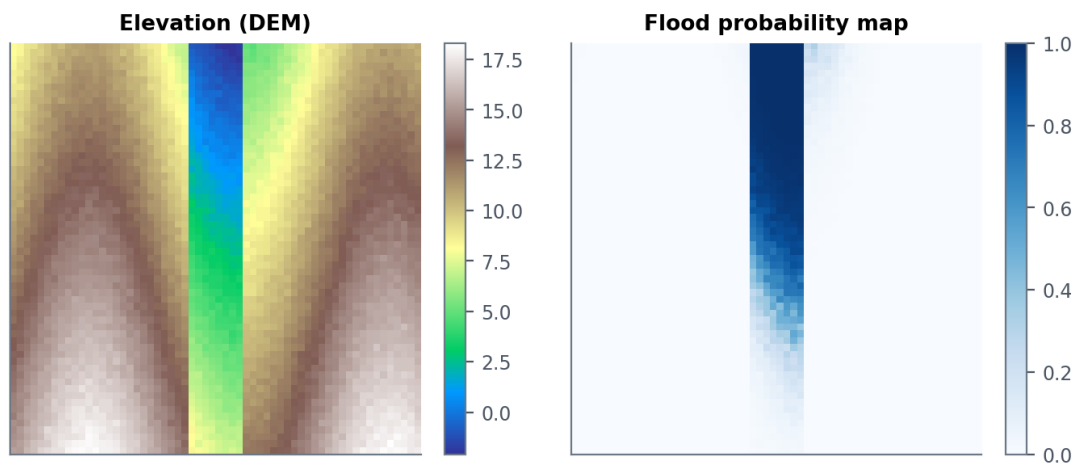


Figure 1. Concept of the final product: elevation (left) constrains where water can accumulate, and the model outputs a probability for every pixel (right). Synthetic data for illustration.

3. Data you can use (all free)

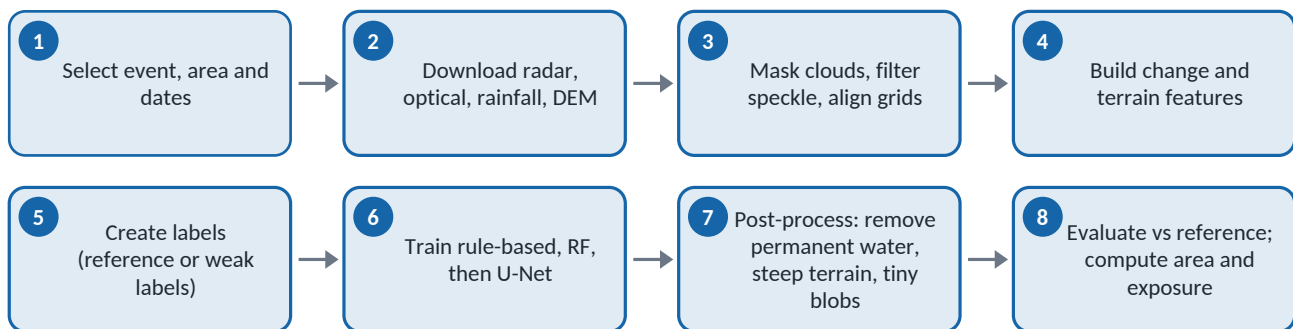
Dataset	Provider and access	Resolution	Role in the project
HLS (HLSL30 and HLSS30 v2.0)	NASA LP DAAC via Earthdata Search, earthaccess or AppEEARS; cloud-optimised GeoTIFF	30 m, about 2–3 day revisit	Optical before/after images; water indices when cloud-free
Sentinel-1 GRD (VV, VH)	Copernicus Data Space, ASF DAAC, Planetary Computer	about 10 m	Cloud-independent flood detection; main input in monsoon
OPERA DSWx products (HLS and Sentinel-1 based)	NASA OPERA project via NASA DAACs	30 m	Ready-made surface-water masks for comparison or weak labels
MODIS / VIIRS near-real-time flood products	NASA LANCE; VIIRS-based flood products also exist	250 m–375 m	Coarse regional context; daily timing of flood onset
GPM IMERG (daily and half-hourly)	NASA GES DISC, Earthdata or Giovanni	0.1°	Event rainfall totals, intensity and antecedent rain
NASADEM or Copernicus DEM GLO-30	LP DAAC, Planetary Computer, Copernicus	30 m	Elevation, slope, HAND, flow accumulation
JRC Global Surface Water	Joint Research Centre; Earth Engine	30 m	Permanent water mask and flood-frequency context
ESA WorldCover	ESA, Earth Engine, Planetary Computer	10 m	Land cover for exposure analysis (cropland, built-up)
Sen1Floods11	Cloud to Street / public repository	10 m chips	Hand-labelled flood masks for training and benchmarking
WorldFloods (ML4Floods)	Oxford / ESA Φ-lab, public	10 m	Labelled Sentinel-2 flood maps from many events

Dataset	Provider and access	Resolution	Role in the project
Global Flood Database; Dartmouth Flood Observatory	Cloud to Street; Dartmouth	event polygons	Event dates and extents for choosing and validating case studies
Copernicus EMS rapid mapping; ISRO / NRSC Bhuvan flood maps	Public portals	event-specific vectors	Independent reference for your chosen event

4. Choosing an event and study window

Pick one well-documented event so that independent reference maps exist: for example the Assam floods of a recent monsoon, the 2018 Kerala floods, the 2015 Chennai floods, or the 2022 Pakistan floods. Define an area of about 40×40 to 60×60 km that contains a river and floodplain. Choose a **pre-event image** from the same season and as close as possible before the event, and a **post-event image** within a few days of the flood peak. Record the dates, the satellite pass, and the cumulative rainfall from IMERG.

5. Pipeline



6. Preprocessing and feature engineering

- **Radar:** convert to decibels; apply a speckle filter (Lee-type or a simple median/multi-temporal filter); use terrain-corrected products; keep both VV and VH.
- **Optical:** apply the HLS Fmask quality layer to remove cloud, cloud shadow and snow; compute NDWI and MNDWI for before and after dates.
- **Change features:** difference and ratio of backscatter (after minus before in dB), difference in water indices, and the pre-event value (a pixel already dark before the event is probably permanent water or shadow).
- **Terrain features:** elevation, slope, HAND, flow accumulation, topographic wetness index, distance to the river channel. Flood water rarely rests on steep slopes or at high HAND values.
- **Rainfall features:** event total, maximum daily rainfall, and 7- and 30-day antecedent totals. IMERG pixels (about 10 km) are far coarser than satellite imagery, so treat rainfall as *context* aggregated by catchment, not as a per-pixel detail, otherwise the model will produce blocky artefacts.
- **Alignment:** reproject everything to one UTM grid at 30 m (or 10 m for radar-only runs) and clip to the area of interest.

7. Machine-learning approach

Stage 0: rule-based baselines

Always start without machine learning. Apply a threshold to radar backscatter (for example Otsu's automatic threshold on the VH or VV histogram) or to MNDWI, and subtract permanent water. This baseline is quick, transparent and often surprisingly strong; any model you train must beat it.

Stage 1: pixel classifier (Random Forest, then gradient boosting)

Each pixel becomes one row: backscatter before and after, difference, water indices, DEM features, catchment rainfall. The label is flood (1) or not flood (0). Random Forest is a good first choice because it handles features on very different scales, copes with correlated inputs and gives feature importances. Gradient boosting often adds a few points of accuracy. Output the **probability**, not only the class, and choose the threshold using the precision–recall curve.

Stage 2: small U-Net on patches

A U-Net takes a stack of channels (for example VV, VH, difference, slope, HAND) cut into 256×256 patches and predicts a flood mask for every pixel, letting the network use spatial context such as the shape of a river-fed inundation or a road embankment. On Colab use a compact encoder (ResNet-18 or MobileNet class), a batch of 8–16, mixed precision and early stopping. Use a loss that combines binary cross-entropy with Dice so that the rare flood class is not ignored. Augment with flips and 90-degree rotations.

Where do labels come from?

- **Gold labels:** manually drawn flood masks from Sen1Floods11 or WorldFloods (best for learning, but from other regions).
- **Silver labels:** operational maps such as Copernicus EMS or OPERA DSWx for your own event (cleaner than nothing; they contain errors that your model can inherit).
- **Weak labels:** thresholded radar or water indices. Cheap and abundant but noisy; train on them, then clean or evaluate on a small hand-checked sample.

WORKED EXAMPLE How many square kilometres were flooded?

Suppose the area of interest is 40 km × 40 km = 1,600 km². At 30 m resolution each pixel covers 900 m², so the area has about 1.78 million pixels. After post-processing, the model classifies 52,000 pixels as flooded. Flooded area = 52,000 × 900 m² = 46.8 km².

Overlaying the ESA WorldCover map shows that 31 percent of those pixels are cropland, so about 14.5 km² of farmland is flooded. If the same overlay shows 4 percent built-up, roughly 1.9 km² of settlements are inundated; combined with a population grid you can estimate exposed people. (All numbers hypothetical.)

Comparing methods on a hand-checked test area, one could imagine: Otsu threshold IoU ≈ 0.55, Random Forest ≈ 0.68, U-Net ≈ 0.76. The point of the exercise is the *ranking and error analysis*, not the exact values.

8. Post-processing

- Remove pixels that were water before the event (permanent water).
- Remove pixels with slope above a threshold or HAND above a plausible flood height.
- Apply a minimum mapping unit: delete isolated blobs smaller than a few pixels; fill tiny holes.
- Use morphological opening and closing to smooth edges (a classical computer-vision step).
- Keep connected components that touch a river or low-lying drainage path.

9. Evaluation

- **Metrics for the flood class:** precision, recall, F1 and IoU. Report omission error (missed flood) and commission error (false flood) separately; emergency managers care about omissions.
- **Spatial and event generalisation:** train on some events or regions and test on a completely different one. This is the real test of whether the model understands floods or only your study area.
- **Reference comparison:** compare with Copernicus EMS, NRSC or a manually interpreted high-resolution image.
- **Error analysis by land cover:** flooded vegetation and flooded urban areas are the hardest; shadows and wet sand are common false positives.
- **Sensitivity analysis:** how does area change with the threshold or with the choice of pre-event date?

10. Colab strategy

Use STAC or AppEEARS to read only your window; keep radar at 10 m only if memory allows, otherwise resample to 30 m to match HLS. Create patches once, save them as compressed arrays, and reuse them across experiments. Random Forest on a few hundred thousand sampled pixels trains in minutes on CPU; reserve the GPU for the U-Net.

11. Extensions

- **Damage assessment:** overlay flood extent with building footprints (Google Open Buildings, Microsoft Footprints, OpenStreetMap), roads and cropland to count affected assets.
- **Flood depth:** use the elevation of the flood boundary to estimate depth with a method such as FWDET.
- **Time series:** map the flood at several dates to show growth and recession as an animation.
- **Forecasting:** predict next-day inundation from rainfall and river gauge data.
- **Transfer test:** apply the Assam-trained model to Bihar and measure the drop in accuracy.

WATCH OUT Common pitfalls

Treating radar shadow or wet sand as water; forgetting to remove permanent water; mixing images from different seasons; using a random pixel split; reporting accuracy instead of IoU or F1; trusting the 10 km rainfall grid as if it were pixel-level truth.

TIP Portfolio deliverables

A notebook series; a map of flood probability with a legend and uncertainty; a table of flooded area by land cover; a comparison against an official map; a short model card explaining where the model fails.

PROJECT 02

Wildfire Spread Predictor

Given today's fire and environment, where will it burn in the next 24 hours?

PROJECT AT A GLANCE

Goal	Predict the probability that each grid cell burns within the next 24 hours, given the current fire and conditions.
Core data	VIIRS and MODIS active fire, MODIS/VIIRS burned area, NDVI and land cover, ERA5 weather, DEM, GPM rainfall
ML / CV	Random Forest or HistGradientBoosting on grid cells, optional small CNN or ConvLSTM, fire-perimeter animation
Difficulty / Colab fit	4/5 – good fit with a 100 × 100 km region and a handful of fire events
Output	Next-day fire probability maps, predicted vs observed spread animation, driver analysis

1. Problem and motivation

Detecting a fire is already solved by satellites; the harder and more valuable question is **where it is going**. Fire managers allocate crews, evacuate villages and plan firebreaks based on predicted spread. India's forest-fire season (roughly February to June, especially in Uttarakhand, Himachal Pradesh, Odisha, Chhattisgarh and the Western Ghats) and crop-residue burning in Punjab and Haryana both produce thousands of satellite detections each season.

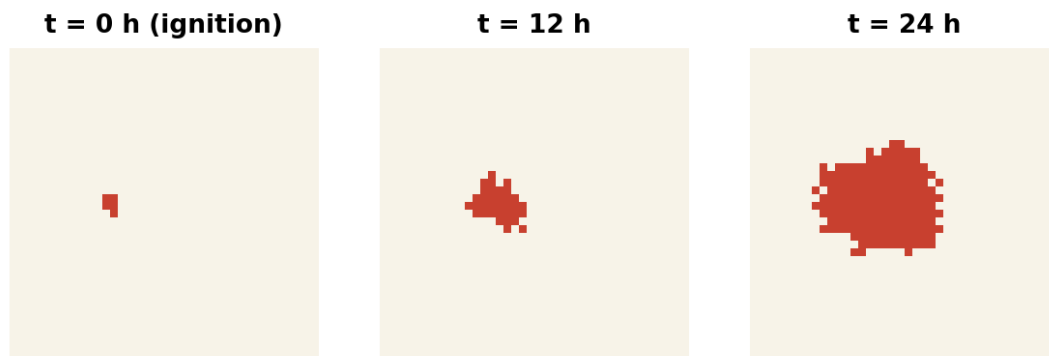
The prediction task: *for every grid cell around the current fire perimeter, estimate $P(\text{fire occurs in this cell in the next 24 hours})$, using vegetation, terrain, weather and the fire's present location.*

2. Earth-science background: the fire behaviour triangle

Fire behaviour is governed by three groups of factors, often called the **fire behaviour triangle**:

- **Fuel:** type, amount and moisture of vegetation. Dry grass and pine litter burn fast; green broadleaf forest resists. Satellites estimate fuel through NDVI, land cover, leaf area index and indices of vegetation water content (NDMI).
- **Weather:** wind speed and direction push the fire front; high temperature and low relative humidity dry fuel; recent rain suppresses spread. Atmospheric stability and fire-induced winds complicate things further.
- **Topography:** fire travels faster uphill because flames preheat the slope above them. Aspect determines how much sun dries a slope; ridges and valleys channel wind.

Fire fronts also depend on **spotting** (embers flying ahead), barriers (rivers, roads, cropland) and suppression. Two terms are useful: the **rate of spread** (metres per hour) and **fire intensity**, which satellites sense indirectly as **fire radiative power (FRP)** in megawatts.



Toy simulation with an eastward wind: spread is elongated downwind. Real models learn this from data.

Figure 1. A toy cellular model with a wind blowing east shows the elongated downwind spread that a good predictor should learn. The actual project replaces the hand-written rule with a machine-learning model trained on real fires.

3. Data you can use (all free)

Dataset	Provider and access	Resolution	Role
VIIRS 375 m active fire (VNP14IMG / NRT)	NASA FIRMS, LANCE, LAADS	375 m, several passes per day	Current fire location, FRP, day/night flag
MODIS active fire (MOD14 / MYD14, MCD14)	NASA FIRMS and LAADS	1 km	Long historical record since 2000
MODIS burned area MCD64A1	LP DAAC	500 m, monthly	Burn dates for training labels and validation
MODIS / VIIRS NDVI, EVI	LP DAAC	250 m–500 m, 8–16 days	Fuel greenness
MODIS land cover MCD12Q1; ESA WorldCover	LP DAAC; ESA	500 m; 10 m	Fuel type and barriers
ERA5 and ERA5-Land	Copernicus Climate Data Store	about 9–31 km, hourly	Wind, temperature, humidity, rainfall
GPM IMERG	GES DISC	0.1°	Recent rainfall (fuel moisture)
NASADEM or SRTM	LP DAAC	30 m	Elevation, slope, aspect
Next Day Wildfire Spread dataset	Published by Google Research; public on Kaggle and Earth Engine	1 km, US fires	Ready-made multi-channel training examples for practising the exact task
Fire perimeters (MTBS, national agencies)	USGS MTBS; state agencies	polygons	Independent validation of fire extent

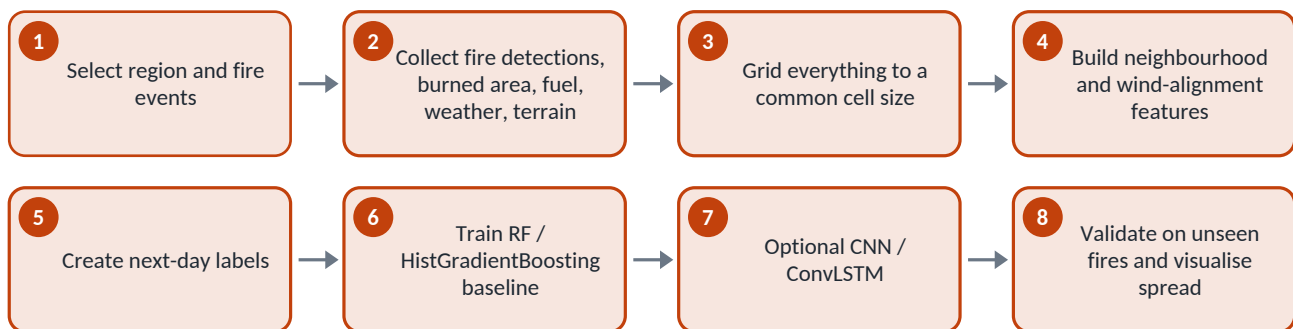
4. Defining the learning problem

Divide the region into a regular grid, for example $1\text{ km} \times 1\text{ km}$ cells (matching VIIRS 375 m aggregated to 1 km, and ERA5 downscaled). For each fire event and each day t :

- **Features at day t :** fire state (cell burning today, distance to nearest burning cell, direction to fire front), fuel (NDVI, land cover, NDMI), terrain (elevation, slope, aspect, uphill direction relative to the fire), weather (wind speed, wind direction, wind component toward the cell, temperature, humidity, rainfall in the last 7 days).
- **Label at day $t+1$:** did the cell contain a fire detection or fall inside the burned-area mask on the next day? (1 or 0)
- **Candidate cells:** only consider cells within a buffer (for example 10 km) of today's fire, since fire does not jump arbitrarily far. This narrows the problem and reduces class imbalance.

A powerful engineered feature is the **wind alignment**: the cosine of the angle between the wind direction and the line from the fire front to the cell. A cell directly downwind with a strong wind scores high; an upwind cell scores low.

5. Pipeline



6. Machine-learning approach

Baselines

- **Persistence:** tomorrow's fire = today's fire (no spread).
- **Distance rule:** cells within a fixed radius of the current fire burn.
- **Wind-weighted rule:** the radius is stretched downwind and uphill.

If the ML model cannot beat these, the features are inadequate or the labels too noisy.

Tabular models

Random Forest and HistGradientBoosting classifiers are ideal: they ingest mixed numerical features, capture interactions (wind $\times$ slope $\times$ dryness) and give probabilities. Because burned cells are very rare among all candidate cells, use class weights, tune the threshold on the precision–recall curve and evaluate with PR-AUC and IoU of the predicted burned area.

Spatial models

A small CNN takes a multi-channel patch (for example 64×64 cells containing today's fire mask, NDVI, slope, wind components, humidity) and predicts tomorrow's fire mask. Adding a time dimension (previous 2–3 days) with ConvLSTM or stacked frames lets the model learn momentum in the fire front. The Next Day Wildfire Spread dataset is an excellent practice set for this exact architecture.

Probabilistic spread simulation

Run the model repeatedly to forecast 12 and 24 hours ahead, feeding each predicted state back as input (autoregressive rollout). Use stochastic sampling to produce an ensemble of plausible fire perimeters and then average them into a probability-of-burn map. Be aware that errors compound with each step.

WORKED EXAMPLE A small prediction example

Imagine a fire front burning at cell A. Cell B lies 3 km downwind (east) on an uphill slope of 15 degrees with NDVI 0.25 (dry grass), wind 8 m/s, humidity 22 percent and no rain for 10 days. Cell C lies 3 km upwind in green forest, NDVI 0.75, on a gentle slope.

A trained model might output $P(B \text{ burns tomorrow}) = 0.71$ and $P(C \text{ burns tomorrow}) = 0.04$. Feature analysis (SHAP) would typically show wind alignment, slope and dryness pushing B up, and NDVI and upwind position pushing C down. With a threshold of 0.35 chosen to favour recall, B is flagged and C is not. (All numbers hypothetical.)

7. Evaluation

- **Hold out entire fires** (leave-one-fire-out) and, ideally, a different year or region; never randomly split cells from the same fire.
- **Metrics:** precision, recall, F1 and IoU between predicted and observed new burned area; PR-AUC for probabilities; calibration (do predicted 70 percent cells burn about 70 percent of the time?).
- **Spatial error:** how far off is the predicted front (mean distance error in km)?
- **Value for operations:** how much of the next-day burned area is covered if you alert only the top 10 percent of cells?
- **Satellite limits:** remember that the observation itself is incomplete – overpass gaps, smoke and cloud hide parts of the fire, and VIIRS sees only an instantaneous snapshot.

8. Colab strategy

Work with a handful of fires, each cropped to about 100×100 km, and with daily frames. Pre-compute the weather and fuel rasters on your grid once. Tabular models run on CPU in minutes; use the GPU for CNN/ConvLSTM experiments with a small batch of patch sequences.

9. Extensions

- Animate predicted versus observed fire fronts at 0, 12 and 24 hours.
- Add fire-weather indices (such as a fuel-moisture index derived from temperature, humidity and rainfall).
- Study crop-residue burning in northwest India as a second use case with strong air-quality links (see Project 9).
- Add uncertainty with an ensemble and show a risk corridor rather than a single line.
- Compare with a physics-based simulator conceptually and discuss where machine learning helps.

WATCH OUT Common pitfalls

Leaking tomorrow's observation into today's features; ignoring that detections are sparse snapshots; evaluating on random cells; using coarse weather without noting its resolution limit; treating every hotspot as a wildfire when agricultural and industrial heat sources exist.

PROJECT 03

Urban Heat Island Mapper

Where will city surfaces become dangerously hot, and why?

PROJECT AT A GLANCE

Goal	Map land surface temperature in a city, explain what drives hot spots, and predict neighbourhood-level heat risk.
Core data	ECOSTRESS LST (about 70 m), Landsat 8/9 surface temperature, NDVI/NDWI, ESA WorldCover, building footprints, DEM, ERA5
ML / CV	Random Forest regression, SHAP explanations, optional CNN, heat-risk index mapping
Difficulty / Colab fit	3/5 – excellent for a single city of 20–60 km across
Output	City heat-risk map, ranked wards or neighbourhoods, what-if analysis for added vegetation

1. Problem and motivation

Cities are hotter than their surroundings because dark, impervious surfaces absorb and store heat while vegetation and water cool by evaporation. This **urban heat island (UHI)** raises energy demand, worsens air quality and increases heat-related illness and mortality, especially for outdoor workers and poor households in dense housing. For Indian cities such as Delhi, Mumbai, Ahmedabad, Hyderabad, Nagpur and the Thane–Kalyan region, heat-wave seasons are already severe, and city governments prepare heat action plans.

You will build a system that does more than display a temperature image: it **explains and predicts** hot spots from their physical drivers and ends with a ranked answer to *which neighbourhoods are most at risk?*

2. Earth-science background

Surface versus air temperature

Satellites measure **land surface temperature (LST)**, the radiative skin temperature of roofs, roads and leaves. Meteorologists use **air temperature** measured about 2 m above ground. The two are related but not identical: on a sunny afternoon an asphalt road may be 25 °C hotter than the air. LST-based UHI is called the **surface urban heat island**; it is strongest by day, whereas the air-temperature UHI is often strongest at night.

The surface energy balance

Net radiation absorbed by the surface is split into **sensible heat** (warming the air), **latent heat** (energy used to evaporate water), and **ground heat storage**. Vegetated and wet surfaces send much of the energy into evaporation (latent heat), keeping themselves cool. Concrete and asphalt have low evaporation and high heat storage, so they become hot and release heat at night. Key controlling properties are **albedo** (reflectivity), **emissivity**, **thermal inertia**, vegetation cover, soil moisture, and urban geometry (tall buildings trap radiation and block wind).

Why ECOSTRESS?

The ECOSTRESS thermal radiometer on the International Space Station observes at around 70 m pixels, much finer than MODIS (1 km), and at different times of the day because the ISS orbit is not sun-synchronous. That lets you study afternoon peaks, evening heat retention and even pre-dawn temperatures. Landsat 8 and 9 give 30 m multispectral data plus thermal data (100 m native) at a fixed morning overpass and are excellent for the vegetation and built-up features.

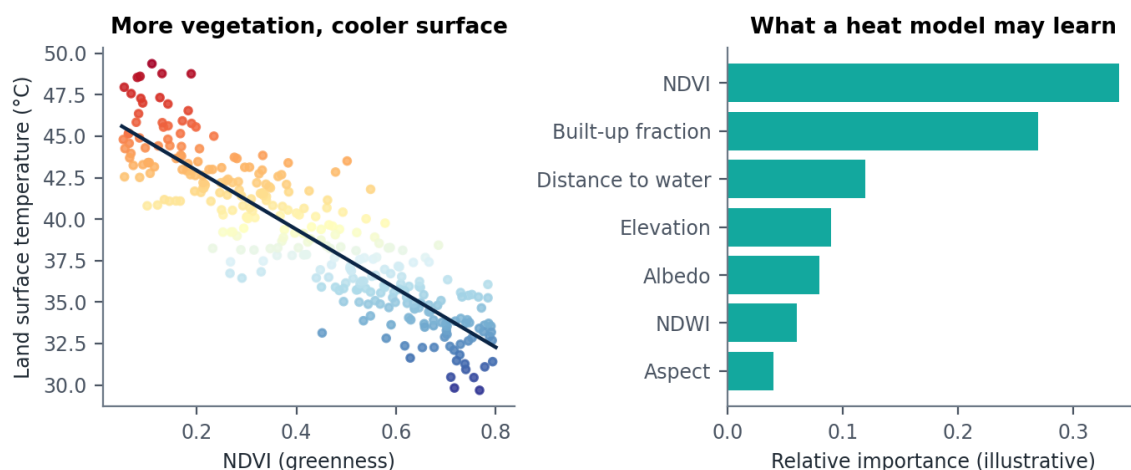


Figure 1. Left: a typical negative relationship between greenness and surface temperature. Right: the kind of driver ranking a Random Forest might produce (illustrative, synthetic).

3. Data you can use (all free)

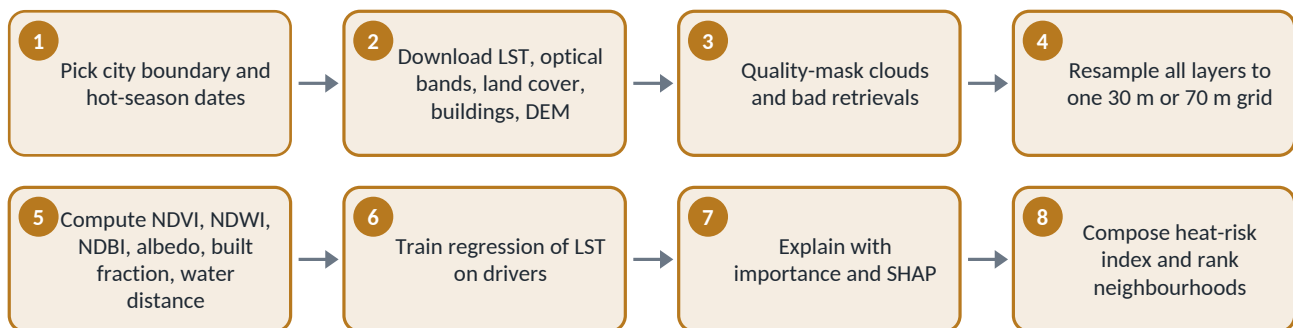
Dataset	Provider and access	Resolution	Role
ECOSTRESS Land Surface Temperature and Emissivity (ECO_L2T_LSTE, tiled)	NASA LP DAAC via Earthdata Search, earthaccess or AppEEARS; check the current version in the catalogue	about 70 m, irregular overpasses	Target variable (LST); diurnal analysis
Landsat 8/9 Collection 2 Level-2 surface temperature and reflectance	USGS EarthExplorer, Planetary Computer, Earth Engine	30 m (thermal resampled)	LST, NDVI, NDWI, NDBI, albedo; cross-check with ECOSTRESS
HLS	LP DAAC	30 m	Frequent optical indices
MODIS / VIIRS LST	LP DAAC	750 m–1 km	Long, consistent seasonal baseline
ESA WorldCover; Dynamic World	ESA; Google (Earth Engine)	10 m	Land cover fractions
Building footprints	Google Open Buildings, Microsoft Building Footprints, OpenStreetMap	vector	Building density, roof area, building height proxies where available
Global Human Settlement Layer (GHSL)	European Commission JRC	10–100 m	Built-up surface and building height estimates
NASADEM / Copernicus DEM	LP DAAC; Copernicus	30 m	Elevation, terrain effects
ERA5-Land	Copernicus CDS	about 9 km, hourly	Air temperature, humidity and wind for context and day matching
WorldPop; SEDAC population	WorldPop; Columbia	100 m–1 km	Exposure: people per heat-risk class

Dataset	Provider and access	Resolution	Role
OpenStreetMap water, parks, roads	OSM	vector	Distance to water and green space, road density

4. Selecting scenes

Heat studies need clean, comparable observations. Choose clear-sky ECOSTRESS and Landsat scenes from the pre-monsoon hot season (April–June in much of India) and, for contrast, a cooler season. Prefer scenes with similar local time. Note the day-of-year, hour of acquisition and ERA5 air temperature. A common design uses 10–30 good scenes over 2–3 years and either averages them (to get a stable *typical hot-day* pattern) or models each scene separately.

5. Pipeline



6. Feature engineering

- **Vegetation and water:** NDVI, NDWI/MNDWI, tree-canopy fraction, park area within 300 m, distance to the nearest water body.
- **Built environment:** NDBI, impervious fraction, building density (roof area per hectare), mean building height or a height proxy, road density.
- **Albedo and brightness:** shortwave albedo from reflectance bands (dark roofs absorb more).
- **Terrain and location:** elevation, slope, aspect, distance to the city centre.
- **Neighbourhood context:** moving-window averages at 100 m, 300 m and 1 km, because heat spreads across adjacent pixels.
- **Meteorological context:** air temperature, humidity and wind on the observation date (especially when training across multiple scenes).

7. Machine-learning approach

Step 1: Descriptive analysis

Map LST, compute its z-score relative to the city mean, and identify statistically significant hot spots (for example with the Getis-Ord G_i^* statistic). Compute the simple relationship between LST and NDVI or built fraction for each land-cover class.

Step 2: Random Forest regression

Predict pixel LST from the driver features. Random Forest handles non-linear thresholds (for example cooling that only appears above 20 percent tree cover) and provides feature importance. Compare with a linear regression baseline and a gradient-boosting model. Evaluate with RMSE, MAE and R^2 on spatially blocked folds.

Step 3: Explanation and what-if analysis

Use permutation importance and SHAP to explain which features drive heat in each neighbourhood. Then run counterfactuals: raise NDVI by 0.1 in selected wards, or convert 10 percent of bare ground to tree cover, and let the model estimate the temperature change. This turns a prediction model into a planning tool, with the caveat that the model learns correlations, not guaranteed causal effects.

Step 4: Optional CNN

A CNN that takes multi-channel patches (NDVI, NDBI, building density, DEM) and outputs the LST of the central area can capture spatial arrangement, such as a park cooling neighbouring streets. Use it as an advanced comparison, not as the starting point.

Step 5: Heat-risk index

Heat risk combines **hazard** (how hot), **exposure** (how many people), and **vulnerability** (how easily harmed: children, elderly, low-income housing, low green space). A transparent approach normalises each component to 0–1 and averages them with documented weights. Aggregate to wards or grid cells and rank the top 10 percent.

WORKED EXAMPLE From pixels to a ward ranking

Suppose Ward 14 has mean afternoon LST 46.5 °C, NDVI 0.12 and 8,500 people per km². Ward 22 has 41.8 °C, NDVI 0.38 and 3,200 people per km².

Normalising each variable across the city to 0–1 gives Ward 14: hazard 0.92, exposure 0.88, vulnerability 0.70; Ward 22: hazard 0.31, exposure 0.35, vulnerability 0.40. With equal weights, risk = $(0.92 + 0.88 + 0.70) / 3 = 0.83$ for Ward 14 and $(0.31 + 0.35 + 0.40) / 3 = 0.35$ for Ward 22.

The what-if model then suggests that raising Ward 14's mean NDVI from 0.12 to 0.22 would reduce predicted LST by about 2 °C, helping to justify a tree-planting priority. (All values hypothetical.)

8. Evaluation

- **Spatial cross-validation:** hold out blocks of the city (for example 5 × 5 km) rather than random pixels.
- **Temporal transfer:** train on 2022–2023 scenes and test on 2024.
- **City transfer:** train in one city and test in another to see whether heat drivers generalise.
- **Independent check:** compare with air-temperature stations (IMD, CPCB or low-cost networks) while acknowledging that surface and air temperature differ.
- **Residual maps:** places where the model under-predicts reveal missing drivers such as industrial heat or sand and rock surfaces.

9. Colab strategy

A city crop of 40×40 km at 30 m is about 1.8 million pixels; with 20 features in float32 that is under 150 MB, which fits comfortably. Sample a few hundred thousand pixels for model training and apply the final model tile-by-tile to the whole city.

10. Extensions

- Nighttime ECOSTRESS or Landsat-night analysis to find places that do not cool down.
- Compare formal and informal settlements and discuss environmental justice.
- Link heat to hospital admissions or heat-stroke statistics where available.
- Test cool-roof and green-corridor scenarios.
- Repeat for several Indian cities and compare their drivers.

WATCH OUT Common pitfalls

Mixing scenes at different times of day; ignoring cloud edges, which produce cold pixels; using too coarse a land cover for a 70 m temperature map; interpreting correlation as proof that planting trees on one street will cool it by a specific amount; forgetting that ECOSTRESS coverage can be limited by instrument operation periods, so check the dates available for your city.

PROJECT 04

Crop Water-Stress Detector

Predicting which fields will be water-stressed a week from now

PROJECT AT A GLANCE

Goal	Estimate a water-stress score for agricultural areas and forecast it 7 days ahead.
Core data	ECOSTRESS temperature and evapotranspiration, Landsat/HLS NDVI and NDMI, SMAP soil moisture, CHIRPS/IMERG rainfall, ERA5-Land weather
ML / CV	Gradient boosting or Random Forest regression on lagged features; optional LSTM; quantile intervals
Difficulty / Colab fit	3/5 – excellent on a district-scale area with 1–3 growing seasons
Output	Weekly stress maps, 7-day forecasts, early-warning list of districts or blocks

1. Problem and motivation

Roughly half of India's cultivated area depends on rain, and even irrigated fields suffer when canal water or groundwater is short. Farmers, insurers and agriculture departments want to know **which areas are running short of water now and which will be next week**, early enough to irrigate, adjust sowing advice or trigger drought-relief protocols.

Traditional greenness maps (NDVI) show damage only *after* leaves have yellowed. Plant water stress shows up earlier in **temperature** and **evapotranspiration**. The project turns these signals into a score between 0 (healthy) and 1 (severely stressed) and, importantly, forecasts it: *predicted stress at day $t + 7$* . That makes it predictive agriculture rather than plain remote sensing.

2. Earth-science background

How a plant responds to water shortage

Plants lose water through tiny pores (stomata) in a process called **transpiration**, which also cools the leaf like sweat cools skin. When soil water is limited, plants close stomata to save water. Transpiration drops, the leaf stops cooling and the canopy gets **warmer** compared with a well-watered neighbour. This temperature rise can be detected by thermal sensors days before the crop visibly wilts or loses chlorophyll.

Key physical quantities

- **Evapotranspiration (ET)**: water returned to the atmosphere from soil evaporation plus plant transpiration.
- **Potential ET (PET)**: the ET that would occur if water were unlimited, driven by radiation, temperature, humidity and wind.
- **Evaporative Stress Index (ESI)**: commonly expressed as actual ET divided by PET (or its standardised anomaly). Values near 1 mean the crop is transpiring as much as the atmosphere demands; low values mean stress. ECOSTRESS provides ET and ESI products.

- **Crop Water Stress Index (CWSI)**: a field-scale idea comparing canopy temperature with a fully transpiring (wet) reference and a non-transpiring (dry) reference. In satellite work, the same idea appears as the position of a pixel inside the NDVI-temperature space (Figure 1).
- **Vapour pressure deficit (VPD)**: how thirsty the air is; high VPD increases water demand.
- **VCI, TCI, VHI**: Vegetation Condition Index, Temperature Condition Index and their combination, the Vegetation Health Index, which rescale NDVI and temperature between historical minimum and maximum for that place and time of year.

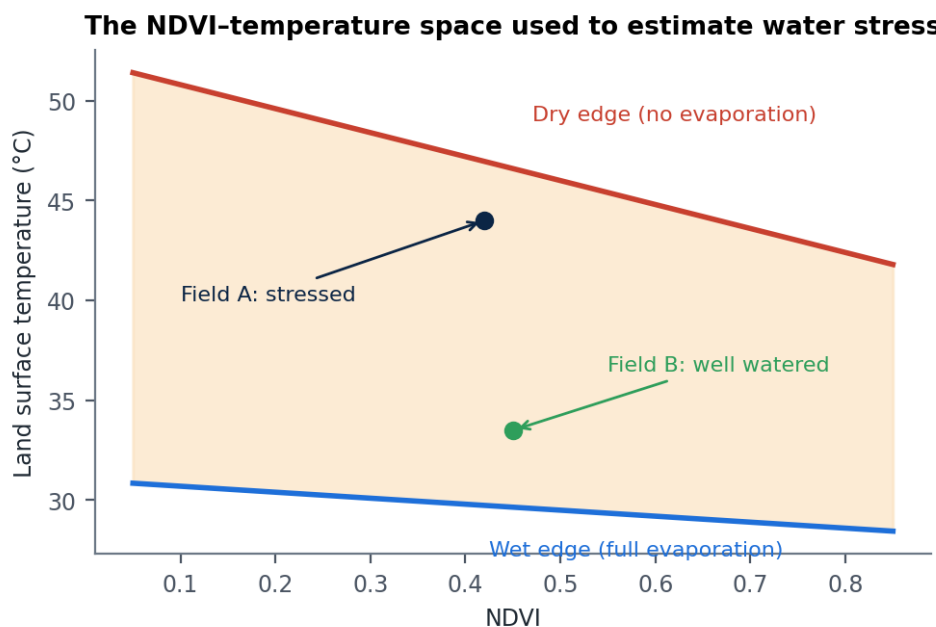


Figure 1. In NDVI-temperature space, stressed fields lie near the hot 'dry edge'; well-watered fields lie near the cool 'wet edge'. The relative position can be turned into a stress score. (Schematic.)

Crop calendar matters

The same temperature anomaly means different things at different growth stages. Stress during flowering and grain filling (for wheat in January–March, for kharif rice and cotton in August–October) damages yield far more than stress during early vegetative growth. India's two main seasons are **kharif** (monsoon-sown, June–October) and **rabi** (winter-sown, November–April). Include growth stage, estimated from the NDVI curve (days since green-up), or at least month, as a feature.

3. Data you can use (all free)

Dataset	Provider and access	Resolution	Role
ECOSTRESS LST and emissivity; ET and Evaporative Stress Index (look for the Level 2-4 products in the ECOSTRESS collection)	NASA LP DAAC via Earthdata Search, earthaccess, AppEEARS	about 70 m, irregular	Stress signal or training target; thermal anomaly
Landsat 8/9 and HLS surface reflectance, surface temperature	USGS, LP DAAC, Planetary Computer	30 m	NDVI, NDMI/NDWI, albedo, LST, time series
MODIS / VIIRS NDVI, EVI, LST, ET (MOD16-type), LAI	LP DAAC	250 m–1 km	Daily/8-day continuity and long baselines for anomalies
SMAP Level 3 (enhanced) and Level 4 root-zone soil moisture	NSIDC DAAC and Earthdata	9 km	Water available to roots, a leading indicator

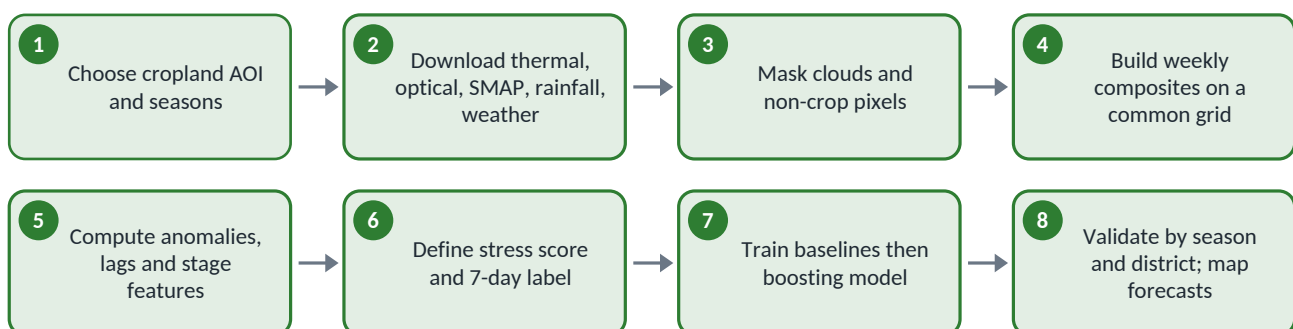
Dataset	Provider and access	Resolution	Role
CHIRPS daily rainfall	UC Santa Barbara Climate Hazards Center; Earth Engine	0.05° (about 5 km)	Rainfall totals and dry-spell counts
GPM IMERG; IMD gridded rainfall	GES DISC; India Meteorological Department	0.1°; 0.25°	Alternative or complementary rainfall
ERA5-Land	Copernicus CDS	about 9 km, hourly	Temperature, humidity, wind, radiation, VPD, PET
Cropland extent (GFSAD30 and ESA WorldCover)	NASA LP DAAC; ESA	30 m; 10 m	Mask to cropland only
Crop statistics and district yields	Ministry of Agriculture (Agricoop), ICRISAT district database, data.gov.in	district	Validation against yield anomalies
International Soil Moisture Network; field stations	ISMN (free for research)	point	Ground truth soil moisture where available

4. Defining the target

You need a number to predict. Choose one and justify it:

- **Satellite ESI as target:** use ECOSTRESS-derived ESI (or a MODIS/VIIRS ET ratio) as the stress variable, and train a model to predict it from more frequently available inputs (rainfall, soil moisture, NDVI, weather). Good first choice.
- **Composite stress score:** combine standardised anomalies of LST (positive = stress), NDVI (negative = stress) and soil moisture (negative = stress) into a 0–1 index, similar to a VHI.
- **Yield or condition anomaly:** train against district yield deviations from trend; powerful but coarse and based on few data points.
- **Forecast target:** define the label as stress at $t + 7$ days using the same variable shifted in time.

5. Pipeline



6. Feature engineering

- **Current state:** NDVI, NDMI, LST, ET/ESI, soil moisture, albedo.
- **Anomalies:** current value minus the multi-year mean for the same week (or divided by standard deviation). Anomalies remove seasonal cycle so the model sees abnormality, not just the calendar.
- **Lags and trends:** values and slopes over the last 1, 2, 4 weeks; rate of NDVI decline.
- **Rainfall history:** totals over 3, 7, 14, 30 days; number of days since last rain greater than 10 mm; consecutive dry days.
- **Atmospheric demand:** VPD, maximum temperature, PET.
- **Management proxies:** irrigated-area indicator, distance to canals or wells, soil water-holding capacity (from a soil dataset).
- **Calendar and stage:** week of year, days since green-up, crop type if known.

7. Machine-learning approach

Baselines

- **Persistence:** the stress forecast for next week equals this week's stress. In slowly changing systems this baseline is hard to beat, so measure skill relative to it.
- **Climatology:** predict the long-term average stress for that week.
- **Linear model** on a few key variables (rainfall deficit, soil moisture).

Main models

Random Forest and gradient boosting regressors (HistGradientBoosting, XGBoost or LightGBM) fit the non-linear, interacting relationships between weather, soil water and crop response. Train them on a table where each row is a pixel (or a block average) at week t , with features computed from data up to week t , and the label is the stress at week $t + 1$. Because weekly samples within a season are strongly correlated, sample pixels and weeks sparsely to avoid duplicates.

Advanced options

- **LSTM or temporal CNN** on short sequences (8–12 weeks) of features, which can learn lagged responses automatically.
- **Quantile regression / quantile forests** to give a plausible range (for example 10th and 90th percentile) for the forecast.
- **Multi-output models** forecasting stress at 7, 14 and 21 days with a single network.
- **Weather-forecast inputs:** if you include a forecast of next week's rainfall as a feature, evaluate with forecasts (not observed values) to avoid optimistic leakage.

WORKED EXAMPLE A field-level forecast

Field A (rainfed cotton, flowering stage) has this week: NDVI anomaly -0.04 , LST anomaly $+3.1$ °C, SMAP soil moisture in the lowest 15 percent for the week, only 6 mm of rain in 14 days, VPD 3.4 kPa.

A trained model predicts stress next week = 0.74 (10th–90th percentile range 0.62–0.85). Persistence predicts 0.58 (this week's value). Observed stress next week turns out to be 0.77, so the model error is 0.03 versus 0.19 for persistence. Averaged over all test pixels, a hypothetical skill score of $1 - (\text{MSE}_{\text{model}} / \text{MSE}_{\text{persistence}}) = 1 - (0.012 / 0.020) = 0.40$ would mean the model removes 40 percent of the error of the naive forecast. (All values hypothetical.)

8. Evaluation

- **Leave-one-season-out** and **leave-one-district-out** cross-validation; never train on a season and test on a later week of the same season at the same pixel.
- **Forecast skill** relative to persistence and climatology at each lead time (7, 14 days).
- **Event detection**: if stress above 0.7 is defined as an alert, report hit rate, false-alarm rate and lead time.
- **Link to outcomes**: correlate seasonal peak stress with district yield anomalies; discuss that irrigation, crop variety and pests also affect yield.
- **Calibration**: do the 80 percent prediction intervals contain about 80 percent of observations?

9. Colab strategy

Aggregate to weekly composites and 30–70 m (or even 250 m) grids over a district. A few thousand sampled pixels $\times$ 40 weeks $\times$ 25 features is a small table. Use AppEEARS to retrieve ECOSTRESS and MODIS/VIIRS time series for your polygon so you never download scenes you will not use.

10. Extensions

- Map irrigated versus rainfed responses and quantify irrigation benefit.
- Combine with a simple water-balance model (rainfall minus PET) as a physics-informed feature.
- Produce block-level advisory maps (traffic-light colours) for extension officers.
- Test transfer from one crop (wheat) to another (rice) or one state to another.

WATCH OUT Common pitfalls

Treating cloud-contaminated or mixed pixels as stress; mixing crop and fallow land; forgetting that irrigation hides rainfall deficits; using a random split of weekly rows; comparing ECOSTRESS acquisitions taken at very different times of day without correction.

PROJECT 05

Forest Carbon Estimator

From sparse space-lidar samples to a wall-to-wall carbon map

PROJECT AT A GLANCE

Goal	Estimate above-ground biomass (AGB) and carbon density across a forested region by combining GEDI lidar with satellite imagery.
Core data	GEDI L2A/L2B/L4A, Landsat or Sentinel-2 composites, Sentinel-1 or ALOS PALSAR radar, NASADEM, land cover, optional field plots
ML / CV	Random Forest or gradient boosting regression, spatial cross-validation, uncertainty by ensemble
Difficulty / Colab fit	3/5 – works well on 50–150 km blocks with filtered GEDI footprints
Output	Biomass and carbon density map, total carbon stock with uncertainty, comparison with published maps

1. Problem and motivation

Forests are one of the planet's largest carbon stores. Knowing how much carbon a forest holds, and how it changes, is essential for climate reporting, carbon-credit verification, forest management and conservation planning, including national commitments such as India's pledge to enlarge its forest and tree carbon sink. Walking through forests with measuring tape is accurate but covers tiny areas.

NASA's **GEDI** lidar on the International Space Station measures forest structure from space but samples only small footprints (about 25 m wide) along narrow tracks, leaving gaps. The project fills the gaps: *use GEDI samples as training data and satellite imagery covering every pixel as predictors, to produce a continuous map of biomass and carbon*. This is a classic **upscaling** problem.

2. Earth-science background

Biomass and carbon

- **Above-ground biomass (AGB)** is the dry mass of living vegetation above the soil, usually given as megagrams per hectare (Mg/ha; 1 Mg = 1 tonne). **AGB density (AGBD)** is that per area.
- About half of dry biomass is carbon. A common default is a carbon fraction of roughly 0.47, so carbon density $\approx 0.47 \times \text{AGBD}$. Converting carbon mass to carbon-dioxide equivalent multiplies by $44/12 \approx 3.67$.
- Below-ground biomass (roots), dead wood and soil carbon are additional pools usually estimated with ratios or separate models; in this project the focus is AGB.

How lidar sees a forest

GEDI emits laser pulses and records the energy returned over time (a **waveform**). Early returns come from the top of the canopy, middle returns from branches and leaves, late returns from the ground. **Relative height (RH) metrics** such as RH50 and RH98 give the height below which that percentage of the returned energy lies. Taller, denser canopies have larger RH values and generally more biomass. Biomass is not measured directly: it is modelled from structure using relationships (**allometry**) calibrated against field plots where trees were measured and weighed through formulas involving trunk diameter, height and wood density.

Why optical and radar are still needed

Lidar gives accurate vertical structure but only at scattered footprints. Optical imagery (Landsat, Sentinel-2) gives spectral and texture information everywhere, and radar gives sensitivity to woody structure (long-wavelength L-band radar from ALOS PALSAR, and from the NISAR mission, penetrates deeper into canopies than C-band Sentinel-1). Combining them lets a model learn: *pixels that look like this and sit on terrain like that typically have this much biomass*.

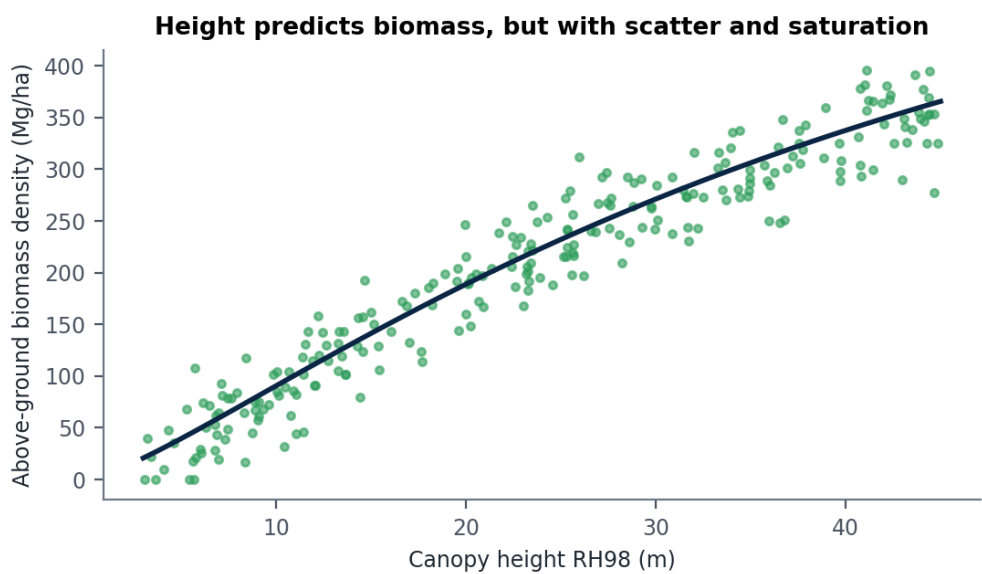


Figure 1. Canopy height is a strong but imperfect predictor of biomass, and the relationship flattens at high values (saturation). Synthetic example.

3. Data you can use (all free)

Dataset	Provider and access	Resolution	Role
GEDI L2A (elevation and height metrics, RH values)	NASA LP DAAC via Earthdata Search or earthaccess	25 m footprints	Canopy height predictors and quality flags
GEDI L2B (canopy cover, plant area index, vertical profiles)	LP DAAC	25 m footprints	Additional structure variables
GEDI L4A footprint above-ground biomass density	NASA ORNL DAAC	25 m footprints	Model-based biomass with uncertainty per footprint; training labels
GEDI L4B gridded biomass; L3 gridded canopy metrics	ORNL DAAC; LP DAAC	1 km	Benchmarks to compare your map with
ICESat-2 ATL08 canopy and terrain height	NSIDC DAAC	100 m segments	Extra lidar samples, particularly across seasons

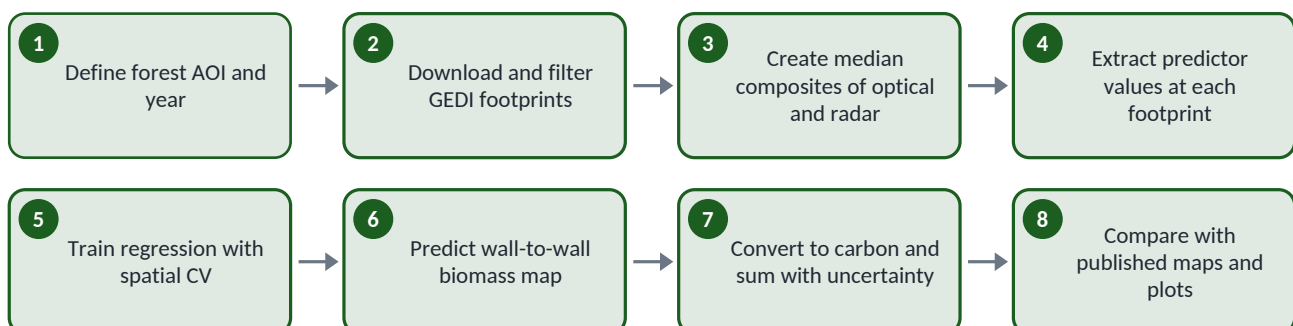
Dataset	Provider and access	Resolution	Role
Landsat 8/9, Sentinel-2, HLS composites	USGS, Copernicus, LP DAAC, Earth Engine	10–30 m	Spectral bands, NDVI, EVI, NBR, tasseled-cap, texture
Sentinel-1 (C-band); ALOS PALSAR-2 yearly mosaics (L-band)	Copernicus; JAXA free mosaics; ASF DAAC	10–25 m	Structure-sensitive backscatter VV, VH, HH, HV
NASADEM / Copernicus DEM	LP DAAC; Copernicus	30 m	Elevation, slope, aspect, TWI
ESA WorldCover; Hansen Global Forest Change	ESA; University of Maryland	10 m; 30 m	Forest mask, tree cover, loss and gain
Climate (WorldClim, CHELSA, ERA5)	Public downloads	1 km	Temperature, rainfall; ecological zones
Published biomass maps (ESA CCI Biomass, ORNL global biomass-carbon maps)	ESA CCI; ORNL DAAC	100 m–300 m	Independent comparison
Field plots and forest inventories	ORNL DAAC compilations, published inventories, Forest Survey of India reports	plots	Independent validation or calibration where accessible

4. Preparing GEDI data: quality first

GEDi data are noisy until filtered. Typical filters (check the product guide for current recommendations):

- Keep footprints with the **quality flag** set to good and the **degrade flag** off.
- Require high **sensitivity** (a measure of whether the laser can penetrate to the ground, often greater than 0.9 for dense forest).
- Prefer **night-time** acquisitions, which have less solar background noise, and full-power beams.
- Exclude steep slopes where ground and canopy returns blur, and exclude footprints in non-forest land cover.
- Remove outliers (negative or implausibly large heights).
- Remember geolocation errors of the order of ten metres: in heterogeneous landscapes, smooth the imagery over a small window at each footprint.

5. Pipeline



6. Feature engineering

- **Spectral:** all bands, NDVI, EVI, NBR, NDMI; dry-season and wet-season composites to capture phenology (deciduous versus evergreen forest).
- **Texture:** standard deviation and entropy in 3×3 and 7×7 windows; texture relates to canopy roughness.
- **Radar:** VV, VH, VH/VV ratio, temporal mean and standard deviation; for L-band HH, HV.
- **Terrain:** elevation, slope, aspect, curvature; forests on valley floors often have more biomass.
- **Climate and ecology:** mean annual rainfall, temperature, ecological zone.
- **Disturbance history:** years since forest loss or fire from Hansen or burned-area data.

7. Machine-learning approach

Choosing labels

Two common designs: (a) predict **GEDI L4A AGBD** at each footprint from imagery (labels are themselves model estimates, so you learn to reproduce them over a wider area), or (b) calibrate against **field-plot biomass** if you can obtain plots, using GEDI RH metrics as additional predictors. Be explicit about which you use and about the uncertainty inherited from the labels.

Models

- **Baseline:** mean biomass per land-cover class; then a linear regression on canopy height or NDVI.
- **Random Forest** regression: robust, handles many correlated predictors, gives feature importance. Known behaviour: it tends to over-predict low biomass and under-predict high biomass (regression toward the mean).
- **Gradient boosting:** often a few percent more accurate; tune depth and learning rate carefully.
- **Ensembles / quantile forests:** provide prediction intervals per pixel.
- **Optional CNN** on image patches to exploit spatial context.

Bias correction and saturation

Optical and C-band radar signals saturate at moderate biomass (roughly 100–150 Mg/ha in many forests), so differences among dense forests are poorly resolved. Evaluate errors in biomass bins, use L-band radar if available, and consider a simple post-hoc calibration that matches the model's predicted distribution to the observed one, while reporting that this is a heuristic.

WORKED EXAMPLE From map to carbon stock

Suppose your map covers 120 km² (12,000 ha) of forest and predicts a mean AGBD of 140 Mg/ha.

Total AGB = 12,000 ha × 140 Mg/ha = 1,680,000 Mg. Carbon = $0.47 \times 1,680,000 \approx 789,600$ Mg C (about 0.79 million tonnes of carbon). In CO₂ equivalent: $789,600 \times 3.67 \approx 2.9$ million tonnes.

If spatial cross-validation shows a relative RMSE of 30 percent at the 1 km scale and an average bias of −8 Mg/ha, report the result as, for example, 0.79 Mt C with a plausible range given by the ensemble, plus the statement that the map probably underestimates the densest stands. (All numbers hypothetical.)

8. Evaluation

- **Spatial block cross-validation** where blocks of at least several kilometres are held out. GEDI footprints along the same track are highly correlated, so random footprint splits are over-optimistic.
- **Metrics:** RMSE, relative RMSE, MAE, bias, R^2 ; compute them on footprints and on aggregated 1 km cells (errors shrink on aggregation).
- **Error by biomass class:** expose saturation and regression-to-the-mean effects.
- **Comparison with independent products:** GEDI L4B, ESA CCI Biomass and plot data; map the differences and explain them.
- **Uncertainty:** combine model error, label error and sampling error rather than quoting a single RMSE.

9. Colab strategy

Download only GEDI granules intersecting your polygon and keep a handful of variables. A region can contain 100,000–1,000,000 good footprints; subsample to 100,000 for training. Build median composites in Earth Engine or from STAC to avoid downloading dozens of scenes. Predict the final map in tiles and write each tile to Drive.

10. Extensions

- Estimate change: compare maps from two years and relate losses to fire or logging.
- Estimate forest canopy height as a separate target, which is less affected by allometric uncertainty.
- Fuse ICESat-2 and GEDI footprints.
- Test NISAR L-band data when available for your area.
- Compare biomes: train in one forest type and test in another.

WATCH OUT Common pitfalls

Using unfiltered footprints; random train/test splits; treating modelled L4A values as ground truth; ignoring forest versus non-forest masks; extrapolating beyond the biomass range seen in training; quoting carbon without stating the carbon fraction and the pools included.

PROJECT 06

Landslide Susceptibility Predictor

Which slopes are most likely to fail and how much does rainfall change that?

PROJECT AT A GLANCE

Goal	Map the spatial likelihood of landslides and build a rainfall 'what-if' simulator for risk.
Core data	NASADEM or Copernicus DEM, rainfall (IMERG/CHIRPS/IMD), soil and lithology, NDVI/land cover, rivers, roads, landslide inventories (NASA COOLR/GLC, GSI/NRSC)
ML / CV	Frequency ratio baseline, logistic regression, Random Forest, XGBoost; SHAP and partial dependence
Difficulty / Colab fit	3/5 - tabular, light, well suited to Colab
Output	Five-class susceptibility map, rainfall-risk curves, ranked risk drivers

1. Problem and motivation

Landslides kill people and destroy roads, houses and farmland every monsoon across the Himalaya, the Northeast, the Western Ghats and the Nilgiris. Notable events include Kedarnath (2013), Malin in Pune district (2014), Kerala (2018 and Wayanad in 2024) and Irshalwadi in Raigad (2023). Geological Survey of India assessments suggest that roughly an eighth of India's land area is landslide-prone to some degree.

Instead of detecting a landslide that has already happened, this project predicts **where one could happen**: a **susceptibility map**. Then it adds the dynamic dimension: *how does the risk of a particular slope change as rainfall increases?* That second step turns the model into a risk simulator.

2. Earth-science background

Why slopes fail

A slope is stable when the **resisting forces** (soil strength, root cohesion, friction) exceed the **driving forces** (weight pulling downslope). Engineers express this as a **factor of safety**: resisting divided by driving force; below 1 the slope fails. Rainfall soaks into the soil, adds weight and raises **pore-water pressure**, which reduces the effective friction between particles, so a slope that was stable in dry conditions can fail during a prolonged or intense storm.

Predisposing factors versus triggers

Group	Examples	Role
Predisposing (slow, static)	Slope angle, lithology and weathering, soil depth and texture, curvature, aspect, land cover and roots, distance to faults and rivers	Make a slope inherently prone to fail
Triggers (fast, dynamic)	Intense or prolonged rainfall, snowmelt, earthquakes, river undercutting	Provide the final push
Human factors	Road cutting, quarrying, deforestation, construction on slopes, poor drainage	Reduce stability; often missing from datasets but critical

Susceptibility, hazard and risk

Susceptibility asks where; **hazard** asks where, when and how big; **risk** multiplies hazard by exposure and vulnerability (people and property). A purely static model produces susceptibility. Hazard needs an event dimension, usually rainfall thresholds. A classic example is the intensity–duration threshold: landslides become likely when rainfall intensity exceeds a power-law function of storm duration (Caine, 1980). NASA's LHASA system combines a susceptibility map with satellite rainfall to produce near-real-time landslide hazard maps; it is a valuable reference.

3. Data you can use (all free)

Dataset	Provider and access	Resolution	Role
NASADEM; Copernicus DEM GLO-30	LP DAAC; Copernicus / Planetary Computer	30 m	Slope, aspect, curvature, TWI, ruggedness, drainage
GPM IMERG; CHIRPS; IMD gridded rainfall	GES DISC; UCSB CHC; IMD	0.1°; 0.05°; 0.25°	Event rainfall, antecedent rainfall, long-term mean
SoilGrids (ISRIC)	ISRIC, open download / API	250 m	Clay, sand, silt, organic carbon, bulk density
Global Lithological Map (GLiM); GSI geological maps (Bhukosh)	University of Hamburg; Geological Survey of India	vector	Rock type, weathering susceptibility
Sentinel-2 / Landsat / HLS NDVI; ESA WorldCover	Copernicus, USGS, LP DAAC, ESA	10–30 m	Vegetation cover, bare ground, land use
Rivers and streams (HydroSHEDS, OpenStreetMap), roads (OSM), faults (GEM Global Active Faults, GSI)	Public downloads	vector	Distance to river, road and fault
NASA Global Landslide Catalog (GLC) and COOLR	NASA Goddard / Earthdata	points with location-accuracy fields	Landslide occurrences and dates, global
Landslide inventories and atlases	GSI National Landslide Susceptibility programme; ISRO-NRSC Landslide Atlas of India; Bhuvan	points / polygons	Detailed national inventories
Landslide4Sense benchmark	Public research dataset	10 m patches	Pre-labelled Sentinel-2 + DEM patches for image-based mapping
Sentinel-1 and ALOS PALSAR	Copernicus; ASF DAAC	10–25 m	Coherence change, ground deformation (advanced)

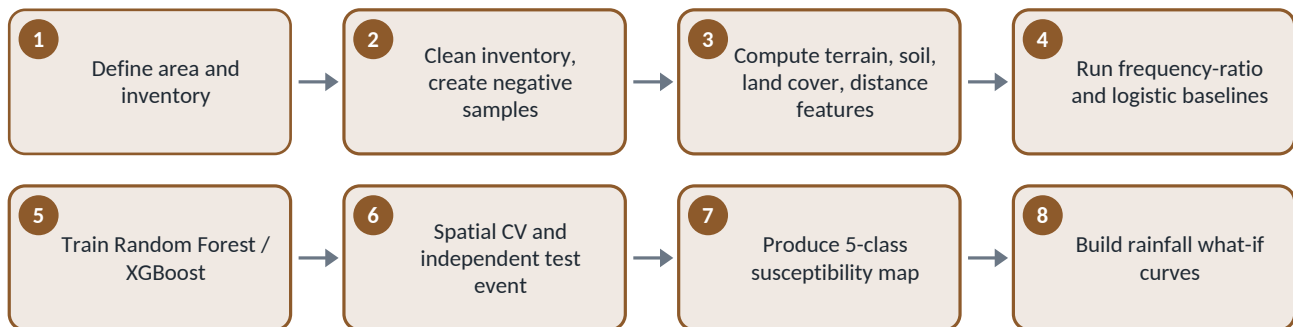
4. The inventory: your labels and your biggest risk

The model learns from a **landslide inventory**: a list of known landslide locations. Inventories are never complete, and they are biased: landslides get recorded where people live or travel (near roads), not in remote forests. If you train on such data, the model may learn "near roads = risky" because of reporting bias and not geology. Practical measures:

- Check the **location accuracy** of each record; discard those with uncertainty larger than a few hundred metres (many global-catalogue records are accurate only to kilometres).
- Sample **non-landslide points** carefully: random points away from known landslides (outside a buffer), preferably from similar accessibility conditions (also near roads) so the model cannot cheat.

- Use roughly balanced classes for training (1:1 to 1:3), but evaluate on realistic prevalence.
- Use one landslide point per slide (the crown or centroid), avoiding many pixels from one large slide that would inflate sample size.
- Document what is missing: small slides, old slides covered by vegetation, slides in remote areas.

5. Pipeline



6. Feature engineering

- **Terrain:** slope (the strongest predictor in most studies), aspect, plan and profile curvature, elevation, relief, topographic wetness index, terrain ruggedness, flow accumulation, stream power index.
- **Hydrology:** distance to streams and river order; drainage density.
- **Surface:** NDVI (mean and minimum), land cover, bare-soil fraction, vegetation loss in the last years.
- **Geology and soil:** lithology class, soil texture, soil depth proxy, distance to faults.
- **Human:** distance to roads, road density, built-up fraction.
- **Rainfall:** long-term mean annual rainfall and extreme-rain indices (static model); event-total and antecedent rainfall (dynamic model).

Categorical features such as lithology and land cover should be one-hot encoded or handled by a model that supports categories. Check multicollinearity (variance inflation factor or correlation matrix) for linear models; tree models tolerate it but importance can be shared among correlated variables.

7. Machine-learning approach

Classical baselines

The **frequency ratio** method is the standard bivariate baseline in landslide literature: for each class of each factor (for example slope 30–40°), compute the share of landslide cells in that class divided by the share of the study area in that class. Values above 1 indicate positive association. Summing frequency ratios over factors gives a susceptibility index. **Logistic regression** is the multivariate version with interpretable coefficients. Any machine-learning model should be compared with these.

Machine-learning models

- **Random Forest classifier:** robust default; probability outputs; importance ranking.
- **Gradient boosting (XGBoost, LightGBM):** typically top accuracy; use early stopping.
- **Calibration:** use Platt scaling or isotonic regression if you want probabilities to be literally interpretable, because training with balanced samples shifts probabilities upward relative to reality.

Making the susceptibility map

Predict the probability for every pixel, then classify into five classes (very low, low, moderate, high, very high) using quantiles or natural breaks. Report what fraction of the area falls in each class and what fraction of the held-out landslides falls in each class. A good map concentrates most landslides in a small high-risk area, for example 80 percent of events inside the top 20 percent of the area (hypothetical).

The rainfall 'what-if' simulator

A purely static model cannot respond to rainfall changes, because rainfall is a constant. To simulate rainfall effects you need an **event-based training set**: each row is a location on a particular date with that event's rainfall (for example 3-day total, 30-day antecedent) and a label saying whether a landslide occurred then. Non-event rows come from the same dates at other locations and from rainy periods without slides. Then keep all terrain features fixed for a chosen pixel and vary only the rainfall to trace its risk curve (an **ICE** or **partial-dependence curve**).

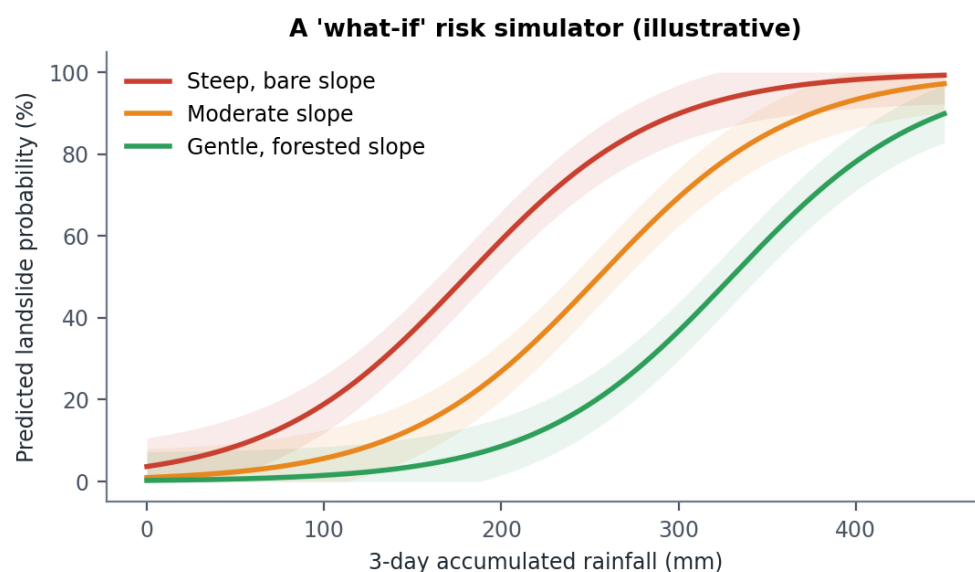


Figure 1. Predicted probability versus rainfall for three slope situations. The curves, shaded bands and values are illustrative; real curves must be derived from an event-based model and shown only within the rainfall range observed in training.

WORKED EXAMPLE Reading a what-if table

For one steep, sparsely vegetated slope the event-based model gives: 100 mm in 3 days → 12 percent; 200 mm → 31 percent; 300 mm → 67 percent; 400 mm → 89 percent. The jump between 200 and 300 mm shows a threshold-like response, typical when soil saturates. If a weather forecast predicts 280 mm for this slope's catchment, the simulator suggests a risk near 55–60 percent, which could trigger an advisory. A gentle, forested slope with identical rainfall might show only 8 percent. (All values hypothetical.)

8. Evaluation

- **Spatial block cross-validation** or hold out entire valleys or districts.
- **Temporal hold-out**: train on landslides before a given year, test on later ones; or train on one monsoon and test on the next.
- **Metrics**: ROC-AUC (standard in the field), PR-AUC, F1, Brier score; **success-rate and prediction-rate curves** showing cumulative landslides captured as area ranked by risk increases.
- **Compare** against frequency ratio, logistic regression and, if possible, an official susceptibility map.

- **Error analysis:** map false positives (steep stable slopes, perhaps well-vegetated or on strong rock) and false negatives (small cut-slope failures along roads).

9. Colab strategy

This is the lightest project in the book: a table of tens of thousands of points or pixel samples with 20–30 features. Train in minutes; spend your effort on inventory cleaning and validation. For map production, predict tile by tile on a 30 m grid.

10. Extensions

- Add Sentinel-1 coherence or InSAR to detect slow-moving landslides.
- Use a U-Net on pre/post-event Sentinel-2 to map landslide polygons (Landslide4Sense).
- Combine with a near-real-time rainfall feed to emulate a simple LHASA-style hazard map.
- Compute exposure by overlaying roads, buildings and population to produce a risk map.
- Examine how susceptibility changes after deforestation or road building.

WATCH OUT Common pitfalls

Random pixel splits (neighbouring pixels from the same slide leak into both sets); ignoring inventory bias; choosing negatives from flat plains (makes the problem trivially easy); interpreting what-if probabilities as physical certainties; extrapolating rainfall values beyond training data.

PROJECT 07

Coastal Erosion Predictor

Measuring how a coastline moves, and forecasting where it will be

PROJECT AT A GLANCE

Goal	Extract shorelines from decades of imagery, calculate erosion or accretion rates along transects, and forecast future shoreline position.
Core data	Landsat (since the 1980s), Sentinel-2, HLS, tide model (FES2014/TPXO), ERA5 waves and wind, GEBCO/DEM, IBTrACS cyclone tracks
ML / CV	Water/land segmentation, thresholding, contours, morphology, transect statistics, regression and gradient boosting
Difficulty / Colab fit	4/5 – ideal on 5–20 km coastal stretches, small image patches
Output	Shoreline time series, rate-of-change maps, 5–10 year forecast with uncertainty

1. Problem and motivation

India has roughly 7,500 km of coastline, home to tens of millions of people, ports, fishing villages, temples, farmland and mangroves. National assessments by coastal research agencies have found that a significant share of the coast, on the order of one third in some analyses, is eroding, while other stretches are accreting (gaining land). Erosion is driven by waves, sea-level rise, cyclones, reduced river sediment supply and coastal construction. Villages in Odisha, Kerala, Tamil Nadu, Puducherry and the Sundarbans have lost homes and land.

This project builds a measurement-and-forecast system: *detect the shoreline in every image, measure how far it moves, and predict where it will be in five to ten years*. It is highly visual and an excellent introduction to classical computer vision applied to Earth observation.

2. Earth-science background

What is a shoreline?

The shoreline looks like an obvious line but is surprisingly hard to define. Satellites capture the **instantaneous waterline**, where water meets land at that moment. This line moves up and down the beach with the tide. Scientists often want a **datum-based shoreline**, such as the position of mean high water, or the vegetation line, or the toe of the dune. A satellite waterline is therefore a proxy that needs tide correction.

WORKED EXAMPLE Why tide correction is not optional

Suppose a beach has a gentle slope of 1:20 (rises 1 m for every 20 m horizontal). If one image is taken at a tide 0.5 m higher than another, the waterline shifts horizontally by $0.5 \text{ m} \times 20 = 10 \text{ m}$, which is one-third of a Landsat pixel. A real erosion rate of 1–2 m per year is therefore easily masked by tide differences unless you correct them. Correction: estimate the beach slope, compute the tide level at each image time from a tide model, and shift the waterline to a common reference level ($\Delta x = \Delta \text{tide} / \tan \beta$).

Processes that move coastlines

- **Waves and longshore drift:** oblique waves push sand along the coast; coasts erode where more sand leaves than arrives.
- **Storms and cyclones:** surge and large waves can remove tens of metres of beach in hours; recovery can take months or years.
- **Seasonal cycles:** on the west coast of India, strong monsoon waves erode beaches in June–September and calmer months rebuild them. Seasonal swings can be larger than the long-term trend, so compare the same season across years or model the seasonal cycle.
- **Sea-level rise and land subsidence:** raise water level relative to land, especially in deltas.
- **Sediment supply:** dams and sand mining reduce river sediment delivered to the coast.
- **Human structures:** harbours, groynes and seawalls interrupt sediment flow, causing accretion on one side and erosion on the other.

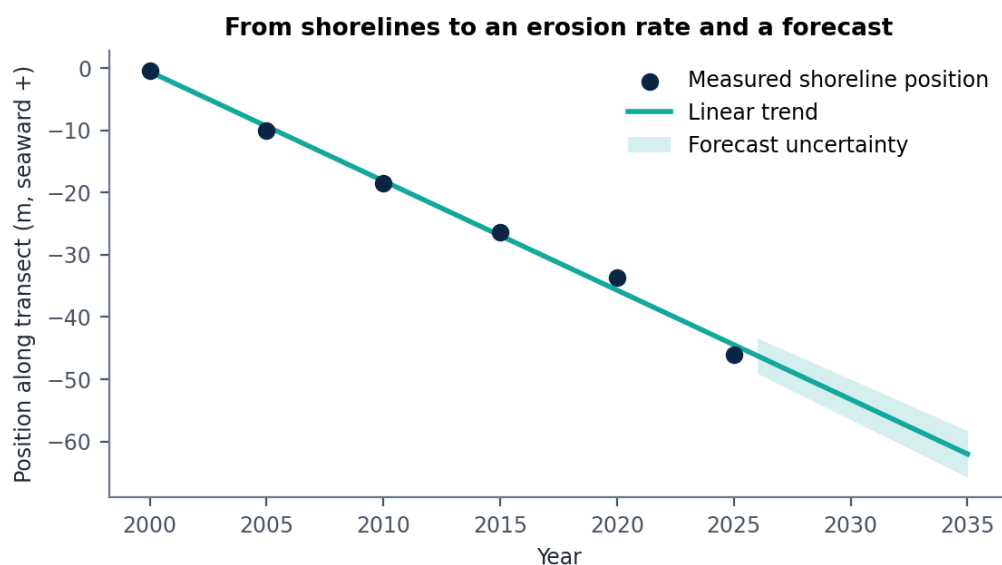


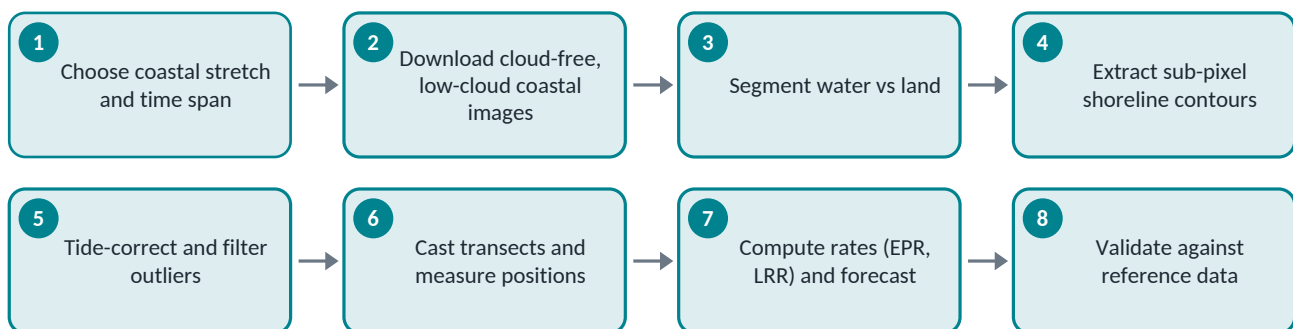
Figure 1. Shoreline position along one transect through time, a linear trend and the widening uncertainty of a forecast. Synthetic example with a retreat of about 1.8 m per year.

3. Data you can use (all free)

Dataset	Provider and access	Resolution	Role
Landsat 5, 7, 8, 9 Collection 2 (Level 2)	USGS EarthExplorer, Earth Engine, Planetary Computer	30 m, since mid-1980s	Long historical shoreline record; panchromatic 15 m on some sensors
Sentinel-2	Copernicus, Earth Engine	10 m, since 2015	Finer, frequent recent shorelines
HLS	LP DAAC	30 m	Dense recent time series
CoastSat open-source toolkit and shoreline-change method literature	Open-source (GitHub), academic papers	n/a	Reference workflow for tide-corrected satellite shorelines
Tide models (FES2014 from AVISO, TPXO, GOT)	Free registration or academic licence	about 1/16°–1/30°	Tide level at image time for tide correction
ERA5 waves and winds; CMEMS wave reanalysis	Copernicus	about 0.5°	Wave height, period, direction as drivers
Sea level (satellite altimetry)	NASA PO.DAAC, Copernicus Marine	0.25°	Regional sea-level trends and anomalies

Dataset	Provider and access	Resolution	Role
IBTrACS tropical cyclone tracks; IMD cyclone records	NOAA; India Meteorological Department	tracks	Dates and intensity of storm events
GEBCO bathymetry; NASADEM / Copernicus DEM	GEBCO; LP DAAC	15 arc-seconds; 30 m	Nearshore depth and beach slope proxies
JRC Global Surface Water	JRC	30 m	Water occurrence masks; checking tidal flats
Published coastal-change atlases and shoreline datasets	NCCR (India), published global sandy-shoreline datasets	vector / tables	Independent comparison

4. Pipeline



5. Water/land segmentation (classical computer vision)

The first task is converting each image to a binary mask of water and land. Options in increasing sophistication:

- **Index thresholding:** compute MNDWI or NDWI (water is high) and apply a threshold. Using **Otsu's method** per image picks an automatic threshold from the histogram, adapting to illumination and turbidity.
- **Supervised classification:** a small classifier (Random Forest, a shallow neural network) that labels pixels into sand, white-water (breaking waves), water and other, trained on a few hundred hand-labelled pixels. This works better than a single index where foam and wet sand confuse thresholds.
- **Segmentation network:** a small U-Net trained on masks; useful when you have labelled data but heavier than needed here.
- **Sub-pixel contouring:** instead of tracing the staircase edge of the binary mask, interpolate the contour where the index crosses the threshold (a marching-squares contour). This can place the shoreline within a fraction of a pixel.

OpenCV-style operations you will rely on

- **Thresholding** (global and adaptive) to create masks.
- **Morphological opening and closing** to remove speckle and fill gaps in the water mask; erosion/dilation to smooth edges.
- **Connected components** to keep the main ocean body and drop ponds and lagoons you are not studying.
- **Edge detection (Canny)** and **contour extraction** to trace the land–water boundary.
- **Polygon simplification** and smoothing to reduce jaggedness without erasing real features.

6. From shorelines to rates

Draw **transects**: straight lines perpendicular to the coast, spaced at 25–100 m, extending from a landward baseline. For each date, record where the shoreline crosses each transect. The result is a table: transect × date → distance from baseline. Rate statistics follow standard coastal-science conventions (as in USGS's Digital Shoreline Analysis System):

Statistic	Meaning
Net Shoreline Movement (NSM)	Distance between the oldest and the newest shoreline along a transect
End Point Rate (EPR)	NSM divided by the number of years between oldest and newest; simple but ignores intermediate dates
Linear Regression Rate (LRR)	Slope of a straight line fitted to all shoreline positions versus time; more robust
Weighted LRR	As LRR but weights each point by its uncertainty
Shoreline Change Envelope (SCE)	Largest distance between any two shorelines along the transect; highlights variability

7. Machine learning and forecasting

Level 1: statistical extrapolation

Fit a regression line of position versus time for each transect, extrapolate to the target year and compute a prediction interval. Include a seasonal term if you have enough images per year. This is surprisingly competitive for steady retreat.

Level 2: driver-based machine learning

Real coasts do not move linearly. A gradient-boosting or Random Forest model can learn shoreline change between successive dates as a function of drivers: wave energy summed over the interval, maximum storm surge or cyclone presence, mean sea level, river discharge, beach slope proxy, presence of structures and previous change. Because training samples are limited (a few dozen dates per transect), you pool many transects together and include transect characteristics as features. Evaluate rigorously with forward-chaining validation.

Level 3: hybrid physics-informed models

Equilibrium shoreline models describe change as a balance between wave-driven erosion and accretion. A machine-learning residual model can correct what the simple physics misses. This is an advanced extension.

WORKED EXAMPLE Computing and forecasting a rate

Suppose a transect has these shoreline positions relative to the 2000 position (negative = landward retreat): 2000: 0 m; 2005: -9; 2010: -17; 2015: -28; 2020: -34; 2025: -47.

$EPR = (-47 - 0) / 25 \text{ years} \approx -1.9 \text{ m/yr}$. A least-squares fit gives $LRR \approx -1.83 \text{ m/yr}$ with a residual standard deviation of about 1.5 m. Extrapolating the fitted line gives about -55 m in 2030 and -64 m in 2035.

Satellite shoreline error (about $\pm 10 \text{ m}$ for Landsat after tide correction, less for Sentinel-2) is much larger than the statistical scatter, so the honest statement is 'retreat of roughly $1.8 \pm 0.5 \text{ m/yr}$, equivalent to about 55–65 m by 2035 if the process continues unchanged', and a caution that a major cyclone could change this abruptly. (All values hypothetical.)

8. Evaluation

- **Shoreline accuracy:** compare with GPS-surveyed beach profiles, drone orthomosaics, or high-resolution historical imagery; report RMSE in metres.
- **Effect of tide correction:** show results with and without it.
- **Hindcast test:** withhold the last 5 years, forecast them from earlier data, and compute forecast error.
- **Uncertainty budget:** combine pixel size, georeferencing error, tide-correction error and statistical error.
- **Consistency between sensors:** check for jumps when switching from Landsat to Sentinel-2.
- **Spatial pattern:** map rates along the coast and compare with known structures (harbours, groynes) to see if the pattern makes sense.

9. Colab strategy

Work with small image patches (for example 20 km × 5 km of coast). Pull only the necessary bands (green, NIR, SWIR). Process each image to a shoreline vector and save only the vector, then discard the raster. This keeps memory and disk use tiny, which is why the project fits so well in Colab.

10. Extensions

- Compare erosion hotspots with population or built-up exposure.
- Quantify the effect of a new harbour or seawall before and after construction.
- Estimate beach slope from images taken at different tides.
- Study mangroves or tidal flats with vegetation indices.
- Build a transect dashboard map showing risk categories (stable, moderate erosion, severe erosion).

WATCH OUT Common pitfalls

Ignoring tide; comparing monsoon and non-monsoon images; clouds and sun glint creating false waterlines; mixing sensors without checking offsets; using EPR on only two dates; treating transects as independent when they are neighbours; extrapolating linear trends beyond coherent physical limits (a beach cannot retreat through a rock cliff).

PROJECT 08

Desertification Early-Warning System

Detecting land degradation before it becomes visible

PROJECT AT A GLANCE

Goal	Detect abnormal vegetation behaviour that suggests degradation, separate it from rainfall effects, and issue a risk score.
Core data	NDVI/EVI/LAI/NPP from MODIS/VIIRS, GIMMS long record, albedo, LST, soil moisture, CHIRPS/TerraClimate rainfall and PET, aridity index
ML / CV	Random Forest and GAM-style climate models, RESTREND residual trends, PCA, Isolation Forest, early-warning indicators
Difficulty / Colab fit	4/5 – time-series heavy but data volume is modest at 250–500 m for a dryland district
Output	Desertification risk index (0–1), hotspot maps, trend and breakpoint diagnostics

1. Problem and motivation

Desertification is land degradation in dry regions caused by climatic variation and human activities such as overgrazing, deforestation, unsustainable irrigation, salinisation and wind erosion. India's national atlas of desertification and land degradation, prepared by ISRO, reports that roughly 30 percent of the country's land area is affected by degradation. The Thar fringe of Rajasthan, Kachchh in Gujarat, rain-shadow districts of Maharashtra, Bundelkhand and the Rayalaseema region are often discussed as vulnerable.

The problem with degradation is that it is **slow and gradual**: when a hillside is visibly bare or dunes are creeping over farmland, recovery is costly. The aim here is an **early-warning system** that spots the subtle change in vegetation behaviour years earlier: *is this ecosystem behaving abnormally compared with what its climate would predict?*

2. Earth-science background

Drylands and the aridity index

Drylands are classified by the **aridity index (AI)**, the ratio of annual precipitation to potential evapotranspiration (P/PET). Common classes: hyper-arid (below 0.05), arid (0.05–0.20), semi-arid (0.20–0.50) and dry sub-humid (0.50–0.65). Degradation in these zones is formally termed desertification. Dryland vegetation is highly sensitive to rainfall variability, which is why a naive vegetation trend is misleading.

Climate versus human degradation

A year with low rain produces low NDVI. That is drought, not degradation. True degradation is a **persistent decline in vegetation productivity relative to what climate would allow**. Three classical tools separate the two effects:

- **Rain-use efficiency (RUE)**: productivity per unit rainfall (NPP or peak NDVI divided by annual rainfall). A falling RUE suggests degrading land, because vegetation uses rain less effectively.

- **Residual trend analysis (RESTREND):** regress vegetation on climate (rainfall, temperature, PET). The residuals are the part of vegetation *not explained by climate*. A significant negative trend of residuals flags degradation.
- **Breakpoint detection (for example BFAST-type methods):** decomposes the time series into trend, season and remainder and finds abrupt changes such as land conversion or overgrazing onset.

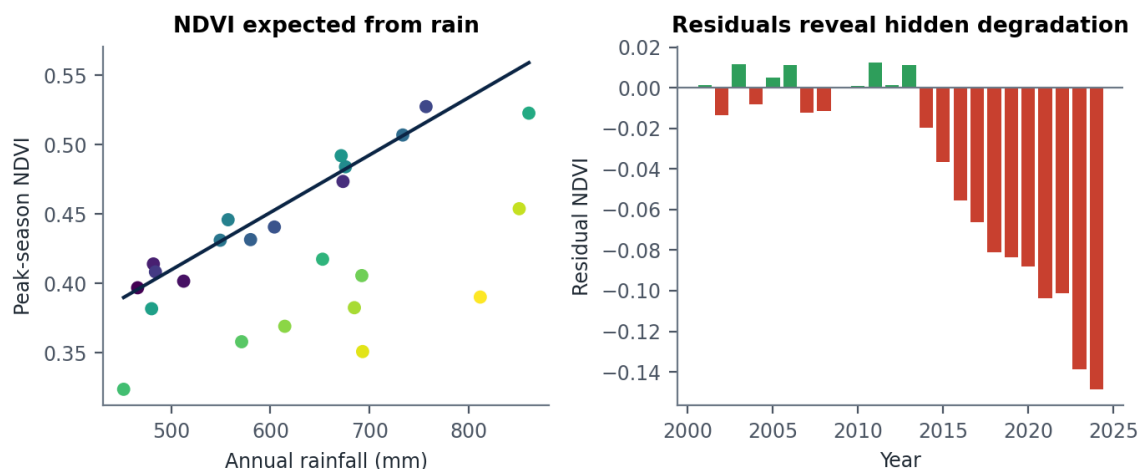


Figure 1. Left: vegetation expected from rainfall (fitted on a baseline period). Right: residuals after 2012 turn persistently negative, revealing degradation that raw NDVI would hide behind wet and dry years. Synthetic example.

Indicators used in global monitoring

The United Nations Convention to Combat Desertification (UNCCD) and Sustainable Development Goal indicator 15.3.1 define land degradation through three sub-indicators: **land cover change**, **land productivity** and **carbon stocks** (soil organic carbon). Open tools such as Trends.Earth compute these. This project concentrates on land productivity and ancillary signals.

Supporting signals

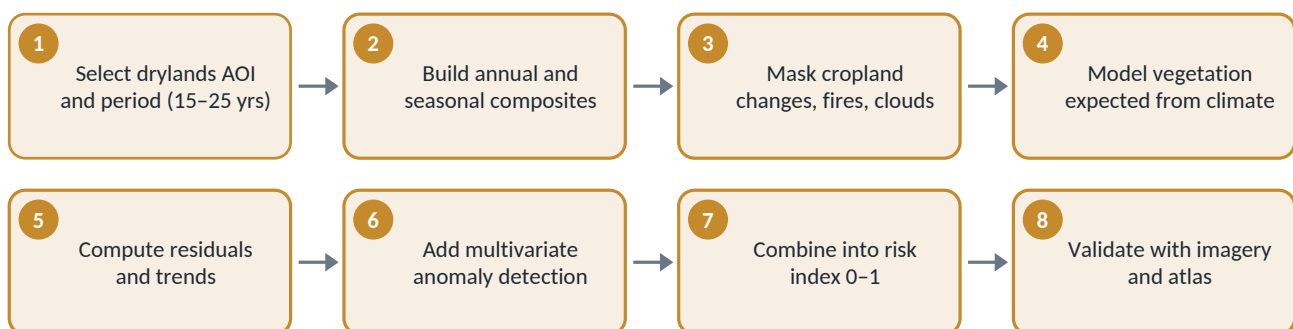
- **Albedo** usually rises as vegetation is lost and bright soil is exposed.
- **Land surface temperature** rises when evaporative cooling and shading decline.
- **Soil moisture** and its variability reflect water availability.
- **Dust and wind:** increased aerosol index or dust events point to wind erosion.
- **Early-warning statistics** from complex-systems theory: before a system tips from one state to another, it recovers more slowly from shocks, so **variance** and **lag-1 autocorrelation** of its time series tend to rise ('critical slowing down'). These are controversial and noisy in practice, but make an excellent advanced extension.

3. Data you can use (all free)

Dataset	Provider and access	Resolution	Role
MODIS NDVI/EVI (MOD13), LAI, GPP/NPP (MOD17), albedo (MCD43), LST (MOD11)	LP DAAC; AppEEARS	250 m–1 km	2000–present archive for baselines (check end-of-mission notes)
VIIRS vegetation indices, LST, albedo	LP DAAC	375 m–1 km	Continuation of MODIS records into the future
GIMMS NDVI3g and NASA LTDR AVHRR record	NASA / NOAA archives	about 5–8 km	Very long (1981 onward) coarse record for decadal trends

Dataset	Provider and access	Resolution	Role
HLS or Landsat time series	LP DAAC; USGS	30 m	Fine-scale validation and local mapping
CHIRPS; GPM IMERG; IMD gridded rainfall	UCSB CHC; GES DISC; IMD	0.05°–0.25°	Precipitation drivers
TerraClimate; ERA5-Land	Univ. of Idaho / Climatology Lab; Copernicus	4 km; 9 km	Monthly PET, temperature, soil moisture, drought indices
Global Aridity Index and PET database	CGIAR-CSI	about 1 km	Dryland classification and stratification
SMAP and ESA CCI soil moisture	NSIDC; ESA	9–36 km; 0.25°	Root-zone and surface water status
Land cover (ESA CCI annual, MCD12Q1, ESA WorldCover, Dynamic World)	ESA; LP DAAC; Google	10–500 m	Exclude cropland rotation; detect land-cover conversion
SoilGrids (soil organic carbon, texture)	ISRIC	250 m	Soil sensitivity to degradation
Gridded Livestock of the World	FAO	about 10 km	Grazing pressure proxy
TROPOMI / MODIS Deep Blue aerosol and UV aerosol index	Copernicus; LAADS DAAC	3.5–10 km	Dust events
Desertification and Land Degradation Atlas of India	ISRO-SAC	map / tables	Independent reference for validation

4. Pipeline



5. Data design

- **Growing-season metrics:** for each pixel and year compute peak NDVI, seasonal integral of NDVI (area under the curve), length of season, start of season. Using these instead of every 16-day value reduces noise and highlights ecological change.
- **Climate variables:** seasonal rainfall total, rain days, longest dry spell, mean maximum temperature, PET, and standardised drought indices (SPI/SPEI).
- **Mask** irrigated cropland, forests with sudden clear-cuts, fire scars and water bodies, which have different dynamics and cause false alarms.
- **Sensor harmonisation:** when linking MODIS and VIIRS or AVHRR, check overlap periods and adjust, since artificial trends are common.

6. Machine-learning approach

Step 1: Expected vegetation from climate

Train a Random Forest regressor (or a generalised additive model) that predicts growing-season NDVI from rainfall, PET, temperature, soil moisture and static variables (soil, elevation, aridity). Train it on a **baseline period** assumed relatively undisturbed (for example 2001–2012) or on stable reference pixels. Evaluate on held-out years.

Step 2: Residuals and trends

For each later year, compute residual = observed minus expected NDVI. Test the residual series for a monotonic trend using the **Mann–Kendall test** and estimate its size with **Theil–Sen slope**. Pixels with significant negative residual trend are degradation candidates; significant positive trends represent greening (regrowth, irrigation or restoration).

Step 3: Multivariate anomaly detection

Vegetation is only one dimension. Build a yearly feature vector per pixel (NDVI metrics, albedo, LST, soil moisture, rainfall-adjusted residual) and standardise it. Use:

- **PCA** to compress correlated indicators; the *reconstruction error* of a new year projected onto baseline components measures how unusual it is.
- **Isolation Forest** (or one-class SVM) trained on baseline years to assign an anomaly score to later years; high scores mean 'ecosystem behaving abnormally'.
- **Mahalanobis distance** as an interpretable statistical alternative.

Step 4: Early-warning signals

Calculate rolling-window variance, lag-1 autocorrelation and post-drought recovery time of the residual series. Rising values may flag ecosystems approaching a threshold. Report them as supporting evidence, not as proof.

Step 5: Desertification risk index

Convert each component to a 0–1 scale (for example percentile ranks): residual trend, anomaly score, albedo rise, LST rise, soil moisture decline, dust frequency. Combine them with transparent weights, or learn weights by regressing against reference degradation labels. Present the result as DESERTIFICATION RISK from 0 (stable) to 1 (severe).

WORKED EXAMPLE Scoring one pixel

A grazing-land pixel in a semi-arid district. For 2022 the climate model, using that year's rainfall and temperature, expects a peak NDVI of 0.31, but the observation is 0.24 (residual –0.07). Over 2013–2022 the residual declines by about 0.006 NDVI units per year (Mann–Kendall $p = 0.02$). Albedo has risen by 0.02 and soil moisture variability has increased. The Isolation Forest anomaly score is 0.71 (on a 0–1 scale, higher is more abnormal).

Percentile ranks across the district are 0.88 for residual trend, 0.71 for anomaly, 0.62 for albedo rise and 0.55 for LST rise. With weights 0.4/0.3/0.15/0.15, $\text{risk} = 0.4 \times 0.88 + 0.3 \times 0.71 + 0.15 \times 0.62 + 0.15 \times 0.55 \approx 0.352 + 0.213 + 0.093 + 0.083 = 0.74$. The pixel falls in the 'high risk' class and is queued for imagery inspection. (All values hypothetical.)

7. Evaluation

- **Temporal validation:** build the climate model on 2001–2012 and test whether it detects degradation in 2013–2023 in places later confirmed.
- **Visual validation:** sample 200–300 flagged and unflagged points and interpret them in high-resolution imagery (Sentinel-2 or historical Google imagery) for bare soil, dunes, abandoned fields or road construction; compute precision and recall of the risk map.
- **Comparison with official maps:** compare with the ISRO atlas and with land-cover change maps.
- **Robustness:** results should not depend on a single wet or dry year, on the sensor (MODIS versus VIIRS) or on the choice of climate dataset.
- **False-alarm review:** fire, harvest of plantations, new irrigation, cloud artefacts and shifting cultivation are the usual culprits.

8. Colab strategy

A 100×100 km district at 250 m has 160,000 pixels; with 23 years $\times$ 10 annual features that is small. Download time series with AppEEARS for your polygon, store as a tidy table or a NetCDF cube, and apply per-pixel trend tests in vectorised form (or on a subsample followed by tile-wise application).

9. Extensions

- Add a 'what-if' climate scenario by perturbing rainfall in the climate model.
- Map recovery after restoration projects (positive residual trends).
- Link degradation to livestock density, groundwater decline or migration statistics.
- Compare Rajasthan, Kachchh and Marathwada to find out which drivers differ.
- Test the approach on a well-documented region such as the Sahel.

WATCH OUT Common pitfalls

Interpreting NDVI trends without rainfall; mixing irrigation-driven greening with recovery; comparing sensors with different overpass times; ignoring that MODIS Terra and Aqua orbits are drifting; using a baseline period that already contains degradation; treating the risk index as a probability when it is a relative ranking.

PROJECT 09

Air Pollution Hotspot Predictor

Estimating ground-level PM2.5 from satellites, weather and geography

PROJECT AT A GLANCE

Goal	Predict PM2.5 (or hotspot probability) at places and days without ground monitors, and test whether a model trained in one city transfers to another.
Core data	MAIAC and VIIRS AOD, OMI/TROPOMI NO2, ERA5 meteorology and boundary-layer height, FIRMS fires, CPCB/OpenAQ ground PM2.5, population and land use
ML / CV	Random Forest and gradient boosting regression, hotspot classification, leave-one-station-out and leave-one-city-out validation
Difficulty / Colab fit	3/5 – tabular and light; the challenge is missing satellite data
Output	Daily or monthly 1 km PM2.5 maps, hotspot maps, transferability report (Delhi to Mumbai)

1. Problem and motivation

Fine particles (PM2.5, particles smaller than 2.5 micrometres) penetrate deep into the lungs and bloodstream and are linked to heart disease, stroke, respiratory disease and premature death. Many Indian cities regularly exceed the national 24-hour standard of $60 \mu\text{g}/\text{m}^3$ (annual $40 \mu\text{g}/\text{m}^3$), and the World Health Organization's 2021 guidelines are much stricter (24-hour $15 \mu\text{g}/\text{m}^3$, annual $5 \mu\text{g}/\text{m}^3$). The National Clean Air Programme targets reductions in cities, but monitoring stations are sparse: a few dozen in a megacity, very few in towns and almost none in rural areas.

Satellites see aerosols everywhere. The scientific challenge is that satellites measure the whole atmospheric column, while people breathe air at the surface. The project learns the mapping from *column and meteorological information* to *surface PM2.5*, producing maps for unmonitored places.

2. Earth-science background

Aerosol optical depth (AOD)

AOD measures how much sunlight is absorbed or scattered by particles throughout the vertical column of air, from the surface to space. It is dimensionless: values below 0.1 indicate very clean air; values above 1 indicate thick haze or smoke. Retrieval algorithms include **Dark Target** (good over vegetated land and ocean), **Deep Blue** (works over bright desert and urban surfaces) and **MAIAC** (Multi-Angle Implementation of Atmospheric Correction), which delivers 1 km AOD from MODIS.

Why AOD is not PM2.5

- **Vertical distribution:** when the boundary layer (the well-mixed lowest layer) is shallow, as on winter mornings in northern India, pollution is trapped near the ground and surface PM2.5 is high for a given AOD. When the layer is deep, the same AOD is spread over a thick column and surface PM2.5 is lower. Boundary-layer height (BLH) is therefore a key predictor.
- **Humidity:** particles swell in humid air (hygroscopic growth), raising AOD without raising dry PM2.5 mass.

- **Composition:** smoke, dust, sulfate and sea salt scatter light differently per unit mass. The UV aerosol index from TROPOMI helps separate absorbing aerosols (smoke, dust).
- **Clouds and bright surfaces** cause retrievals to fail. During the monsoon and during dense winter fog, AOD is often missing precisely when pollution matters most.
- **Dispersion:** wind speed and direction, temperature inversions and rainfall wash-out. A common summary is the ventilation coefficient, boundary-layer height multiplied by wind speed.

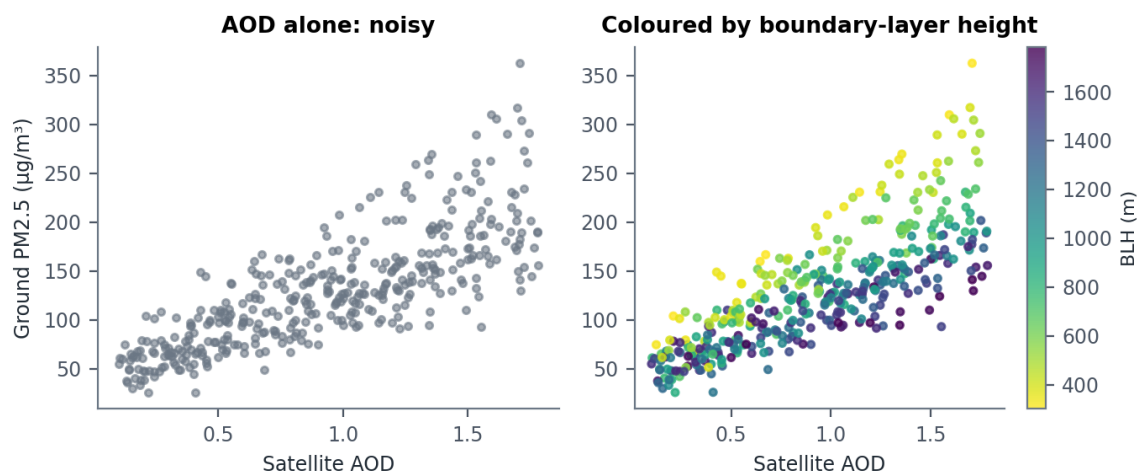


Figure 1. Left: AOD alone explains PM2.5 only loosely. Right: colouring the same points by boundary-layer height shows that meteorology organises the scatter; adding it as a feature improves prediction. Synthetic data.

Pollution sources and seasonality

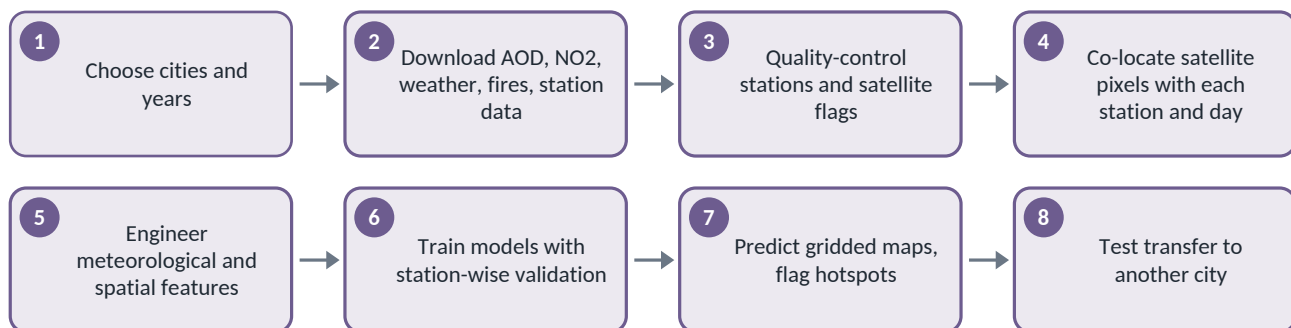
In the Indo-Gangetic Plain, winter brings temperature inversions, low wind, vehicle and industrial emissions, biomass burning for heating, and in October–November regional crop-residue burning smoke. Pre-monsoon (April–June) brings dust. In coastal cities such as Mumbai, sea breezes, higher humidity, sea salt and the monsoon wash-out change the relationship. Therefore a model trained on Delhi cannot be assumed to work in Mumbai: this is precisely the interesting experiment.

3. Data you can use (all free)

Dataset	Provider and access	Resolution	Role
MAIAC AOD (MCD19A2)	NASA LP DAAC; AppEEARS	1 km, daily	Main satellite aerosol predictor (check MODIS status notes)
VIIRS Deep Blue / Dark Target aerosol products	NASA LAADS DAAC	about 6 km (also finer products)	MODIS continuity and cross-check
OMI NO ₂ (OMNO ₂); TROPOMI NO ₂ , CO, UV aerosol index	NASA GES DISC; Copernicus Data Space	13×24 km; 5.5×3.5 km	Combustion tracers and aerosol type
MERRA-2 aerosol reanalysis; CAMS reanalysis	NASA GES DISC; Copernicus ADS	0.5°, hourly	Gap-filling when AOD is missing; model PM2.5 diagnostic
ERA5 and ERA5-Land	Copernicus CDS	about 9–31 km, hourly	Temperature, humidity, wind, BLH, rain, pressure
FIRMS active fire counts	NASA FIRMS	375 m–1 km	Upwind biomass-burning intensity

Dataset	Provider and access	Resolution	Role
Ground PM2.5 and PM10 (CPCB continuous monitors, state boards, US embassy monitors)	CPCB portal; OpenAQ (free API and archive)	stations, hourly	Training and test labels
Population (WorldPop), land use (WorldCover), roads and industry (OpenStreetMap), power plants (WRI Global Power Plant Database)	Public downloads	100 m–1 km; vector	Emission-related spatial features
Elevation (NASADEM)	LP DAAC	30 m	Terrain and valley trapping
Published satellite-derived PM2.5 surfaces (Washington University ACAG, NASA SEDAC)	Public downloads	about 1–10 km	Independent benchmark

4. Pipeline



5. Building the training table

- **Station data:** compute 24-hour means (or daytime means matching the satellite overpass) and require a minimum number of valid hours per day. Remove obvious sensor spikes and stuck values.
- **Co-location:** for each station–day, take the mean of valid AOD pixels within a small radius (for example 1–3 km) and record the number of valid pixels as a quality feature.
- **Features:** AOD, NO2, UV aerosol index, temperature, relative humidity, wind speed and direction, BLH, ventilation coefficient, precipitation (same day and previous day), surface pressure, upwind fire counts (sum within 200–300 km in the wind direction), elevation, population density, road density, distance to major industry, land-use fractions, day of year, day of week, season.
- **Temporal:** lagged meteorology; optionally lagged PM2.5 if the goal is forecasting at monitored sites (omit it if the goal is mapping unmonitored places).
- **Missing AOD:** add a flag, fill from MERRA-2 or CAMS AOD, or train a two-stage model (stage 1 predicts AOD from reanalysis and meteorology where missing).

6. Machine-learning approach

Baselines

- Simple linear regression $\text{PM}_{2.5} \sim \text{AOD}$.
- Linear regression $\text{PM}_{2.5} \sim \text{AOD} + \text{BLH} + \text{humidity}$ (physically motivated).
- Seasonal climatology per station.

Main models

Random Forest and gradient boosting (XGBoost, LightGBM, HistGradientBoosting) regressors are the standard workhorses for satellite $\text{PM}_{2.5}$ estimation. They capture the non-linear interplay between AOD, BLH, humidity and wind. Use log-transformed $\text{PM}_{2.5}$ if the distribution is heavy-tailed, or optimise a loss that emphasises high concentrations if hotspots are the goal.

Hotspot formulation

Alternatively, define a binary target: $\text{PM}_{2.5}$ above the national 24-hour standard of $60 \mu\text{g}/\text{m}^3$. Train a classifier, tune the threshold for the recall you need, and evaluate with precision, recall, F1 and PR-AUC. For mapping, apply the regression model to every grid cell and then identify clusters of high values using a hot-spot statistic (Getis-Ord G_i^*) or density-based clustering (DBSCAN).

Interpretability

SHAP values reveal what drives high estimates on a polluted day: typically low BLH, low wind, high NO_2 , high AOD, dry air and nearby fires. Compare these with the physics above to make sure the model is learning the right relationships.

WORKED EXAMPLE Two days, same AOD, different air

At a monitoring station on a winter morning: AOD 1.1, BLH 450 m, wind 1.2 m/s, relative humidity 40 percent, 85 upwind fire detections. The ventilation coefficient is $450 \times 1.2 = 540 \text{ m}^2/\text{s}$, indicating very poor dispersion. The model predicts $119 \mu\text{g}/\text{m}^3$ and the monitor shows 128 (error -9).

On a spring afternoon at the same station: AOD 1.1 again, BLH 1,500 m, wind 4.0 m/s, no fires. The ventilation coefficient is $6,000 \text{ m}^2/\text{s}$, and the model predicts $55 \mu\text{g}/\text{m}^3$; the monitor records 61. The same satellite AOD leads to a very different ground concentration, which is exactly why the meteorological features matter. (All values hypothetical.)

7. Validation: the transferability experiment

Test	Design	What it tells you
Random row split	Mix all station-days randomly	Overoptimistic; do not report as the main result
Leave-one-station-out	Hold out all days from one station	Can the model predict at an unmonitored location in the same city?
Leave-one-year-out / forward chaining	Train on earlier years, test later	Can it forecast future days?
Leave-one-city-out	Train on Delhi stations, test on Mumbai stations	Does it generalise to a different climate and aerosol mix?
Fine-tuned transfer	Pretrain on Delhi, then add a few Mumbai stations	How little local data is needed?

Metrics: RMSE, MAE, bias, R^2 , and for hotspots precision, recall and F1. Report performance by season and by concentration range, because models usually underestimate the highest peaks and perform worst in monsoon months with few valid AOD retrievals. A typical honest finding is a noticeable drop in accuracy for the cross-city test; the interesting part of the project is explaining why (humidity, sea salt, aerosol composition, different sources, fewer stations) and showing how local features or fine-tuning narrow the gap.

8. Colab strategy

Use AppEEARS to extract MAIAC and other daily layers at station coordinates (point sampling) instead of downloading tiles. For maps, build a 1 km grid over each city (about 2,500–10,000 cells) and predict only on valid days. The entire project can be done on CPU.

9. Extensions

- Build a one-day-ahead forecast using weather forecasts and lagged PM2.5.
- Attribute pollution events to crop-residue burning with fire counts and back-trajectories.
- Estimate population exposure (people $\times$ concentration) and rank wards.
- Quantify COVID-19 lockdown effects as a natural experiment.
- Compare your map with published ACAG PM2.5 surfaces.

WATCH OUT Common pitfalls

Treating AOD as PM2.5; random splits that leak neighbouring stations; ignoring missing data patterns (missing AOD is not random); mixing hourly and daily means; using meteorology from the wrong time of day; extrapolating to rural areas where emissions differ.

PROJECT 10

Glacier Retreat & Melt Predictor

Automatically measuring glacier area over time and projecting its future

PROJECT AT A GLANCE

Goal	Delineate glaciers automatically from multi-year imagery, compute area loss per year, relate it to climate, and project future area.
Core data	Landsat, Sentinel-2, ASTER, MODIS/VIIRS snow cover, RGI 7.0 and GLIMS outlines, ITS_LIVE velocities, NASADEM, ERA5-Land, NEX-GDDP-CMIP6
ML / CV	NDSI and band-ratio thresholding, OpenCV contours and morphology, regression with climate drivers, optional lightweight U-Net
Difficulty / Colab fit	4/5 – start with one or two glaciers or a small catchment
Output	Glacier area time series, area-loss rate, climate-driven projection with uncertainty

1. Problem and motivation

The Himalaya, Karakoram and Hindu Kush hold more ice than anywhere outside the polar regions and are often called the Third Pole. Glacier and snow melt feeds the Indus, Ganga and Brahmaputra systems that support hundreds of millions of people, supplies hydropower, and shapes hazards: glacial lake outburst floods and ice-rock avalanches, such as the Chamoli disaster in Uttarakhand (February 2021) and the South Lhonak lake outburst in Sikkim (October 2023). Most Himalayan glaciers are losing mass, and the pace has quickened in recent decades according to multiple studies, although the Karakoram shows more complex behaviour.

The core task is measurement: *automatically map a glacier's outline in many years and measure how fast it shrinks*. The second task is projection: *relate retreat to temperature and snowfall and estimate where the glacier may be in 2030, 2035 or 2040*.

2. Earth-science background

Mass balance in one paragraph

A glacier gains mass by **accumulation** (snowfall, avalanching, refreezing) in its upper part and loses mass by **ablation** (melting, sublimation, calving) in its lower part. The annual difference is the **mass balance**. The line separating the accumulation zone from the ablation zone at the end of the melt season is the **equilibrium line**, and its altitude is the **ELA**. A glacier in balance has an **accumulation area ratio (AAR)** of roughly 0.5–0.7; as climate warms, the ELA rises, the accumulation zone shrinks and the glacier thins and retreats. Large glaciers respond with a delay of years to decades, so retreat today partly reflects past climate.

Area, length and volume are different

- **Area change:** easiest to measure from imagery but changes slowly and lags climate.
- **Terminus (snout) retreat:** change in length along the flowline; visually intuitive but depends on glacier geometry and debris cover.
- **Elevation and mass change:** measured with repeat DEMs (ASTER stereo, SRTM, ICESat-2); these are the most direct signal of mass loss.

Special difficulties in the Himalaya

- **Seasonal snow** covers glacier surroundings and hides outlines. Use end-of-ablation-season images (typically September–October).
- **Debris-covered glaciers:** rock debris spectrally resembles the surrounding moraine, so simple snow/ice indices fail. Additional information is needed: slope and curvature from DEM, surface temperature (debris is cooler than surrounding bedrock in some settings), velocity (the glacier moves, rock slopes do not), and manual expert editing.
- **Shadows** in steep terrain mimic dark ice or water. Apply illumination masks and topographic correction.
- **Clouds** are frequent in the monsoon; select clear-sky scenes.
- **Glacial lakes:** proglacial lakes appear where the ice used to be; lake growth is a hazard indicator and should be mapped separately.

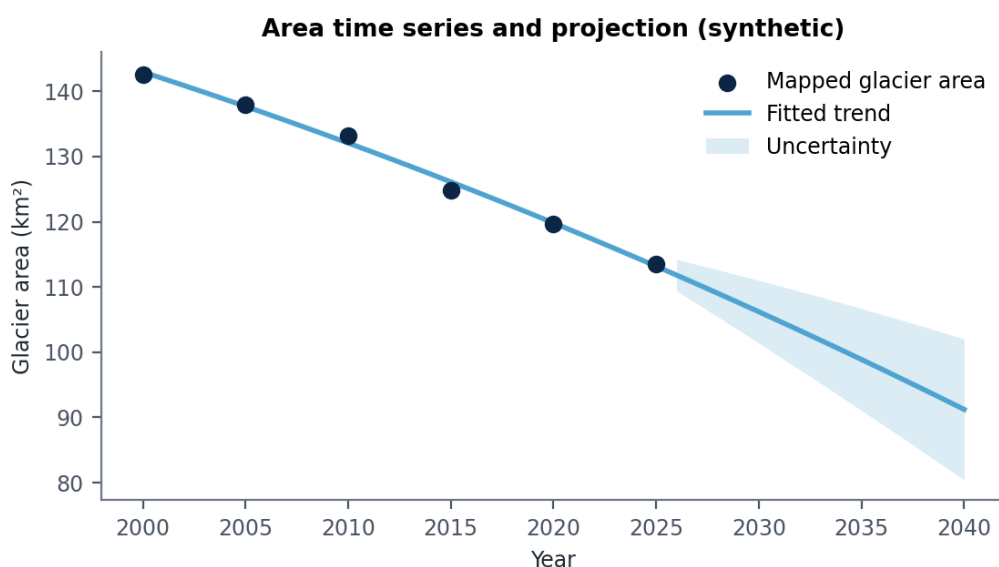


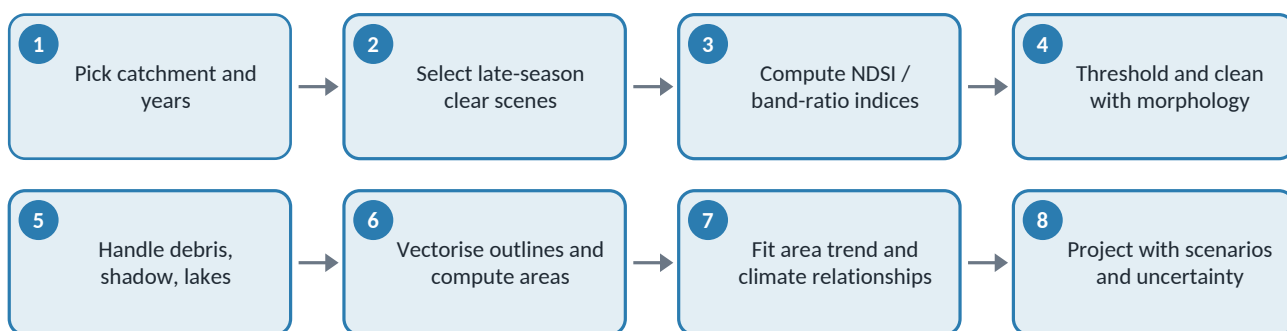
Figure 1. Mapped glacier area through time (dots), a fitted trend and a widening projection band. All values are synthetic, for demonstrating the workflow.

3. Data you can use (all free)

Dataset	Provider and access	Resolution	Role
Landsat 5, 7, 8, 9 (Collection 2)	USGS EarthExplorer, Earth Engine, Planetary Computer	30 m, from mid-1980s	Long outline time series; SWIR for ice/snow
Sentinel-2	Copernicus, Earth Engine	10–20 m, from 2015	High-quality recent outlines
ASTER (L1T, GDEM, AST_L1A stereo DEMs)	NASA LP DAAC	15–30 m	Multispectral and elevation change since 2000
MODIS / VIIRS snow cover (MOD10, VNP10)	NSIDC DAAC	375–500 m, daily	Snow-line altitude, snow-cover duration
Randolph Glacier Inventory (RGI 7.0); GLIMS glacier database	RGI Consortium; NSIDC / GLIMS	polygons	Reference outlines and glacier IDs
GAMDAM Asian glacier inventory; ICIMOD glacier and glacial lake inventories	Published open datasets; ICIMOD	polygons	Regional high-mountain inventories, lakes

Dataset	Provider and access	Resolution	Role
ITS_LIVE glacier velocity	NASA-supported open data (NSIDC / cloud)	120 m–240 m	Distinguish moving ice from stagnant debris
ICESat-2 ATL06 land-ice height	NSIDC DAAC	along-track	Surface elevation change
Glacier elevation-change and mass-balance maps (ASTER-based global studies); NASA HiMAT products	Published open data; NSIDC	100 m–1 km	Benchmarks for mass change
NASADEM, Copernicus DEM, ALOS	LP DAAC; Copernicus; JAXA	30 m	Slope, aspect, hypsometry
ERA5-Land; NASA NEX-GDDP-CMIP6	Copernicus; NASA NCCS	9 km; 0.25°	Historical climate and future scenarios

4. Pipeline



5. Computer-vision delineation

Spectral separation of ice and snow

Clean ice and snow are bright in the visible and strongly absorbing in SWIR. Two classical measures: the **Normalised Difference Snow Index (NDSI)** = $(\text{Green} - \text{SWIR1}) / (\text{Green} + \text{SWIR1})$ and the **Red/SWIR ratio** (Landsat TM band 3 / band 5), with thresholds commonly around NDSI above 0.4 or ratio above 2. Choose thresholds per scene (Otsu's method on the histogram) and verify visually.

Typical OpenCV-style steps

- **cv2.threshold**: convert the index image into a binary glacier mask.
- **cv2.morphologyEx** (opening and closing): remove small snow patches and fill small holes.
- **cv2.findContours**: trace the outlines of each glacier polygon; filter by minimum area (for example 0.02 km²).
- **cv2.Canny** edge detection: helps locate the terminus and moraine edges and can serve as a check on the contour.
- **Mask by DEM**: remove pixels with very low elevation or very steep slopes where glaciers cannot exist.
- **Polygon area**: convert pixel counts to km² using the correct projected pixel size.

Dealing with debris and lakes

Combine the spectral mask with slope (glacier surfaces rarely exceed about 25–30 degrees), curvature, and velocity. Treat the combined result as a first draft and refine with manual editing or a lightweight U-Net trained on RGI outlines. Map glacial lakes separately by NDWI so that you do not count lake pixels as ice or as bare ground.

6. Time series, drivers and projection

Area change

Compute glacier area for each year (or every 5 years) and fit a trend. Report the rate in km² per year and as a percentage of the initial area per year. Always propagate mapping uncertainty, commonly estimated as the glacier perimeter multiplied by about half a pixel size.

Climate drivers

- **Positive degree days (PDD):** the sum of daily mean temperatures above 0 °C during the melt season; a classical, physically reasoned predictor of melt.
- **Winter snowfall and total precipitation:** control accumulation.
- **Snowline altitude** from MODIS/VIIRS snow maps as a proxy for the ELA.
- **Cloud and radiation** variables as secondary factors.

Models

- **Baseline:** linear (or quadratic) trend in area versus time.
- **Temperature-index model:** melt = degree-day factor × PDD. A simple but powerful physical baseline.
- **Regression:** predict annual area change or snowline altitude from PDD, snowfall and previous change using Ridge, Random Forest or gradient boosting; the small sample size (a few dozen years) means simple models and regularisation work best.
- **Scenario projection:** feed future temperature and precipitation from NEX-GDDP-CMIP6 (for example a moderate and a high emissions scenario) into the model to produce a range of outcomes.
- **Cautious hybrid:** a statistical model alone cannot predict that a glacier will vanish; a glacier cannot lose more area than it has, and geometry (thick ice, steep valleys) limits retreat. Use physical bounds.

WORKED EXAMPLE A small glacier, a melt estimate and a projection

Mapped area: 12.4 km² (2010), 11.6 (2015), 10.9 (2020), 10.1 (2025). Loss 2010–2025 = 2.3 km², about 18.5 percent of the 2010 area, or 0.153 km² per year (roughly 1.2 percent per year). Linear projection for 2035: $10.1 - 0.153 \times 10 \approx 8.6$ km². If the rate accelerates to 0.18 km² per year, 2035 area would be about 8.3 km².

A temperature check: an extra +1 °C over a 100-day melt season adds about 100 positive degree days. With a degree-day factor for ice of 6 mm water equivalent per °C per day, that is about 600 mm w.e. of extra melt, or 0.6 m. Over 10 km² of ice, $0.6 \text{ m} \times 10,000,000 \text{ m}^2 = 6$ million m³ of water. (All numbers hypothetical.)

7. Evaluation

- **Outline accuracy:** compute IoU between your outlines and RGI or manually digitised high-resolution outlines for several glaciers.
- **Season and sensor consistency:** check that retreat is not an artefact of mapping in different seasons or using different sensors.
- **Hindcast:** hold out the most recent 5 years and compare the forecast with mapped areas.
- **Uncertainty:** report mapping uncertainty and projection spread across scenarios and models.
- **Cross-check:** compare area change with elevation change or mass-balance estimates from published datasets.

8. Colab strategy

A single glacier catchment of 30×30 km at 30 m is only a million pixels per scene. Keep only SWIR, green and NIR bands, clip early, and save outlines as vector files so you can free up memory. For a regional extension, process glaciers in batches by catchment.

9. Extensions

- Track snout position along a flowline and compute terminus retreat in metres per year.
- Add velocity from ITS_LIVE to separate active from stagnant debris-covered ice.
- Map glacial lake growth and assess outburst susceptibility.
- Compare Himalayan, Karakoram and other glaciers to discuss regional contrasts.
- Estimate glacier mass loss from DEM differencing.

WATCH OUT Common pitfalls

Mapping in snow season; ignoring debris cover (leads to underestimating area and retreat); mixing sensors without overlap tests; counting lakes as ice; extrapolating linear trends past physical limits; omitting uncertainty.

PROJECT 11

Harmful-Algal-Bloom Detector

Finding abnormal patches of ocean life from space

PROJECT AT A GLANCE

Goal	Detect and map anomalous ocean-colour signatures that could indicate harmful algal blooms.
Core data	MODIS-Aqua and VIIRS ocean colour, NASA PACE OCI, Sentinel-3 OLCI, MUR SST, ERA5/CCMP winds, in situ bloom records (HABSOS, HAEDAT, SeaBASS)
ML / CV	PCA, k-means, DBSCAN, Gaussian mixtures, climatology-based anomaly detection; optional supervised classifier
Difficulty / Colab fit	4/5 – moderate data volume if you use Level-3 or small Level-2 subsets
Output	Bloom-likelihood maps, bloom-event timeline, cluster interpretation, validation against records

1. Problem and motivation

Phytoplankton are microscopic ocean plants that form the base of the marine food web and produce roughly half of the oxygen we breathe. Occasionally they multiply rapidly into **blooms**. Many blooms are harmless or beneficial, but some are **harmful algal blooms (HABs)**: they release toxins that poison fish, shellfish and people, or they die and decay, consuming oxygen and causing fish kills. HABs hurt fisheries, aquaculture and tourism. Indian waters experience winter blooms of the dinoflagellate *Noctiluca scintillans* in the northern Arabian Sea and episodic red tides and fish kills along several coasts; marine research institutions and ocean-information services monitor them.

The project is framed as **unsupervised and semi-supervised learning**: *identify areas of abnormal biological activity from their spectral and spatial signatures, and decide which of them are potential blooms*. It is unusual because you will often have little or no labelled data.

2. Earth-science background

How ocean colour works

Water appears blue because it absorbs red light and scatters blue. Phytoplankton contain chlorophyll, which absorbs blue and red light and reflects green, so productive water shifts toward green. Satellite ocean-colour sensors measure the faint light leaving the water (**water-leaving radiance**), usually expressed as **remote-sensing reflectance (Rrs)**, at several wavelengths. Over the ocean, about 90 percent of the light reaching the sensor comes from the atmosphere, so **atmospheric correction** is the most important and most error-prone step. Errors are larger in turbid coastal water, near dust and absorbing aerosols, and in sun glint.

Open ocean versus coastal waters

In open-ocean (**Case 1**) water, optical properties are dominated by phytoplankton and the simple blue/green band-ratio chlorophyll algorithms (the OCx family) work well. In coastal (**Case 2**) water, suspended sediment and coloured dissolved organic matter (CDOM) from rivers also change colour, and those algorithms overestimate chlorophyll. Distinguishing a river plume from a bloom is a central challenge, and spectral shape, temperature and location help.

Indicators used for bloom detection

Indicator	Idea	Comment
Chlorophyll-a concentration (mg/m ³)	Proxy for phytoplankton biomass	Compare with climatology; absolute thresholds depend on region and season
Fluorescence line height (FLH)	Chlorophyll fluoresces near 678 nm; the peak height above a baseline signals bloom	Less affected by CDOM and sediment than blue/green ratios
Red band difference (RBD) and related indices	Contrast of red bands used for specific species such as <i>Karenia brevis</i>	Species- and region-specific; validate locally
Spectral shape / ratio features	Green/blue and red/green ratios, spectral derivatives, hyperspectral shape (PACE)	Hyperspectral data can distinguish phytoplankton groups, an exciting new capability
Sea surface temperature and stratification	Blooms often follow warm, calm, nutrient-fed conditions	Contextual predictor, also helps classify bloom type
Wind and upwelling	Upwelling brings nutrients; weak wind allows surface accumulation	Use ERA5 or satellite winds

Note that satellite ocean colour sees only the surface layer and cannot identify the species or measure toxins. A satellite flag is a **screening tool** that tells scientists where to sample.

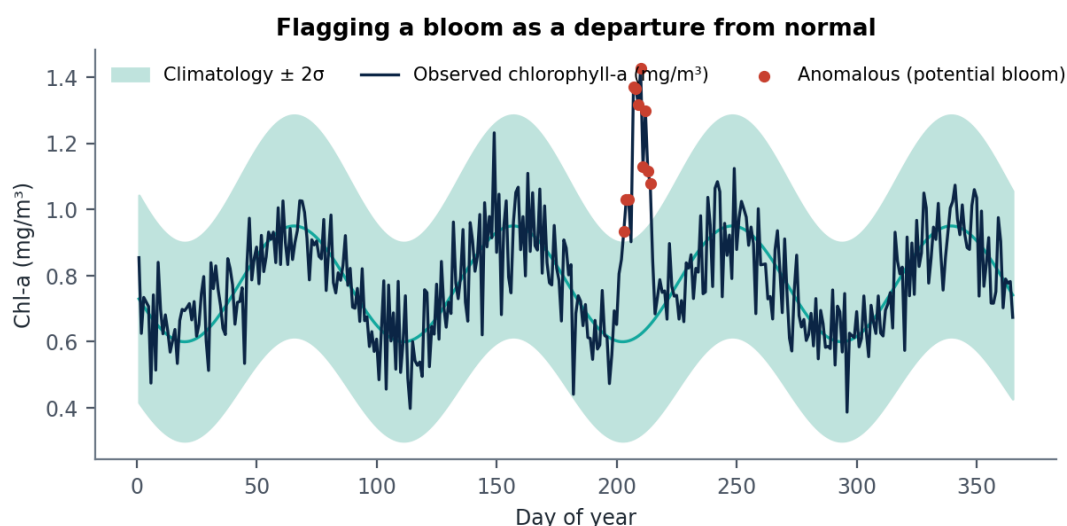


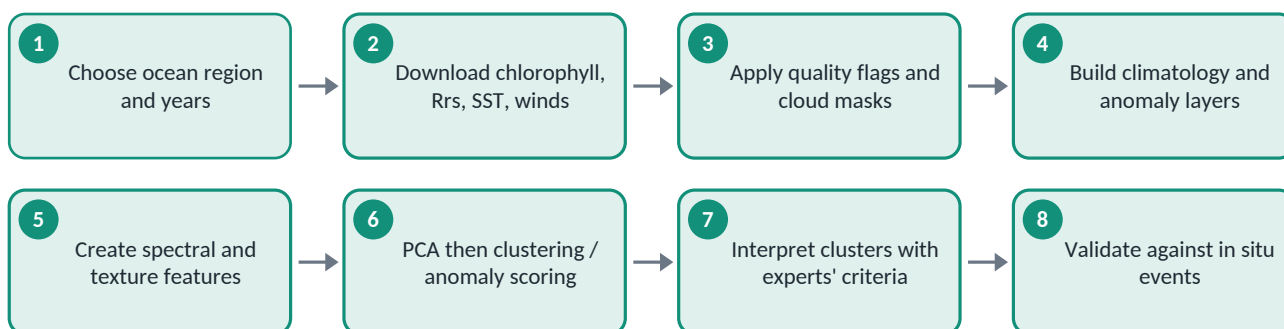
Figure 1. A bloom flagged as chlorophyll exceeding its seasonal climatology by more than two standard deviations. Synthetic time series.

3. Data you can use (all free)

Dataset	Provider and access	Resolution	Role
MODIS-Aqua ocean colour (L2 and L3: chlorophyll-a, Rrs, flags)	NASA OB.DAAC / OceanColor Web	1 km (L2), 4–9 km (L3)	Long archive since 2002 (check end-of-mission notes)
VIIRS ocean colour (SNPP, NOAA-20/21)	NASA OB.DAAC	750 m	MODIS continuation
PACE OCI (hyperspectral ocean colour) and phytoplankton products	NASA OB.DAAC	about 1 km, near-daily	Spectral shape and phytoplankton-group information (launched 2024)
Sentinel-3 OLCI	Copernicus Data Space	300 m	Finer coastal detail

Dataset	Provider and access	Resolution	Role
Copernicus Marine (CMEMS) merged chlorophyll	Copernicus Marine Service	300 m–4 km	Gap-filled multi-sensor time series
MUR SST (GHRSSST); VIIRS and MODIS SST	NASA PO.DAAC	1 km–4 km	Temperature and fronts
ERA5 winds; CCMP satellite winds; OSCAR currents	Copernicus; PO.DAAC	0.25°	Wind and circulation drivers
SeaBASS in situ bio-optical archive	NASA OB.DAAC	stations	Validation of reflectance and chlorophyll
HABSOS (Gulf of Mexico cell counts); Florida FWC red tide data; HAEDAT harmful-algae events	NOAA NCEI; FWC; IOC-ICES-PICES	points / events	Labelled bloom observations for validation or training
Indian bloom reports and advisories	INCOIS, CMFRI and scientific literature	events	Local validation
GEBCO bathymetry	GEBCO	15 arc-seconds	Shelf/coastal masks

4. Pipeline



5. Preparation

- **Quality flags:** Level-2 files contain flags for land, cloud/ice, atmospheric-correction failure, high glint, high radiance, stray light and others. Exclude flagged pixels; the flags are essential for ocean colour.
- **Heavy missing data:** expect large gaps from cloud and the monsoon. Use composites (3-day, 8-day) or a gap-tolerant approach; keep a record of how many valid observations each composite contains.
- **Log transform** chlorophyll before computing statistics because it is approximately log-normally distributed.
- **Climatology:** for every pixel (or small region) and day of year (or month) compute a multi-year median and robust spread (median absolute deviation). Anomaly = (observation – climatology) / spread.
- **Spatial features:** local standard deviation and gradient of chlorophyll and SST (fronts), area of connected high-anomaly regions.
- **Temporal features:** change from the previous composite and persistence over 2–3 composites.

6. Machine-learning approach

Approach A: statistical anomaly detection

Flag pixels where the log-chlorophyll anomaly exceeds 2 (or 3) robust standard deviations. Keep only connected regions larger than a minimum size that persist for at least two days. This is simple, transparent and an excellent baseline.

Approach B: unsupervised clustering of water types

- **Standardise** a feature set per pixel: Rrs at several bands (or ratios), chlorophyll, FLH, SST, local texture, anomaly.
- **PCA** to reduce correlated spectral features to a few components (typically 3–6 explaining 90 percent or more of the variance). Inspect the loadings: the first component often tracks overall brightness/turbidity, later ones spectral shape.
- **k-means** groups pixels into k water types; choose k using silhouette scores and interpretability. Plot the mean Rrs spectrum of each cluster.
- **DBSCAN** or **Gaussian mixtures** identify dense groups and, importantly, label sparse outliers as unusual water signatures without forcing them into a cluster.
- **Interpretation**: label clusters as clear oceanic, productive upwelling, river plume/turbid, bloom-like, and artefacts by examining spectra, chlorophyll, SST and location.

Approach C: semi-supervised or supervised classification

If you have bloom records (for example cell counts above an agency's threshold), match them with satellite pixels within a day and a few kilometres. Train Random Forest or gradient boosting on features from that matchup to predict bloom/non-bloom. Because positive records are few and clustered in time, evaluate by leaving out entire events or years and report precision-recall performance.

WORKED EXAMPLE Interpreting one flagged patch

On day 41 of a bloom season, a patch has log-chlorophyll mean 2.1 mg/m^3 versus the climatological mean of 1.2 mg/m^3 (SD 0.30), giving $z = (2.1 - 1.2) / 0.30 = 3.0$. FLH is positive, SST is 0.8°C above the 8-day mean, wind speed is 2 m/s (calm), and the green/blue Rrs ratio is 1.4 against a typical 0.7 . After PCA and k-means the pixels fall in the cluster whose mean spectrum has a green peak and low blue reflectance, and the patch contains 140 pixels (about 140 km^2 at 1 km). The patch persists for 3 days and sits 12 km offshore, away from the river mouth.

The verdict: 'high-probability bloom candidate, recommend field sampling'. If the patch hugged the coast right at a river mouth with a high CDOM signature, the same chlorophyll value would instead be flagged as likely river plume. (All numbers hypothetical.)

7. Evaluation

- **Event matchups**: with HABSOS or similar records, compute hit rate (probability of detection), false-alarm ratio and lead/lag between first satellite flag and first in situ report.
- **Case studies**: compare maps for well-documented blooms with published studies.
- **Cluster validity**: silhouette score and stability across random starts and across years; expert review of cluster spectra.
- **Sensitivity analysis**: how do results change with the anomaly threshold, minimum patch size and compositing window?

- **Failure modes:** river plumes, resuspended sediment, dust deposition, sun glint, thin cloud edges and atmospheric correction failure are the most common sources of false alarms; map them explicitly.

8. Colab strategy

Use Level-3 daily or 8-day products at 4 km for the wide region, then drill down to Level-2 swaths (1 km) around events. Store climatologies as compact arrays (day-of-year $\times$ lat $\times$ lon). K-means and PCA on a few hundred thousand pixels run in seconds on CPU.

9. Extensions

- Use PACE hyperspectral reflectance and phytoplankton-group products to separate bloom types.
- Forecast bloom likelihood seven days ahead from SST, wind, lagged chlorophyll and river discharge.
- Study freshwater cyanobacterial blooms in lakes and reservoirs with Sentinel-2/OLCI.
- Link bloom timing to fish catch or aquaculture mortality records.
- Apply a time-series model (for example Isolation Forest on the temporal features of each region).

WATCH OUT Common pitfalls

Treating every chlorophyll peak as harmful; ignoring atmospheric-correction flags; using coastal pixels with open-ocean algorithms; interpreting clusters without checking spectra; random splits of matchup points from the same event; expecting satellites to detect toxins.

PROJECT 12

Urban Expansion Forecaster

Learning how cities grow and simulating where they will grow next

PROJECT AT A GLANCE

Goal	Map built-up area through time, learn the drivers of conversion, and forecast the location of new urban land over 5–10 years.
Core data	Landsat/HLS, Sentinel-2, GHSL, WSF Evolution, VIIRS nighttime lights, WorldCover, OSM roads, WorldPop, DEM
ML / CV	Land-cover classification, transition-potential modelling (RF/GBM), cellular automata or ConvLSTM/U-Net, Figure-of-Merit validation
Difficulty / Colab fit	4/5 – needs care with consistency across years; work at 30–90 m on a 40–60 km window
Output	Built-up maps for several epochs, growth-probability map, scenario simulation of future growth

NOTE Note on this chapter

Project 12 appears in the original project list (Landsat/HLS with CNN and temporal ML) but without a written description. This chapter develops it in the same format as the others.

1. Problem and motivation

India is urbanising quickly. Cities expand into farmland, wetlands and floodplains, creating pressure on housing, transport, water and drainage, increasing heat and flood exposure, and consuming agricultural land. Planners need to know not only how big a city will become but **where** growth is most likely, so that roads, schools, drainage and green belts can be planned before development occurs.

The project has two halves: **measurement** (map urban land in several years from satellite imagery) and **forecasting** (learn from past conversion patterns and simulate future growth). The forecasting half is a classic problem in land-change science and is well suited to a combination of tabular machine learning, CNNs and temporal models.

2. Earth-science and urban-science background

How cities grow

- **Infill:** development of vacant land inside the existing urban area.
- **Edge expansion:** growth outward from the urban boundary, the most common type.
- **Leapfrog (outlying) growth:** new patches appear away from the existing city, often along highways, near industrial parks, or around new rail and metro stations.
- **Corridor growth:** development stretching along transport routes.

Drivers of conversion

Driver group	Examples	Direction of effect
Accessibility	Distance to roads, highways, rail/metro stations, city centre, employment hubs	Closer means higher probability
Neighbourhood	Share of built-up cells within 300 m, 1 km, 3 km	More built-up neighbours means higher probability (agglomeration)
Physical	Slope, elevation, floodplain, water bodies	Steep or flooded land is less likely to be built
Socio-economic	Population density and growth, nighttime-light intensity and its growth	Higher growth means higher pressure
Policy and constraints	Protected areas, forest, planned zones, special economic zones	Protected land is unlikely to convert; planned zones are likely
Land cover	Current cover: farmland, scrub, bare land, wetland	Bare/scrub converts more easily than forest

Remote sensing of built-up land

Built-up surfaces have high SWIR reflectance and low vegetation signals. Indices such as NDBI ($(\text{SWIR1} - \text{NIR})/(\text{SWIR1} + \text{NIR})$) and NDVI help, but in India's dry season bare soil and fallow fields often mimic buildings, so index thresholding alone is unreliable. Supervised classifiers using multi-season composites and texture work better. Nighttime lights from VIIRS (NASA Black Marble) measure economic and settlement intensity independent of daytime reflectance.

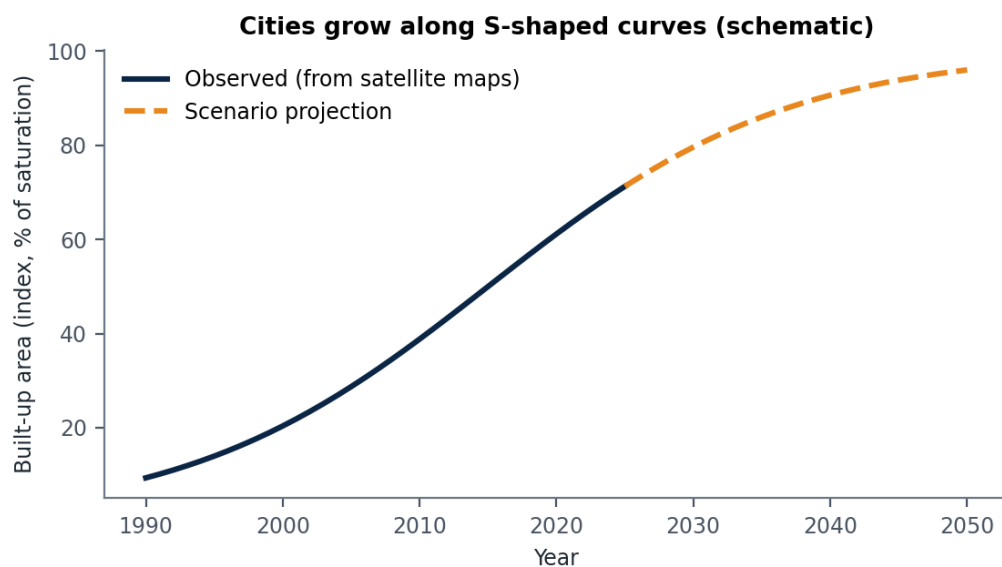
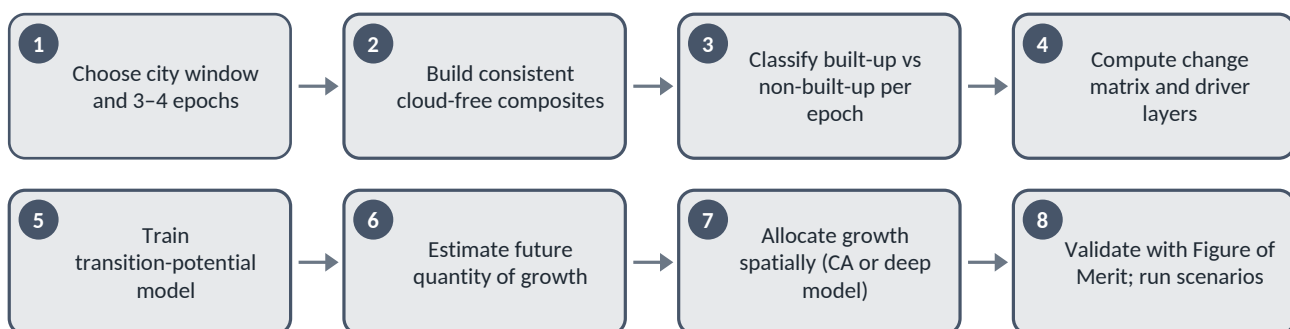


Figure 1. Urban areas typically grow along an S-shaped (logistic) curve: slow start, rapid middle phase, saturation. Schematic. Forecasting the 'quantity' of growth and the 'location' of growth are separate sub-problems.

3. Data you can use (all free)

Dataset	Provider and access	Resolution	Role
Landsat 5/7/8/9; HLS	USGS, Earth Engine, LP DAAC	30 m	Multi-epoch imagery for your own classification
Sentinel-2	Copernicus	10 m	Recent, high-quality built-up mapping
Global Human Settlement Layer (GHS-BUILT-S, GHS-POP, GHS-SMOD)	European Commission JRC	10–100 m, epochs 1975–2030	Ready built-up time series for labels or comparison; includes projections
World Settlement Footprint Evolution; Global Artificial Impervious Area (GAIA)	DLR; Tsinghua University	30 m, 1985 onward	Alternative historical settlement maps
ESA WorldCover 2020/2021; Dynamic World; Esri 10 m land cover	ESA; Google; Esri	10 m	Current land cover; static constraints
VIIRS nighttime lights (NASA Black Marble); DMSP-OLS (older)	NASA LAADS DAAC; NOAA	500 m; 1 km	Economic activity and growth intensity
WorldPop; GHS-POP	WorldPop; JRC	100 m	Population density and change
OpenStreetMap roads, rail, stations, protected areas; WDPA	OSM; Protected Planet	vector	Accessibility and constraints
NASADEM / Copernicus DEM; JRC Surface Water	LP DAAC; JRC	30 m	Slope, floodplain, water
Master plans and municipal zoning maps	Local planning authorities, city portals, Bhuvan	maps	Policy constraints (digitise a small area manually)

4. Pipeline



5. Mapping the past reliably

- **Composite design:** build dry-season cloud-free median composites for each epoch (for example 2000, 2010, 2020) from Landsat/HLS, keeping the same season each time to avoid false change.
- **Training labels:** sample built-up and non-built-up points from GHSL/WorldCover and high-resolution imagery. Use these (or GHSL directly) to train a Random Forest classifier on bands, indices (NDBI, NDVI, MNDWI), texture and nighttime lights.
- **Temporal consistency:** urban land rarely reverts to non-urban. Enforce irreversibility (if a pixel is built-up in a later epoch, it must be built-up in all subsequent epochs, with checks against classification error).

- **Accuracy assessment:** for each epoch, evaluate classification with independent samples and report overall accuracy, user's and producer's accuracy for the built-up class.

6. Machine-learning approach

Component 1: transition potential (where?)

Take cells that were non-built-up at the start of an epoch. The label is 1 if the cell became built-up by the end. Predictors are the driver layers measured at the start of the epoch, including the share of built-up neighbours. Train logistic regression (interpretable baseline), Random Forest and gradient boosting. The model output is a map of **transition potential**: the probability that each non-built-up cell converts. Because only a small fraction of cells convert, use class weights and evaluate with AUC and PR-AUC.

Component 2: quantity (how much?)

Decide how many cells will convert in the forecast period. Options: extrapolate the historical growth rate; fit a logistic growth curve; link to population projections and a per-capita built-up area; or take the quantity from GHSL projections. Scenarios (compact growth, business-as-usual, sprawl) change the quantity or the weights.

Component 3: allocation (simulation)

- **Cellular automata (CA):** step by step, convert the cells with the highest transition potential, adjusting for neighbourhood effects and constraints, until the target quantity is reached. CA-Markov and SLEUTH-style models are classic examples.
- **Deep spatiotemporal models:** a CNN, U-Net or ConvLSTM that takes a stack of past built-up maps plus driver layers and outputs the built-up probability for the next epoch, learning local growth patterns directly. Train on pairs such as 2000→2010 and validate on 2010→2020.
- **Stochastic simulation:** run the allocation many times with random draws to produce a probability-of-development map and an uncertainty range.

WORKED EXAMPLE Quantity, allocation and the Figure of Merit

A city had 180 km² built-up in 2010 and 262 km² in 2020. Growth was +82 km², a compound annual growth rate of $(262 / 180)^{(1/10)} - 1 \approx 3.8$ percent. If the same rate persisted, 2030 built-up area would be about $262 \times 1.456 \approx 381$ km² (+119 km²); a logistic slowdown might give a smaller number such as 340 km². At 30 m resolution 82 km² equals about 91,000 pixels.

Validation on 2010→2020: the model is told to convert 91,000 pixels. Observed change: 91,000 pixels. Hits (predicted change and observed change agree) = 38,000; misses (observed change not predicted) = 53,000; false alarms (predicted change not observed) = 53,000. Figure of Merit = hits / (hits + misses + false alarms) = 38,000 / 144,000 ≈ 0.26 . A null model (no change) has FoM = 0 by definition, so even a modest FoM represents real skill; land-change models commonly achieve modest FoM values because the exact location of development is partly random. (All numbers hypothetical.)

7. Evaluation

- **Hold-out in time:** calibrate on 2000→2010, simulate 2010→2020 and compare with the real 2020 map.
- **Figure of Merit and its components:** hits, misses, false alarms; also quantity disagreement versus allocation disagreement (Pontius' approach).
- **AUC of transition potential** on held-out spatial blocks.
- **Comparisons:** against a null model (no change), a random-allocation model and a simple distance-to-built-up rule.
- **Plausibility checks:** does growth respect protected areas and water? Are patch sizes and shapes similar to the observed pattern (landscape metrics)?

8. Colab strategy

Process a 50×50 km window at 30 m (2.8 million cells), or aggregate to 60–90 m for modelling. Sample 200,000 cells for training the transition model, but predict on all non-built-up cells in tiles. Vectorised array operations make CA simulations fast; for deep models use 64×64 or 128×128 patches.

9. Extensions

- Overlay forecast growth with flood hazard (Project 1), heat hotspots (Project 3) and landslide susceptibility (Project 6) to find risky growth zones.
- Compare three policy scenarios: business as usual, compact growth, and protected-green-belt.
- Estimate loss of farmland or wetland under each scenario.
- Compare several cities and discover whether growth drivers differ.
- Use nighttime light trends as an early indicator of where growth is accelerating.

WATCH OUT Common pitfalls

Creating fake change through inconsistent classification across years; mixing seasons; allowing urban pixels to revert; predicting with drivers measured at the end rather than the beginning of the period; ignoring policy constraints; reporting overall accuracy for a problem where nearly all cells do not change.

PART III

Appendices

Dataset directory, index formulas, skills map, glossary, responsible use and further learning.

Appendix A: Dataset Directory

Where to find the data used across the twelve projects

This appendix lists the principal datasets by theme, with the provider and the projects that use them. Product names and versions evolve; always confirm the current version, coverage dates and status on the provider's page before you build a study around a dataset.

A.1 Optical and thermal imagery

Dataset	Provider	Access route	Projects
HLS (HLSL30, HLSS30)	NASA LP DAAC	Earthdata Search, earthaccess, AppEEARS	1, 4, 7, 8, 12
Landsat Collection 2 (Level-1 and Level-2)	USGS	EarthExplorer, Earth Engine, Planetary Computer, AWS	3, 4, 5, 7, 10, 12
Sentinel-2 MSI	ESA / Copernicus	Copernicus Data Space, Earth Engine, AWS	1, 5, 7, 10, 12
ECOSTRESS LST/emissivity, ET, ESI	NASA LP DAAC	Earthdata Search, AppEEARS	3, 4
ASTER	NASA LP DAAC	Earthdata Search	10

A.2 Daily and coarse-resolution land, atmosphere and ocean products

Dataset	Provider	Access route	Projects
MODIS and VIIRS land products (NDVI/EVI, LST, albedo, LAI, NPP, burned area, land cover)	NASA LP DAAC	AppEEARS, Earthdata Search, Earth Engine	2, 4, 8, 9, 10
MODIS/VIIRS active fire	NASA FIRMS / LANCE	FIRMS downloads and API	2, 9
MAIAC AOD (MCD19A2); VIIRS aerosol	NASA LP DAAC / LAADS	AppEEARS, Earthdata Search	9
OMI and TROPOMI trace gases	NASA GES DISC; Copernicus	Earthdata Search; Copernicus Data Space	9
MODIS/VIIRS snow cover	NASA NSIDC DAAC	Earthdata Search	10
Ocean colour (MODIS-Aqua, VIIRS, PACE)	NASA OB.DAAC	OceanColor Web, earthaccess	11
VIIRS Black Marble nighttime lights	NASA LAADS DAAC	Earthdata Search	12

A.3 Radar and lidar

Dataset	Provider	Access route	Projects
Sentinel-1 GRD/SLC	ESA / Copernicus; ASF DAAC	Copernicus Data Space, ASF Vertex, Planetary Computer	1, 5, 6, 10
ALOS PALSAR mosaics	JAXA; ASF DAAC	JAXA free downloads, Earth Engine	5
OPERA surface-water and disturbance products	NASA / JPL OPERA	Earthdata (ASF, LP DAAC)	1
GED1 L2A, L2B, L4A, L4B	NASA LP DAAC / ORNL DAAC	Earthdata Search, earthaccess	5
ICESat-2 ATL06 and ATL08	NASA NSIDC DAAC	Earthdata Search	5, 10

A.4 Precipitation, climate and soil

Dataset	Provider	Access route	Projects
GPM IMERG	NASA GES DISC	Earthdata Search, Giovanni	1, 2, 4, 6, 8
CHIRPS	UC Santa Barbara Climate Hazards Center	Direct download, Earth Engine	4, 6, 8
IMD gridded rainfall and temperature	India Meteorological Department	IMD Pune data portal	4, 6, 8
ERA5 and ERA5-Land	Copernicus Climate Data Store	CDS API, Earth Engine	2, 3, 4, 7, 8, 9, 10
MERRA-2	NASA GES DISC	Earthdata Search	9
SMAP soil moisture	NASA NSIDC DAAC	Earthdata Search	4, 8
TerraClimate; WorldClim; CHELSA	Public research repositories	Direct download, Earth Engine	5, 8
NEX-GDDP-CMIP6	NASA NCCS	Public cloud bucket	10
SoilGrids	ISRIC	Download or API	6, 8

A.5 Terrain, land cover, population and vector layers

Dataset	Provider	Access route	Projects
NASADEM; SRTM; Copernicus DEM GLO-30	NASA LP DAAC; ESA	Earthdata, Planetary Computer, Earth Engine	1, 2, 3, 5, 6, 7, 10, 12
ESA WorldCover; Dynamic World; MCD12Q1	ESA; Google; NASA	Earth Engine, Planetary Computer, LP DAAC	1, 3, 4, 5, 8, 12
JRC Global Surface Water	European Commission JRC	Earth Engine, direct download	1, 7, 10
GHSL (built-up, population, degree of urbanisation)	European Commission JRC	Direct download, Earth Engine	3, 12
WorldPop	University of Southampton	Direct download	1, 3, 9, 12

Dataset	Provider	Access route	Projects
Google Open Buildings; Microsoft Building Footprints	Google; Microsoft	Direct download	1, 3
OpenStreetMap	OSM community	Overpass API, extracts, OSMnx	1, 3, 6, 9, 12
HydroSHEDS	WWF / HydroSHEDS	Direct download	1, 6

A.6 Labels, inventories and ground truth

Dataset	Provider	Used for	Project
Sen1Floods11; WorldFloods	Cloud to Street; Oxford / ESA	Flood masks	1
Global Flood Database; Dartmouth Flood Observatory	Cloud to Street; Dartmouth	Flood events	1
Next Day Wildfire Spread dataset; MTBS perimeters	Google Research; USGS	Fire spread labels and perimeters	2
ISMN soil-moisture network	ISMN	Soil-moisture truth	4
GEDI L4A; ORNL biomass maps; ESA CCI Biomass	NASA ORNL DAAC; ESA	Biomass labels and benchmarks	5
NASA COOLR / Global Landslide Catalog; Landslide4Sense; GSI and NRSC inventories	NASA; research community; GSI; ISRO-NRSC	Landslide points and patches	6
CPCB continuous monitoring; OpenAQ	CPCB; OpenAQ	PM2.5 truth	9
RGI 7.0; GLIMS; ITS_LIVE	RGI Consortium; NSIDC; NASA-supported	Glacier outlines and velocity	10
SeaBASS; HABSOS; HAEDAT	NASA OB.DAAC; NOAA; IOC-ICES-PICES	Ocean validation and bloom records	11
ISRO Desertification and Land Degradation Atlas	ISRO-SAC	Degradation reference	8

Appendix B: Index Formulas and Metric Reference

Compact formulas for quick lookup

Formulas use reflectance values for the named bands. Band names are generic (Green, Red, NIR, SWIR1, SWIR2); check the band numbering of your specific sensor (for example Landsat 8/9 OLI: Green = B3, Red = B4, NIR = B5, SWIR1 = B6, SWIR2 = B7; Sentinel-2: Green = B3, Red = B4, NIR = B8, SWIR1 = B11, SWIR2 = B12; HLS follows the same convention within each product, so read the product guide).

Index	Formula	Purpose	Projects
NDVI	$(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$	Vegetation greenness and vigour	2, 3, 4, 5, 6, 8, 12
EVI	$2.5 \times (\text{NIR} - \text{Red}) / (\text{NIR} + 6 \times \text{Red} - 7.5 \times \text{Blue} + 1)$	Dense canopy vegetation	4, 5, 8
SAVI	$(\text{NIR} - \text{Red}) \times (1 + L) / (\text{NIR} + \text{Red} + L)$, L about 0.5	Vegetation in sparse cover (soil-adjusted)	8
NDWI (McFeeters)	$(\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$	Open water	1, 3, 7
MNDWI	$(\text{Green} - \text{SWIR1}) / (\text{Green} + \text{SWIR1})$	Water in urban and coastal scenes	1, 7, 12
NDMI	$(\text{NIR} - \text{SWIR1}) / (\text{NIR} + \text{SWIR1})$	Vegetation moisture	2, 4
NBR	$(\text{NIR} - \text{SWIR2}) / (\text{NIR} + \text{SWIR2})$	Burned areas; dNBR = pre-fire NBR minus post-fire NBR	2
NDBI	$(\text{SWIR1} - \text{NIR}) / (\text{SWIR1} + \text{NIR})$	Built-up land	3, 12
NDSI	$(\text{Green} - \text{SWIR1}) / (\text{Green} + \text{SWIR1})$	Snow and ice	10
Red/SWIR ratio	Red / SWIR1	Glacier ice mapping	10
VCI	$(\text{NDVI} - \text{NDVI}_{\min}) / (\text{NDVI}_{\max} - \text{NDVI}_{\min})$ for that place and period	Vegetation condition relative to history	4, 8
TCI	$(\text{LST}_{\max} - \text{LST}) / (\text{LST}_{\max} - \text{LST}_{\min})$	Temperature condition	4
VHI	$a \times \text{VCI} + (1 - a) \times \text{TCI}$, commonly $a = 0.5$	Combined vegetation health / drought	4
ESI	Actual ET / Potential ET (or its standardised anomaly)	Evaporative stress	4
Rain-use efficiency	Productivity (NPP or peak NDVI) / annual rainfall	Land degradation indicator	8
Aridity index	Annual precipitation / potential evapotranspiration	Dryland classes	8
Ventilation coefficient	Boundary-layer height $\times$ mean wind speed	Dispersion capacity of air	9
Positive degree days	Sum of daily mean temperature above 0 °C	Melt potential	10
Figure of Merit	Hits / (hits + misses + false alarms)	Skill of land-change simulation	12
Carbon from biomass	Carbon $\approx 0.47 \times \text{biomass}$; CO ₂ e = Carbon $\times 44/12$	Forest carbon	5

B.2 Metrics cheat sheet for confusion-matrix problems

Let TP be true positives, FP false positives, FN false negatives and TN true negatives.

Metric	Formula	Notes
Precision	$TP / (TP + FP)$	Reliability of positive predictions
Recall	$TP / (TP + FN)$	Share of real events found
F1	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$	Balanced summary
IoU	$TP / (TP + FP + FN)$	Overlap for segmentation; $F1 = 2 \times \text{IoU} / (1 + \text{IoU})$
Accuracy	$(TP + TN) / \text{total}$	Misleading when events are rare
False-alarm ratio	$FP / (TP + FP)$	Used in early-warning verification
Probability of detection	$TP / (TP + FN)$	Same as recall
RMSE	Square root of mean squared error	Penalises large errors
Bias	$\text{Mean}(\text{prediction} - \text{observation})$	Systematic over- or under-prediction

Appendix C: Skills Map and Capstone Plan

Turning a project chapter into a finished piece of work

C.1 Skills map: what each project teaches

Skill	Projects where it is central
Raster handling, CRS, resampling, masking	All
Radar processing and thresholding	1, 5, 6, 10
Thermal remote sensing and land surface temperature	3, 4
Classical computer vision (thresholds, contours, morphology)	1, 7, 10, 12
Tabular ML (Random Forest, gradient boosting)	2, 3, 4, 5, 6, 9
Deep learning for imagery (CNN, U-Net)	1, 2, 3, 6, 10, 12
Time-series analysis and forecasting	4, 7, 8, 10, 12
Anomaly detection and unsupervised learning	8, 11
Spatial validation and transferability	All, especially 5, 6, 9
Uncertainty and interpretation	All, especially 5, 7, 10
Scenario analysis and decision support	3, 6, 10, 12

C.2 A twelve-week capstone plan

Week	Activity	Deliverable
1	Pick one project; read its chapter twice; write the question in one sentence and define what success means	One-page project brief
2	Choose region and period; create the Earthdata account; explore data in Earthdata Search/Worldview	Study-area map and data inventory table
3	Download a small subset; inspect quality flags, scale factors, projections	Data-quality notes
4	Build the preprocessing pipeline and a tidy analysis table or patch set	Analysis-ready dataset and data card
5	Exploratory analysis: maps, histograms, correlations, seasonal behaviour	EDA figures and findings
6	Implement rule-based and simple statistical baselines	Baseline results
7	Train the main ML model with spatial/temporal validation	Cross-validated metrics
8	Feature importance, error maps, sensitivity analysis	Error analysis report
9	Advanced model or extension	Comparison against baselines
10	Final maps, uncertainty, scenarios	Publication-quality figures
11	Write the report and model card; review limitations and ethics	Draft report
12	Polish, share notebooks and README, present	Final report and repository

C.3 Project brief template

Heading	Prompts
Question	What decision does this support, and who would use it?
Study area and period	Why this area? Which dates? Which events or seasons?
Data	Which datasets and versions? What resolution and coverage? What are their known limits?
Target and labels	What exactly is predicted? Where do labels come from and how reliable are they?
Method	Which baseline? Which main model? Why?
Validation	How are train and test separated (space, time, event)? Which metrics?
Risks	What could invalidate the result (leakage, bias, sensor change, missing data)?
Deliverables	Which maps, tables and documents will you produce?

Appendix D: Glossary

Key terms used throughout the book

Term	Meaning
Albedo	Fraction of incoming sunlight reflected by a surface; high for snow, low for dark roofs
Anomaly	Difference between an observation and its typical (climatological) value for that place and time
Antecedent condition	Wetness or state of the land before an event; for example soil moisture before a storm
AOD	Aerosol optical depth; how much light particles in the atmospheric column remove
Backscatter	Fraction of radar energy returned to the sensor, usually in decibels
Boundary layer	Lowest part of the atmosphere, in contact with the surface and well mixed
Brightness temperature	Temperature of an ideal emitter that would produce the measured radiance
CDOM	Coloured dissolved organic matter, which tints water yellow-brown
Class imbalance	Rare positive class compared with a very large negative class
CNN	Convolutional neural network; learns spatial filters from images
COG	Cloud-optimised GeoTIFF; supports reading small windows over the network
CRS	Coordinate reference system; defines how coordinates relate to places on Earth
DAAC	Distributed Active Archive Center; NASA data centre specialised by discipline
DEM	Digital elevation model; a raster of terrain height
Emissivity	Efficiency with which a surface emits thermal radiation
Evapotranspiration	Combined evaporation and plant transpiration of water to the atmosphere
Feature	An input variable used by a model
Figure of Merit	Agreement measure for land-change simulations focusing on changed cells
Footprint	Area on the ground illuminated or sampled by a lidar pulse
Granule	Single file or unit of a NASA data product (a scene, orbit segment or tile)
HAND	Height above nearest drainage; elevation above the closest river channel along the flow path
Hyperspectral	Measuring many narrow, contiguous wavelength bands
IoU	Intersection over union; overlap measure for masks
Isolation Forest	Anomaly detection method that isolates unusual points with random splits
Label	The true outcome or class that a supervised model learns to predict
Leakage	Information from outside the training data (often from the test set) that makes validation unrealistically good
Level 1/2/3/4	NASA data-processing levels from calibrated radiance to gridded or modelled products
LST	Land surface temperature; radiative skin temperature of the ground
Mass balance	Annual gain minus loss of glacier mass
NDVI	Normalised Difference Vegetation Index
Overfitting	Model memorises training data and fails on new data

Term	Meaning
PCA	Principal component analysis; rotates data into uncorrelated components ordered by variance
Phenology	Seasonal timing of vegetation growth stages
PM2.5	Particulate matter smaller than 2.5 micrometres in diameter
Radiance / reflectance	Energy measured by a sensor / fraction of incoming light reflected by the surface
Random Forest	Ensemble of many decision trees that vote or average
Revisit time	How often a sensor observes the same location
Rrs	Remote-sensing reflectance used in ocean colour
SAR	Synthetic aperture radar; active microwave imaging that works through cloud
SHAP	Method that attributes a prediction to feature contributions
Spatial autocorrelation	Tendency of nearby locations to have similar values
Speckle	Grainy noise inherent to radar images
STAC	SpatioTemporal Asset Catalog; standard for describing and searching geospatial data
Susceptibility / hazard / risk	Where it could happen / where, when and how severe / hazard combined with exposure and vulnerability
Transect	Straight line along which measurements are taken, for example across a beach
U-Net	Encoder-decoder CNN for pixel-wise segmentation
VPD	Vapour pressure deficit; the drying power of the air

Appendix E: Responsible Use and Failure Modes

Ethics, limitations and a debugging checklist

E.1 Responsible use of Earth-observation models

- **Communicate uncertainty.** Every map should show or state its uncertainty. A flood or landslide map presented as certain can mislead emergency managers.
- **Prefer recall for safety-critical warnings, precision for costly interventions.** Be explicit about the trade-off you chose and why.
- **Know the bias in your labels.** Inventories, station networks and field plots reflect where people looked, not where the phenomenon exists.
- **Do not over-claim causality.** A model that links tree cover with lower temperature does not prove that planting trees on a particular street will cool it by a specific amount.
- **Respect privacy.** High-resolution exposure maps can identify individual households. Aggregate to wards or grid cells when sharing.
- **Consider equity.** Poorer neighbourhoods often have fewer monitors, fewer trees and less data. Models trained on well-instrumented areas may perform worst where need is greatest.
- **Separate research from operational warning.** A student project is not an official forecast. Point users to the authorities (IMD, CWC, GSI, INCOIS, CPCB, state disaster management authorities) for decisions.
- **Keep it reproducible.** Share code, data references and versions so others can check and improve your work.

E.2 Common failure modes checklist

Failure mode	Symptom	Remedy
Spatial leakage	Near-perfect test scores that collapse on a new region	Block or leave-one-region-out validation
Temporal leakage	Using future information in features	Forward-chaining validation; audit feature timestamps
Label noise	Model plateaus; errors cluster around unreliable labels	Clean labels, estimate label quality, use robust losses
Sensor drift or change	Step changes at sensor transitions	Overlap analysis, harmonisation, change-point checks
Cloud and quality artefacts	Spurious change in cloudy months	Use quality flags; composites; radar where appropriate
Distribution shift	Model fails in a new season or city	Test transfer; fine-tune; include context features
Over-interpretation of importance	Importance of correlated features misleading	Permutation importance with grouped features; SHAP; domain reasoning
Extrapolation	Predictions outside training range	Restrict to the domain of applicability; report it

Appendix F: Further Learning

Free courses, communities and search terms

F.1 Where to learn more (free resources)

- **NASA Earthdata** (earthdata.nasa.gov): data discovery, tutorials, webinars, and the Earthdata Forum. Look for the LP DAAC, GES DISC and OB.DAAC tutorial repositories of Jupyter notebooks.
- **NASA ARSET** (Applied Remote Sensing Training): free online courses on land, water, air quality, disasters and SAR, with recordings and slides.
- **Copernicus Data Space Ecosystem** and **ESA Echoes / EO College**: Sentinel data guides and courses.
- **Google Earth Engine tutorials** and **Open Geospatial Solutions** course material for cloud-based processing.
- **SERVIR** (NASA-USAID) resources for applied satellite monitoring, including South Asia hubs at ICIMOD.
- **ISRO Bhuvan, NRSC and IIRS e-learning** for India-focused remote-sensing content and data.
- **scikit-learn user guide** for machine-learning fundamentals and model evaluation; **PyTorch and TorchGeo documentation** for deep learning on geospatial data.
- **Spatial cross-validation literature**: search for 'spatial cross-validation' and 'area of applicability' to understand why random splits mislead.

F.2 Selected topics to search for (by project)

Project	Search terms
1 Flood	Sen1Floods11; WorldFloods; ML4Floods; OPERA DSWx; HAND flood mapping; FwDET
2 Wildfire	Next Day Wildfire Spread; fire behaviour triangle; FRP; ConvLSTM fire spread
3 Urban heat	surface urban heat island; ECOSTRESS LST; SHAP urban temperature; Getis-Ord Gi*
4 Crop stress	Evaporative Stress Index; CWSI; VHI; ECOSTRESS WUE; crop stress forecasting
5 Forest carbon	GEDI L4A biomass; GEDI upscaling; allometric equations; saturation of optical biomass
6 Landslides	frequency ratio; LHASA; COOLR; Landslide4Sense; rainfall intensity-duration thresholds
7 Coast	CoastSat; DSAS; tide correction of satellite shorelines; EPR and LRR
8 Desertification	RESTREND; rain-use efficiency; BFAST; Trends.Earth; SDG 15.3.1; critical slowing down
9 Air pollution	MAIAC AOD PM2.5 estimation; ventilation coefficient; spatial transferability of PM2.5 models
10 Glaciers	RGI 7; NDSI glacier mapping; debris-covered glacier mapping; ITS_LIVE; degree-day model
11 Algal blooms	OCx chlorophyll; fluorescence line height; PACE OCI phytoplankton; Noctiluca Arabian Sea; HABSOS
12 Urban growth	GHSL; cellular automata; SLEUTH; Figure of Merit; ConvLSTM land change

TIP A final word

The best projects are not those with the highest accuracy but those that ask a precise question, use data honestly, validate in a way that matches how the model will be used, and explain clearly where the answer can and cannot be trusted. Pick one project, keep the region small, and finish it end to end before starting the next.